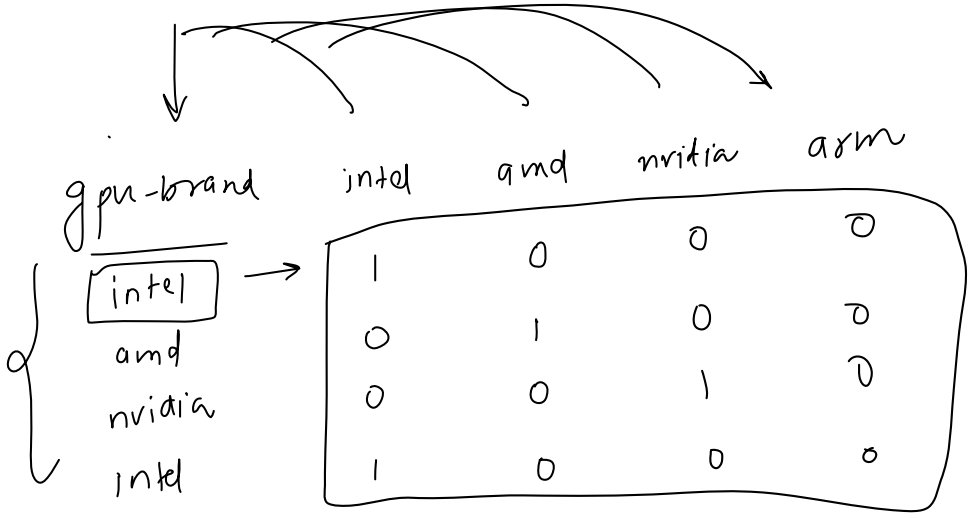
EDA

11 March 2023 15:51

1. head - tail - sample
2. for numerical cols
   - 8 number summary[count,min,max,mean,std,q1,q2,q3]
   - missing values
   - outliers
   - horizontal/vertical histograms
3. for categorical cols
   - value counts
   - pie chart
   - missing value
4. numerical - numerical
   - side by side 8 number analysis--
   - scatterplot
   - correlation
5. categorical-categorical
   - contingency table
   - stacked bar chart
6. numerical-categorical
   - compare distribution across categories
8. missing value treatment
9. feature engineering
   - ppi
   - screen_size_bracket
10. one hot encoding



What is Statistics

09 March 2023 14:56

Statistics is a branch of mathematics that involves collecting, analysing, interpreting, and presenting data. It provides tools and methods to understand and make sense of large amounts of data and to draw conclusions and make decisions based on the data.

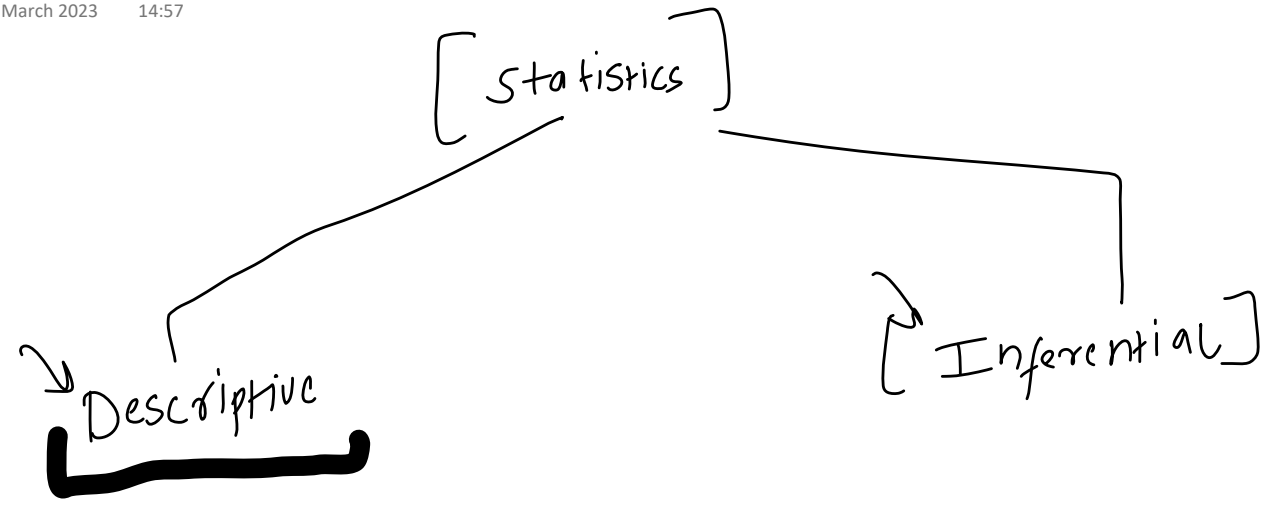
In practice, statistics is used in a wide range of fields, such as business, economics, social sciences, medicine, and engineering. It is used to conduct research studies, analyse market trends, evaluate the effectiveness of treatments and interventions, and make forecasts and predictions.

Examples:

1. Business - Data Analysis(Identifying customer behavior) and Demand Forecasting
2. Medical - Identify efficacy of new medicines(Clinical trials), Identifying risk factor for diseases(Epidemiology)
3. Government & Politics - Conducting surveys, Polling
4. Environmental Science - Climate research

Types of Statistics

09 March 2023 14:57



Descriptive statistics deals with the collection, organization, analysis, interpretation, and presentation of data. It focuses on summarizing and describing the main features of a set of data, without making inferences or predictions about the larger population.

Inferential statistics deals with making conclusions and predictions about a population based on a sample. It involves the use of probability theory to estimate the likelihood of certain events occurring, hypothesis testing to determine if a certain claim about a population is supported by the data, and regression analysis to examine the relationships between variables

Population Vs Sample

09 March 2023 14:57

Population refers to the entire group of individuals or objects that we are interested in studying. It is the complete set of observations that we want to make inferences about. For example, the population might be all the students in a particular school or all the cars in a particular city.

A **sample**, on the other hand, is a subset of the population. It is a smaller group of individuals or objects that we select from the population to study. Samples are used to estimate characteristics of the population, such as the mean or the proportion with a certain attribute. For example, we might randomly select 100 students.

Examples

1. All cricket fans vs fans who were present in the stadium
2. All students vs who visit college for lectures

Things to be careful about which creating samples

1. Sample Size
2. Random
3. Representative

Parameter Vs Statistics

A parameter is a characteristic of a population, while a statistic is a characteristic of a sample. Parameters are generally unknown and are estimated using statistics. The goal of statistical inference is to use the information obtained from the sample to make inferences about the population parameters.

Inferential Statistics

09 March 2023 14:57

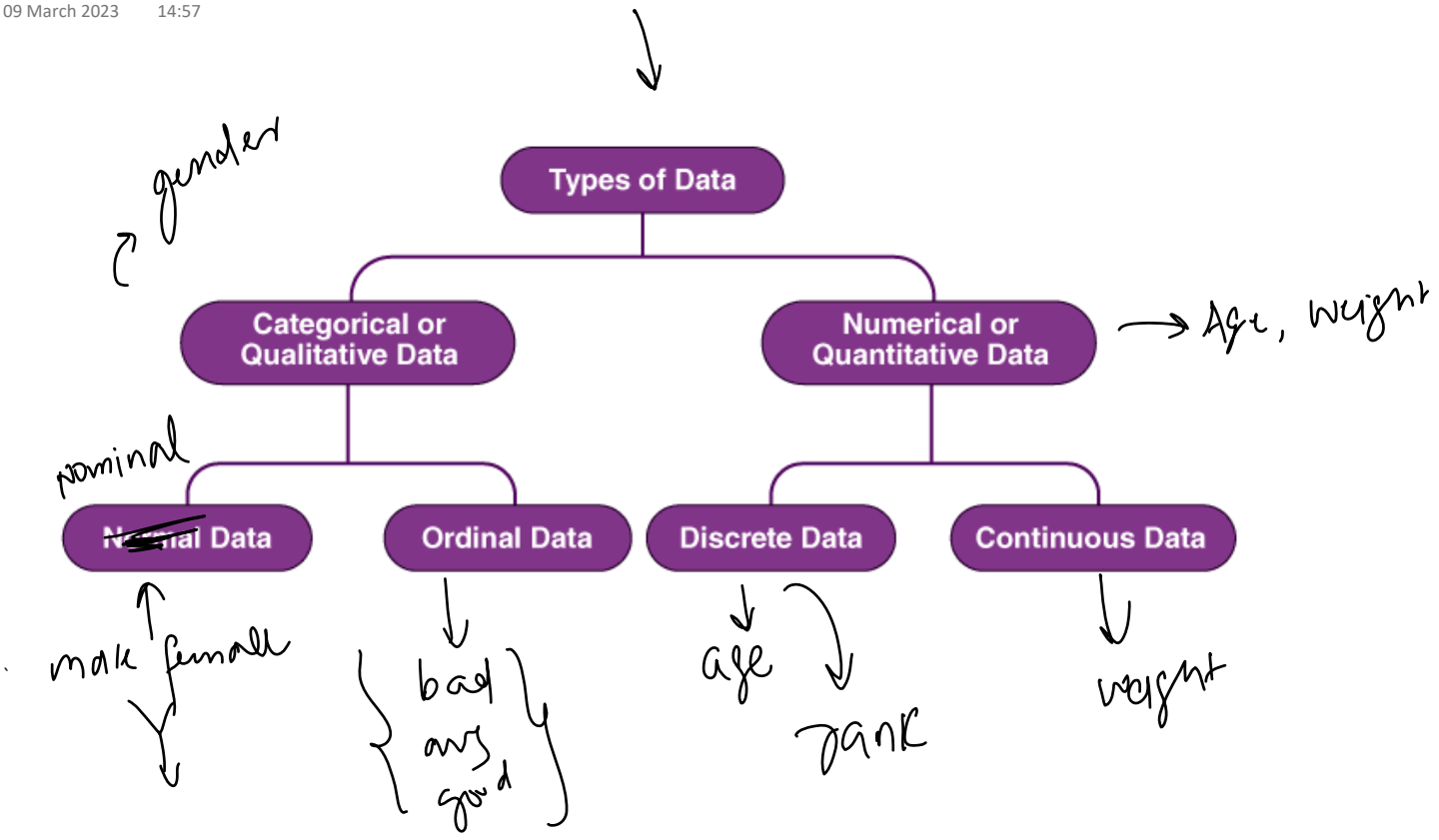
Inferential statistics is a branch of statistics that deals with making inferences or predictions about a larger population based on a sample of data. It involves using statistical techniques to test hypotheses and draw conclusions from data. Some of the topics that come under inferential statistics are:

1. **Hypothesis testing**: This involves testing a hypothesis about a population parameter based on a sample of data. For example, testing whether the mean height of a population is different from a given value.
2. **Confidence intervals**: This involves estimating the range of values that a population parameter could take based on a sample of data. For example, estimating the population mean height within a given confidence level.
3. **Analysis of variance (ANOVA)**: This involves comparing means across multiple groups to determine if there are any significant differences. For example, comparing the mean height of individuals from different regions.
4. **Regression analysis**: This involves modelling the relationship between a dependent variable and one or more independent variables. For example, predicting the sales of a product based on advertising expenditure.
5. **Chi-square tests**: This involves testing the independence or association between two categorical variables. For example, testing whether gender and occupation are independent variables.
6. **Sampling techniques**: This involves ensuring that the sample of data is representative of the population. For example, using random sampling to select individuals from a population.
7. **Bayesian statistics**: This is an alternative approach to statistical inference that involves updating beliefs about the probability of an event based on new evidence. For example, updating the probability of a disease given a positive test result.

Why ML is closely associated with statistics?

Types of Data

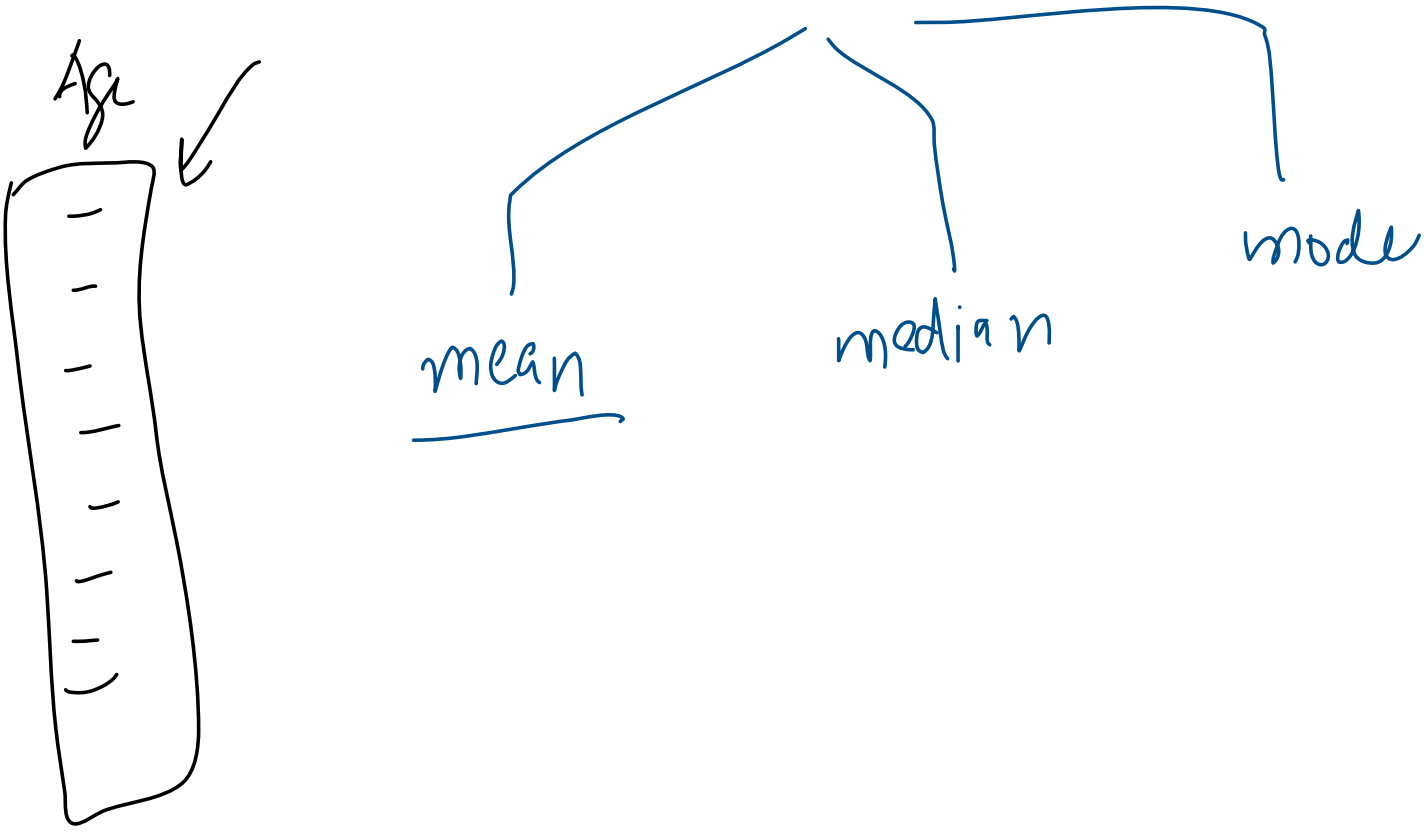
09 March 2023 14:57



Measure of Central Tendency

09 March 2023 14:58

A measure of central tendency is a statistical measure that represents a typical or central value for a dataset. It provides a summary of the data by identifying a single value that is most representative of the dataset as a whole.



1. Mean

09 March 2023 16:35

Mean: The mean is the sum of all values in the dataset divided by the number of values.

3
4
1
2
5

$= \frac{15}{5} = 3$

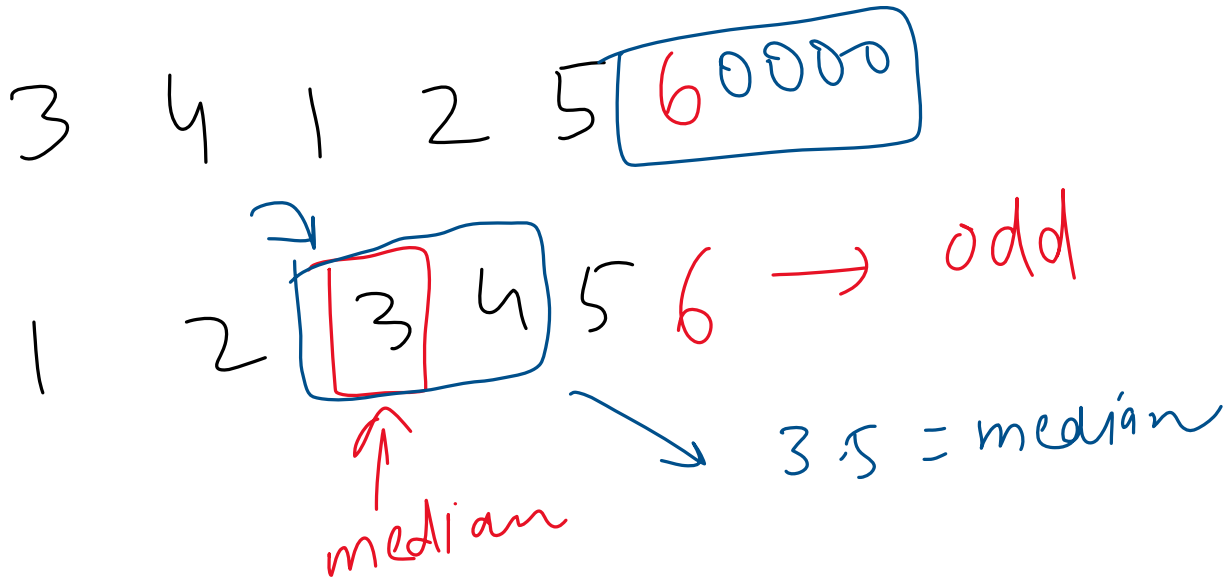
$\sum_{i=1}^{N} x_i$ / N $\qquad$ $\sum_{i=1}^{n} x_i$ / n

Population Mean	Sample Mean
$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$	$\overline{X} = \frac{\sum_{i=1}^{n} x_i}{n}$
N = number of items in the population	n = number of items in the sample

2. Median

09 March 2023 16:36

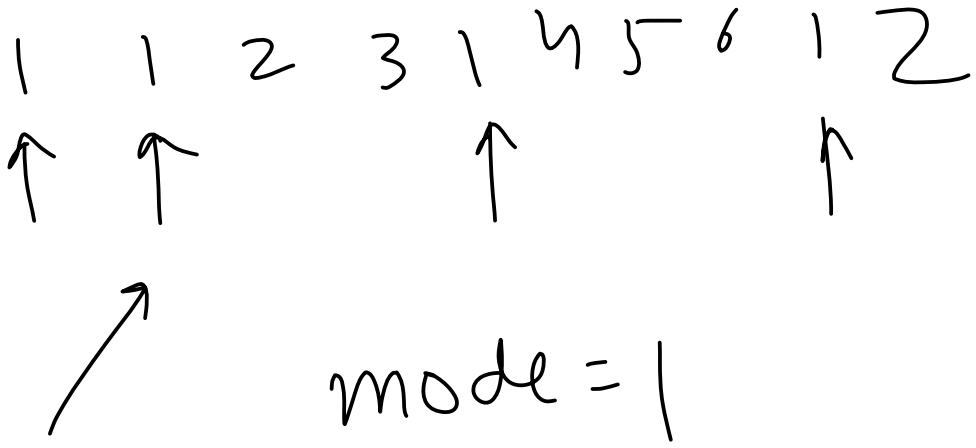
Median: The median is the middle value in the dataset when the data is arranged in order.



3. Mode

09 March 2023 16:36

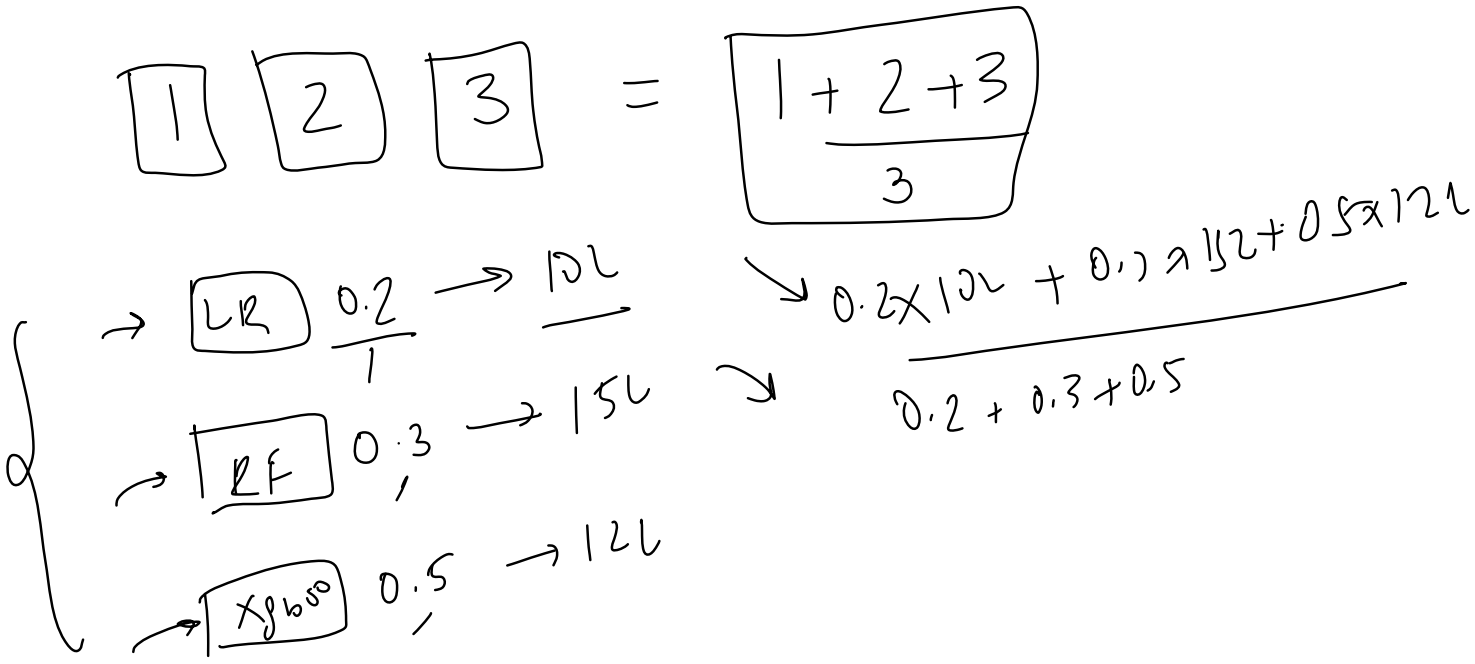
Mode: The mode is the value that appears most frequently in the dataset.



4. Weighted Mean

09 March 2023 16:39

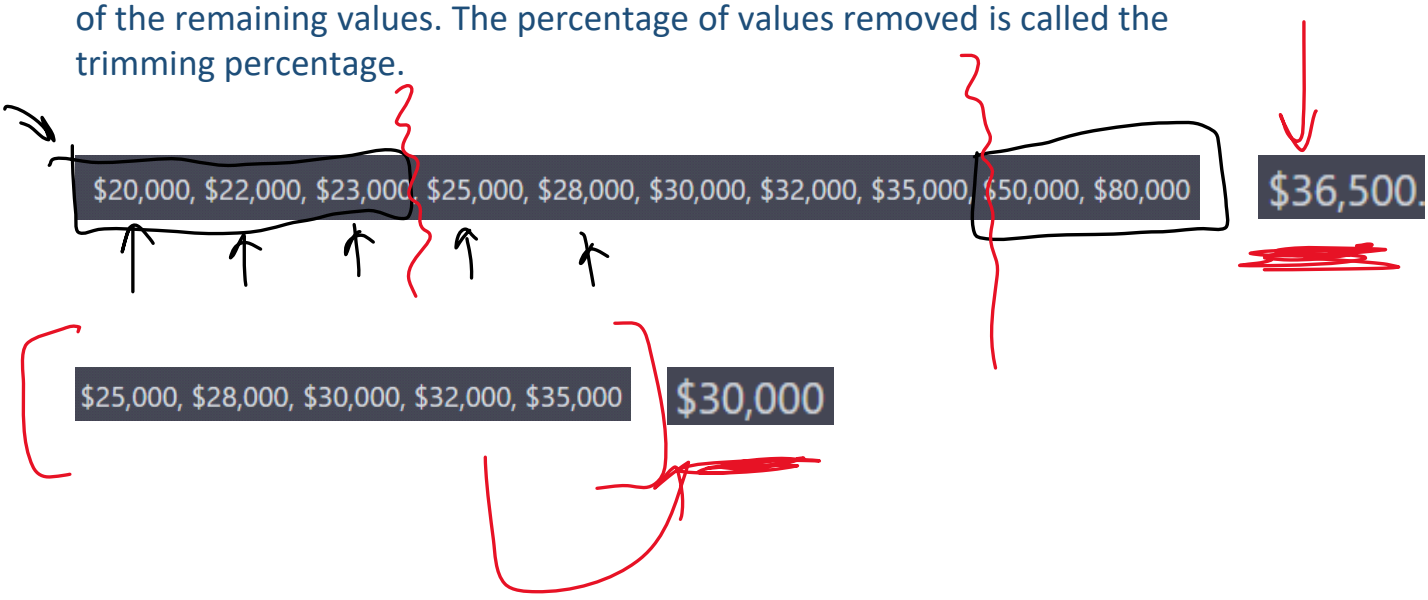
Weighted Mean: The weighted mean is the sum of the products of each value and its weight, divided by the sum of the weights. It is used to calculate a mean when the values in the dataset have different importance or frequency.



5. Trimmed Mean

10 March 2023 09:37

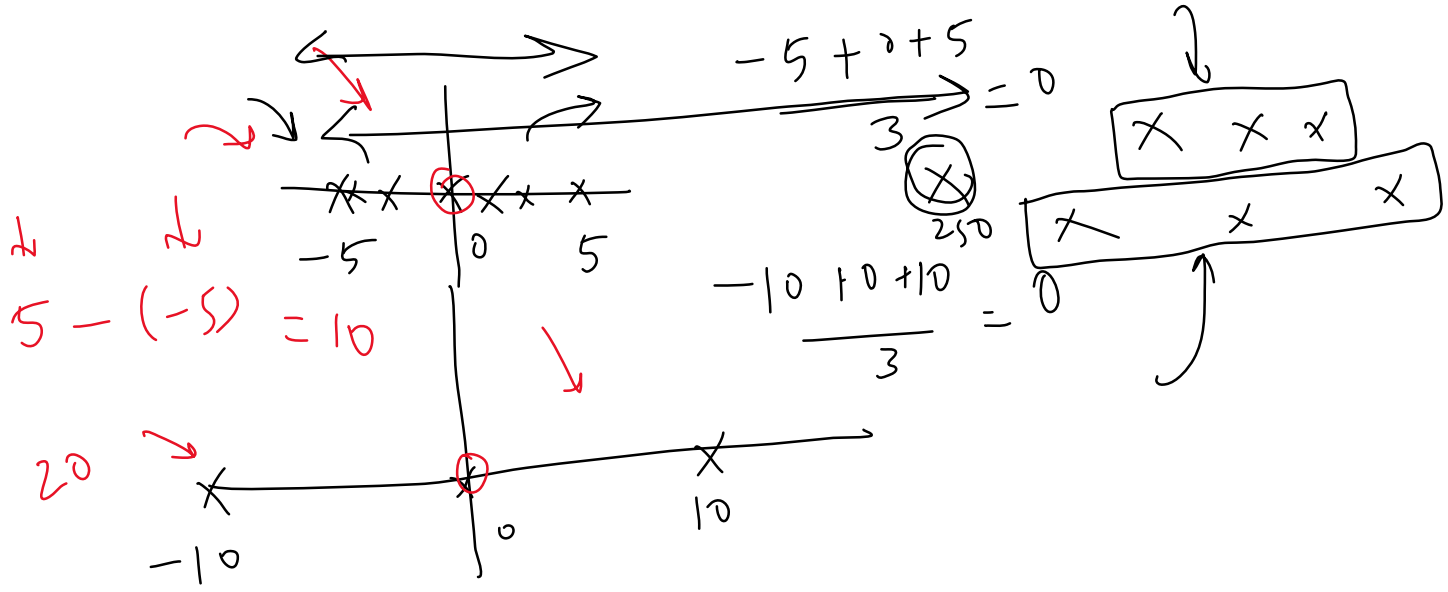
A trimmed mean is calculated by removing a certain percentage of the smallest and largest values from the dataset and then taking the mean of the remaining values. The percentage of values removed is called the trimming percentage.



Measure of Dispersion

09 March 2023 14:58

A measure of dispersion is a statistical measure that describes the spread or variability of a dataset. It provides information about how the data is distributed around the central tendency (mean, median or mode) of the dataset.



1. Range

09 March 2023 16:36

Range: The range is the difference between the maximum and minimum values in the dataset. It is a simple measure of dispersion that is easy to calculate but can be affected by outliers.

2. Variance

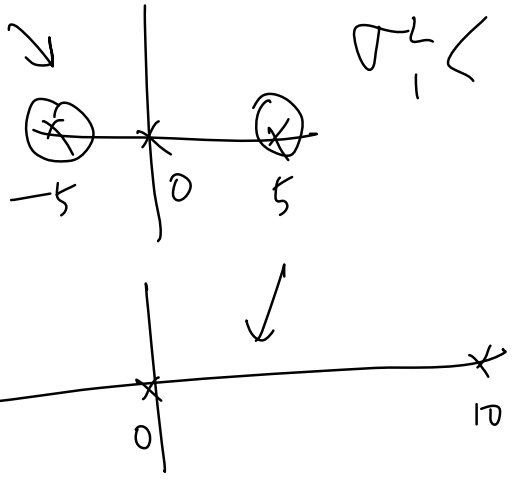
09 March 2023 16:36

Variance: The variance is the average of the squared differences between each data point and the mean. It measures the average distance of each data point from the mean and is useful in comparing the dispersion of datasets with different means.

	X-mean	(X-mean)^2
3	3-3	0
2	2-3	1
1	1-3	4
5	5-3	4
4	4-3	1

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$ Population Variance

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$ Sample Variance



Mean Absolute Deviation

$$MAD = \frac{\sum |x_i - \bar{x}|}{n}$$

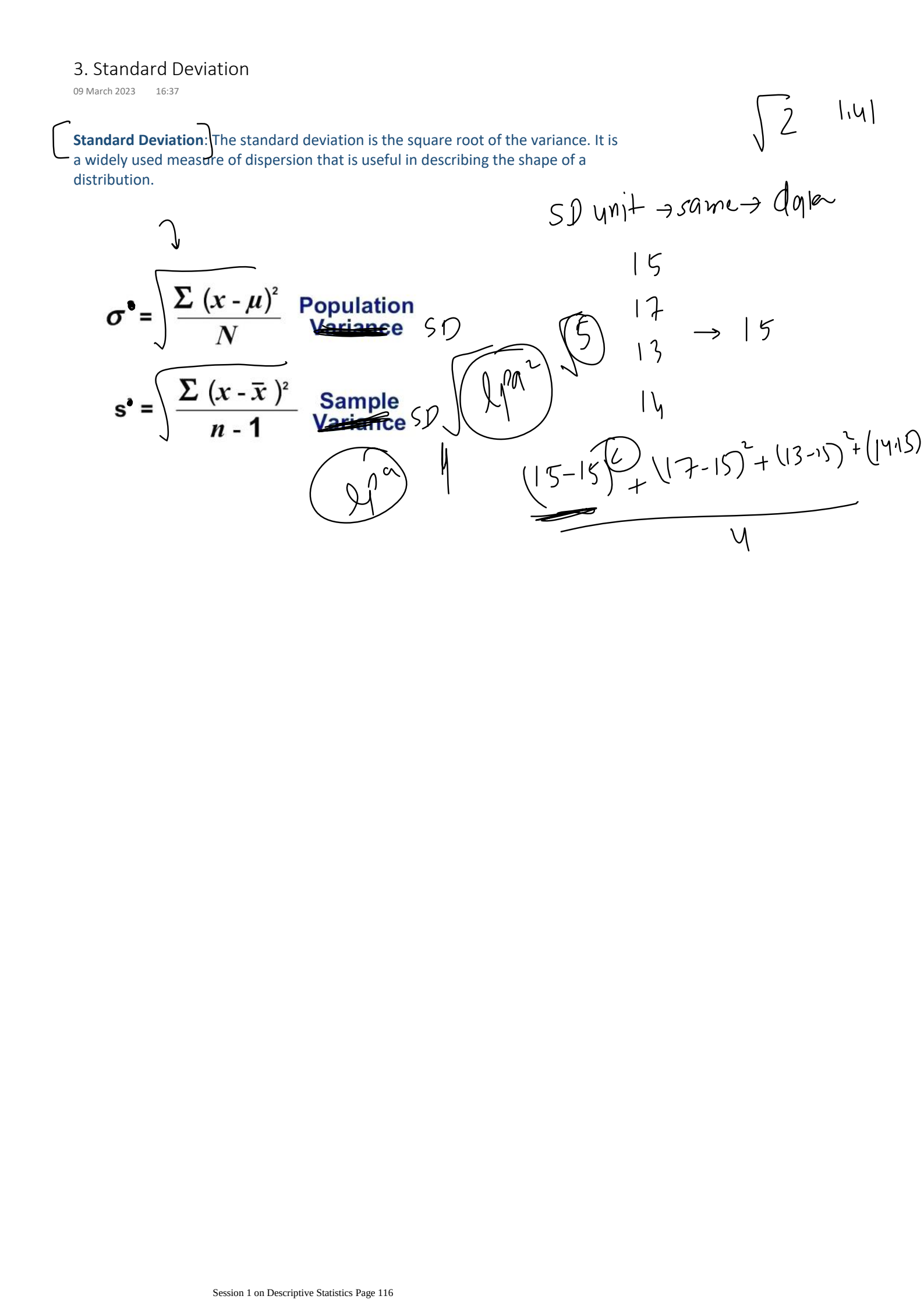
3. Standard Deviation

09 March 2023 16:37

Standard Deviation: The standard deviation is the square root of the variance. It is a widely used measure of dispersion that is useful in describing the shape of a distribution.

$$\sigma = \sqrt{\frac{\Sigma (x - \mu)^2}{N}}$$ Population Variance

$$s = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n - 1}}$$ Sample Variance



4. Coefficient of Variation

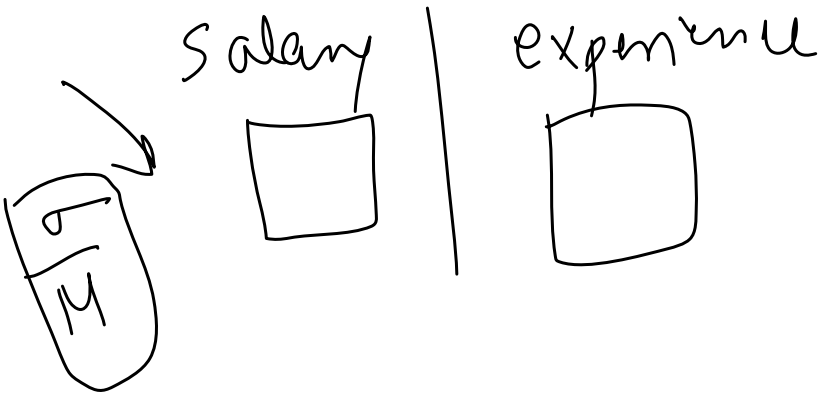
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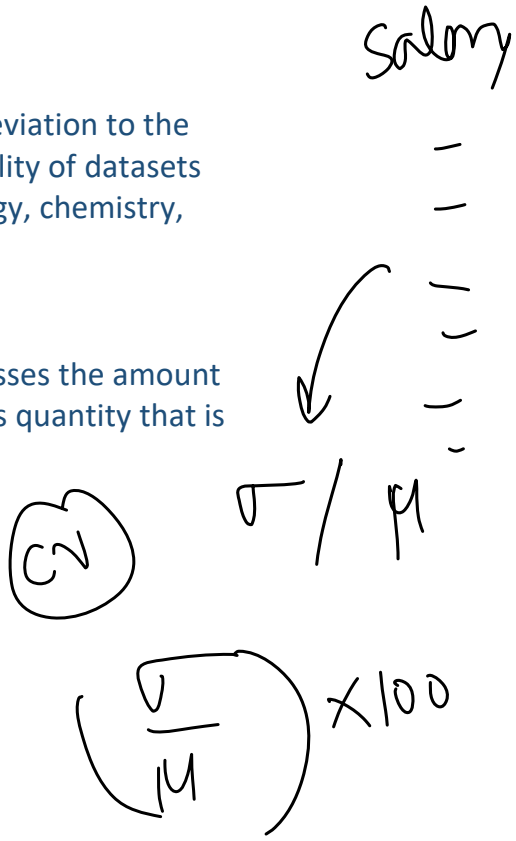
Coefficient of Variation (CV): The CV is the ratio of the standard deviation to the mean expressed as a percentage. It is used to compare the variability of datasets with different means and is commonly used in fields such as biology, chemistry, and engineering.

The coefficient of variation (CV) is a statistical measure that expresses the amount of variability in a dataset relative to the mean. It is a dimensionless quantity that is expressed as a percentage.

The formula for calculating the coefficient of variation is:

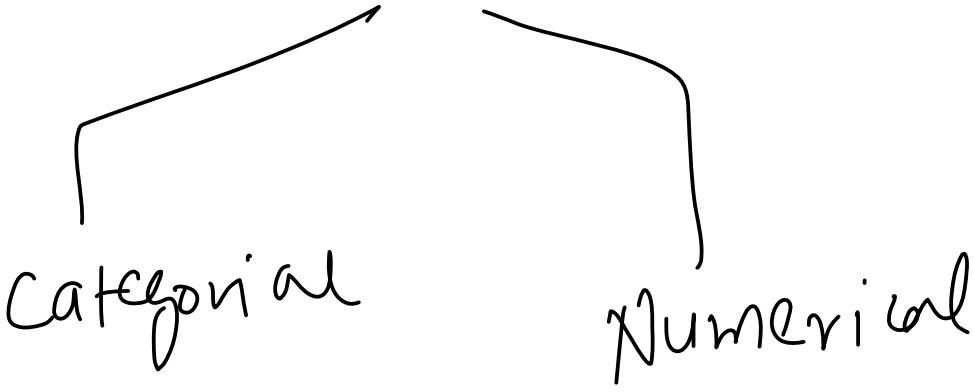
CV = (standard deviation / mean) x 100%





Graphs for Univariate Analysis

09 March 2023 14:58



1. Categorical - Frequency Distribution Table & Cumulative Frequency

09 March 2023 16:50

A **frequency distribution table** is a table that summarizes the number of times (or frequency) that each value occurs in a dataset.

Let's say we have a survey of 200 people and we ask them about their favourite type of vacation, which could be one of six categories: Beach, City, Adventure, Nature, Cruise, or Other

Type of Vacation	Frequency
Beach	60
City	40
Adventure	30
Nature	35
Cruise	20
Other	15

Relative frequency is the proportion or percentage of a category in a dataset or sample. It is calculated by dividing the frequency of a category by the total number of observations in the dataset or sample.

Type of Vacation	Frequency	Relative Frequency
Beach	60	0.3
City	40	0.2
Adventure	30	0.15
Nature	35	0.175
Cruise	20	0.1
Other	15	0.075

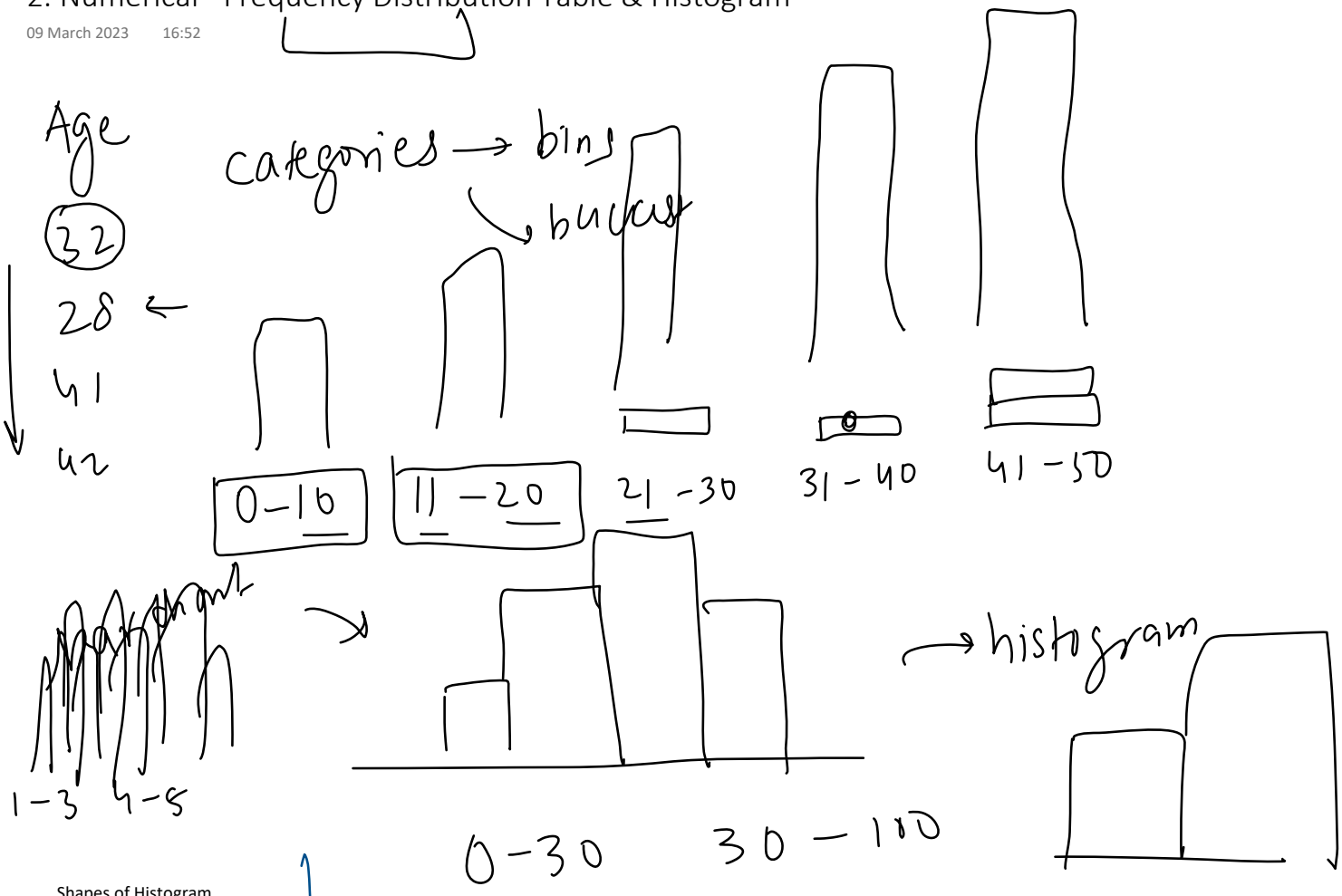
Cumulative frequency is the running total of frequencies of a variable or category in a dataset or sample. It is calculated by adding up the frequencies of the current category and all previous categories in the dataset or sample.

Type of Vacation	Frequency	Relative Frequency	Cumulative Frequency
Beach	60	0.3	60
City	40	0.2	100
Adventure	30	0.15	130
Nature	35	0.175	165
Cruise	20	0.1	185
Other	15	0.075	200

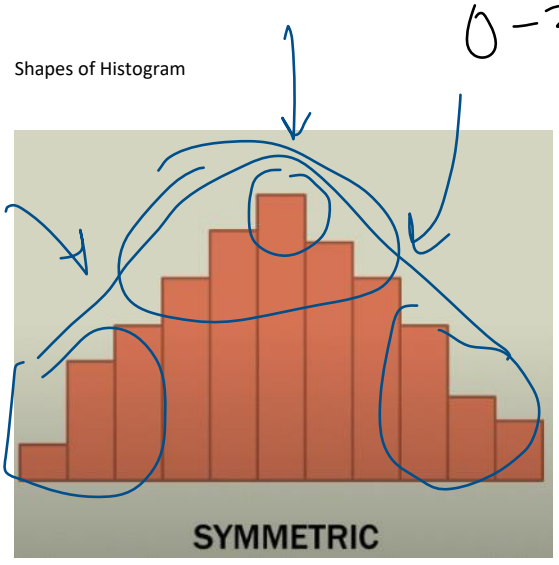


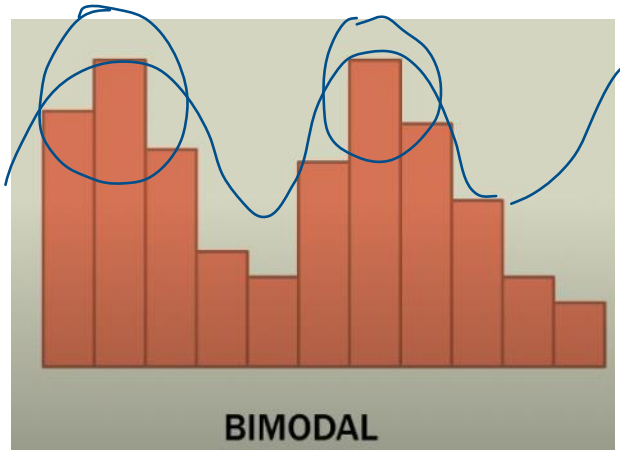
2. Numerical - Frequency Distribution Table & Histogram

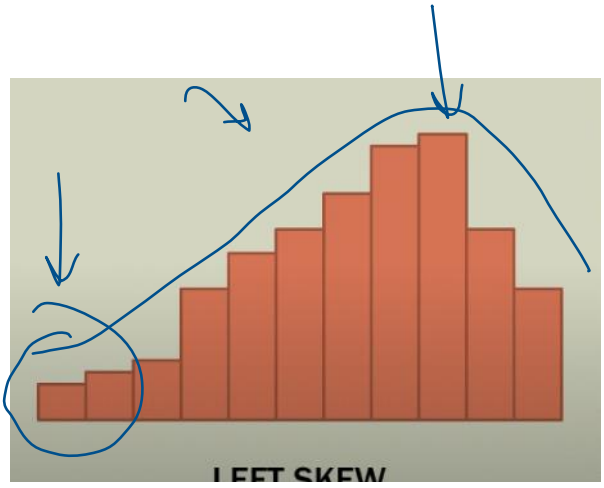
09 March 2023 16:52

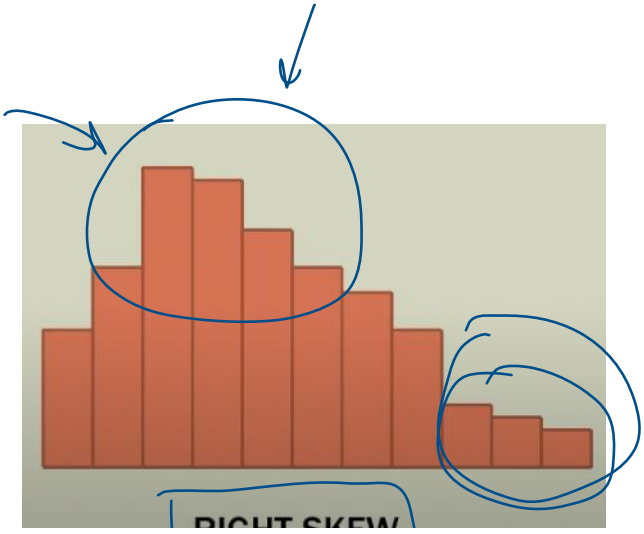


Shapes of Histogram



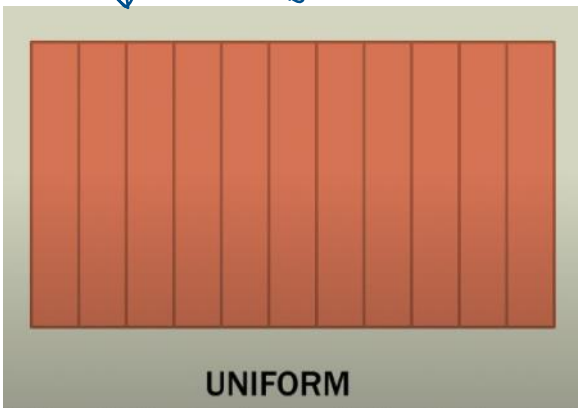


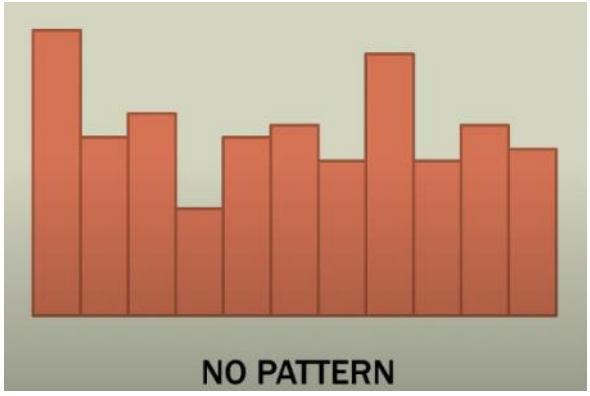





LEFT SKEW

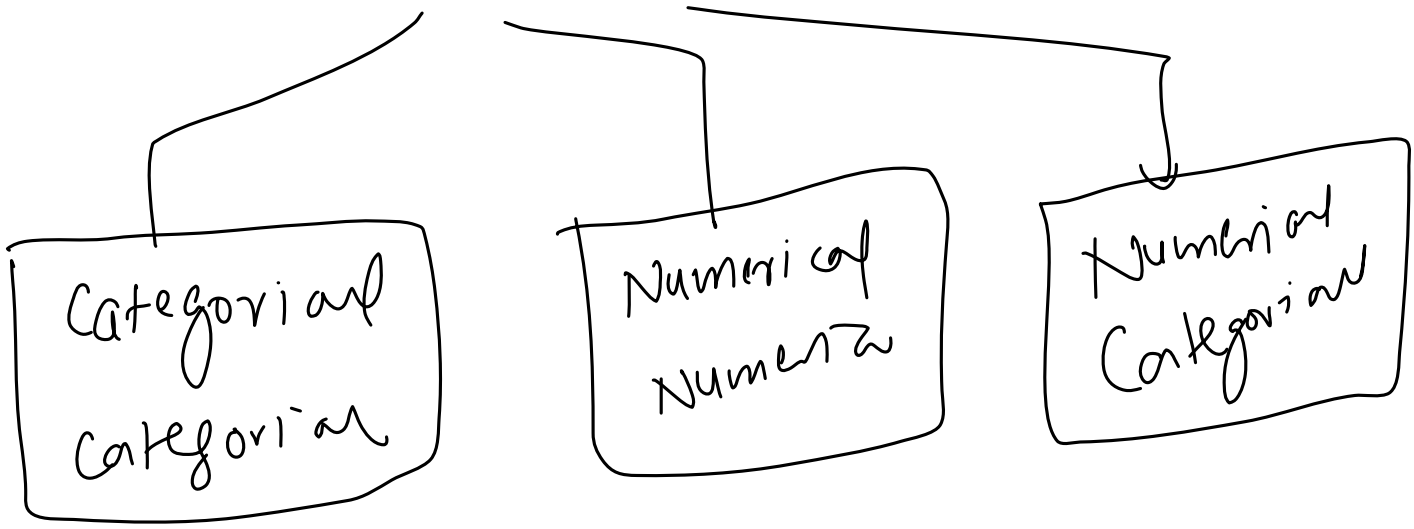

RIGHT SKEW


UNIFORM


NO PATTERN

Graphs for Bivariate Analysis

09 March 2023 14:59

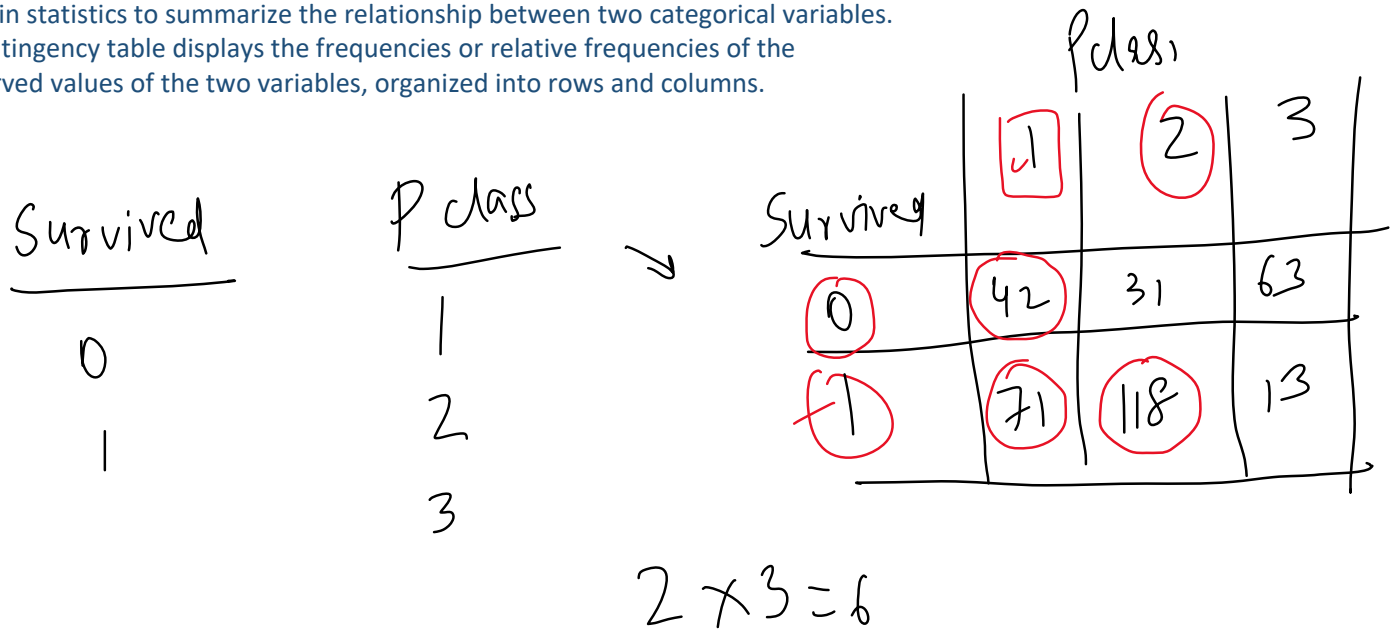


1. Categorical - Categorical

09 March 2023 16:58

Contingency Table/Crosstab

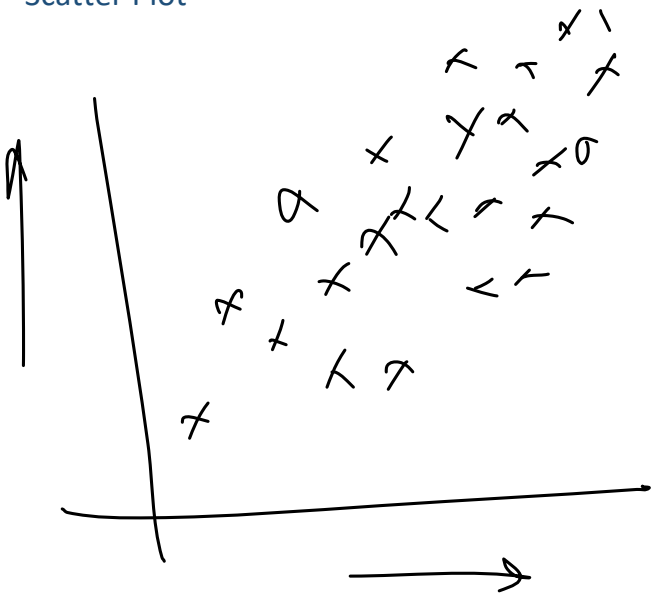
A contingency table, also known as a cross-tabulation or crosstab, is a type of table used in statistics to summarize the relationship between two categorical variables. A contingency table displays the frequencies or relative frequencies of the observed values of the two variables, organized into rows and columns.

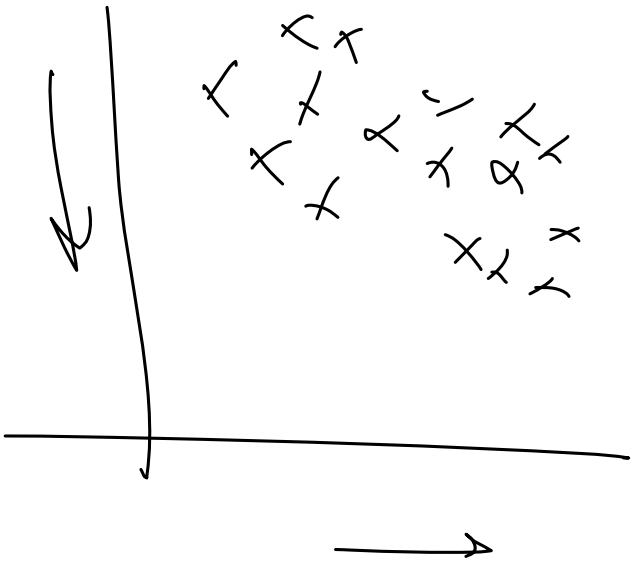


2. Numerical - Numerical

09 March 2023 16:58

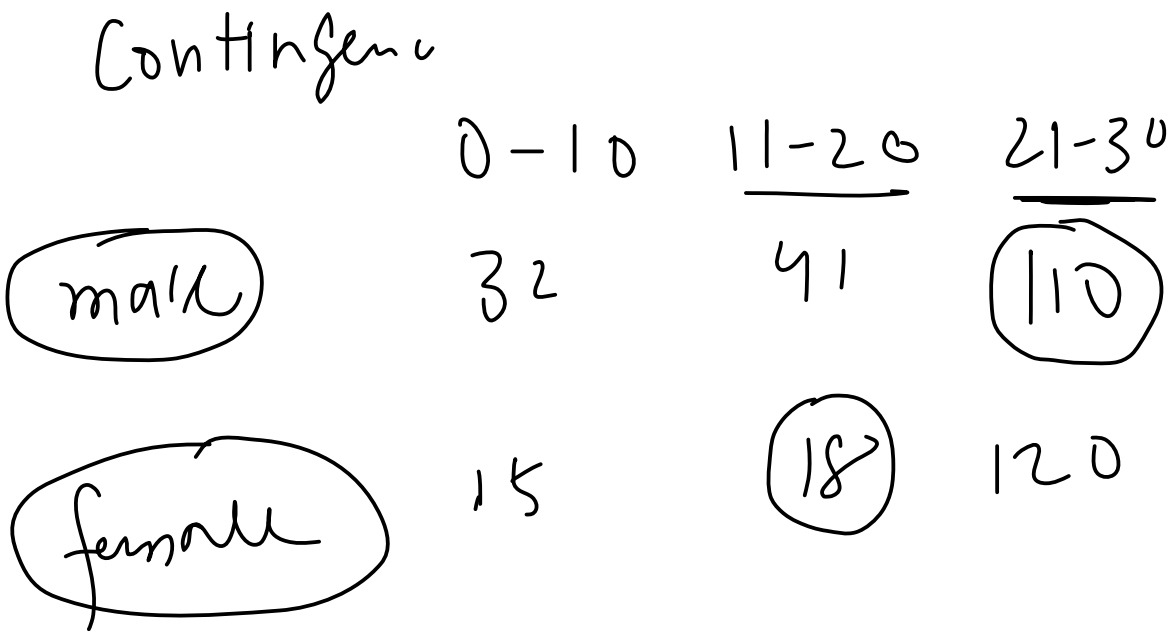
Scatter Plot





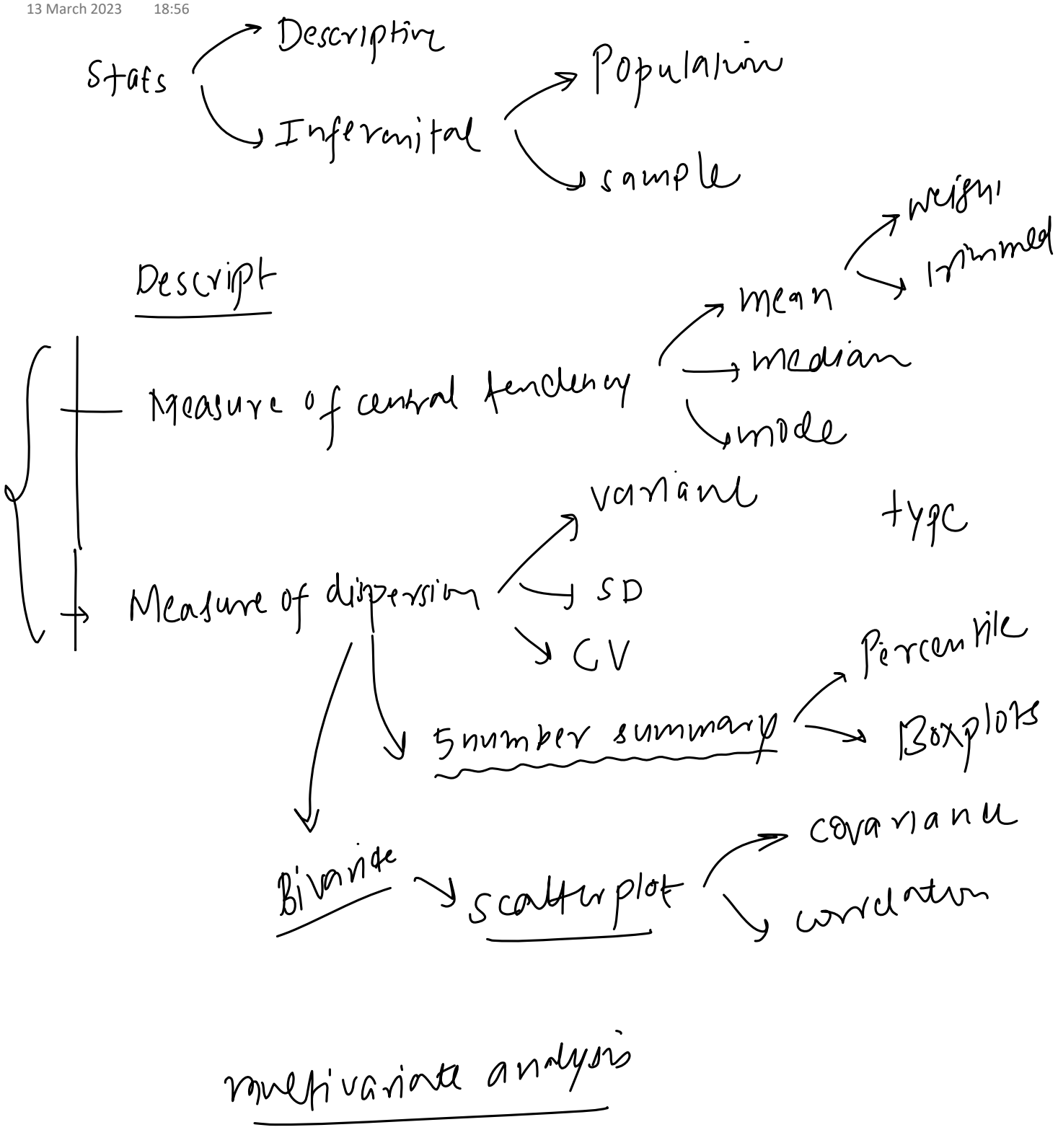
3. Categorical - Numerical

09 March 2023 16:58



Recap

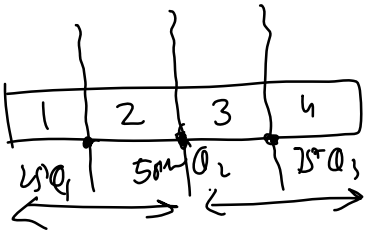
13 March 2023 18:56



Quantiles and Percentiles

13 March 2023 06:57

Quantiles are statistical measures used to divide a set of numerical data into equal-sized groups, with each group containing an equal number of observations.



Quantiles are important measures of variability and can be used to: understand distribution of data, summarize and compare different datasets. They can also be used to identify outliers.

There are several types of quantiles used in statistical analysis, including:

a. Quartiles: Divide the data into four equal parts, Q1 (25th percentile), Q2 (50th percentile or median), and Q3 (75th percentile).

b. Deciles: Divide the data into ten equal parts, D1 (10th percentile), D2 (20th percentile), ..., D9 (90th percentile).

c. Percentiles: Divide the data into 100 equal parts, P1 (1st percentile), P2 (2nd percentile), ..., P99 (99th percentile).

d. Quintiles: Divides the data into 5 equal parts

Things to remember while calculating these measures:

1. Data should be sorted from low to high
2. You are basically finding the location of an observation
3. They are not actual values in the data
4. All other tiles can be easily derived from Percentiles

Percentile

A percentile is a statistical measure that represents the percentage of observations in a dataset that fall below a particular value. For example, the 75th percentile is the value below which 75% of the observations in the dataset fall.

Formula to calculate the percentile value:

$$PL = \frac{p}{100}(N+1)$$

where:

- PL = the desired percentile value location
- N = the total number of observations in the dataset
- p = the percentile rank (expressed as a percentage)

Example:

Find the 75th percentile score from the below data

78, 82, 84, 88, 91, 93, 94, 96, 98, 99

Step1 - Sort the data (Asc)

78, 82, 84, 88, 91, 93, 94, 96, 98, 99

1 2 3 4 5 6 7 8 9 10

$$PL = \frac{75}{100}(10+1) = \frac{3}{4} \times 11 = \frac{33}{4} = 8.25$$

96 —— 98 Q75 91 —— 93

100

96 ——— 98 (0.75) 91 ——— 93

8 ——— 9 5 6

↑

$96 + 0.25(98 - 96) = 96 + 0.25 \times 2 = 96.5$

↘

75th percentile = 96.5

$91 + 0.5(93 - 91)$

$91 + 0.5 \times 2 = 92$

$P_L = \frac{50}{100}(10+1) = \frac{1}{2} \times 11 = 5.5$

Percentile of a value

Percentile rank = $\frac{x + 0.5y}{n}$ $\left(\frac{n}{N}\right)$

X = number of values below the given value

Y = number of values equal to the given value

N = total number of values in the dataset

78, 82, 84, 88, 91, 93, 94, 96, 98, 99

1 2 3

$= \frac{3 + 0.5 \times 1}{10} = \frac{3.5}{10}$

0.35 → 35%

$\frac{9 + 0.5 \times 1}{10} = \frac{9.5}{10} = 0.95$

$\frac{9}{10}$ 0.9

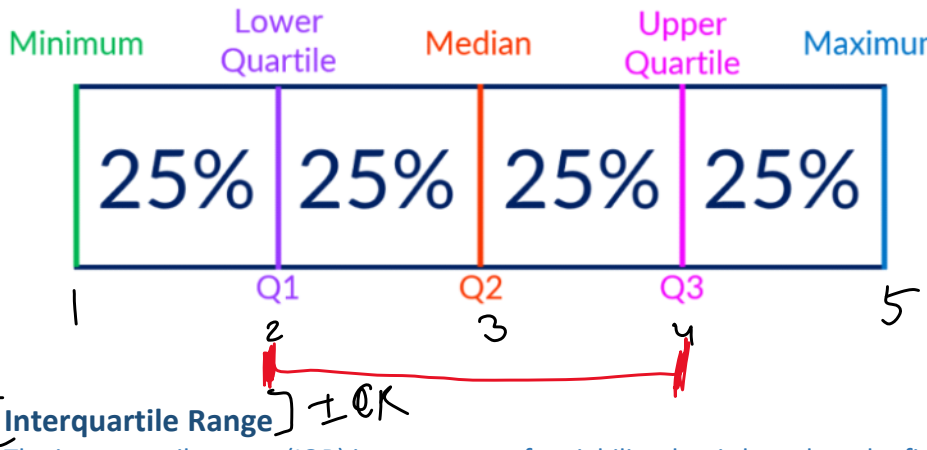
5 number summary

13 March 2023 06:57

The five-number summary is a descriptive statistic that provides a summary of a dataset. It consists of five values that divide the dataset into four equal parts, also known as quartiles. The five-number summary includes the following values:

1. **Minimum value**: The smallest value in the dataset.
2. **First quartile (Q1)**: The value that separates the lowest 25% of the data from the rest of the dataset.
3. **Median (Q2)**: The value that separates the lowest 50% from the highest 50% of the data.
4. **Third quartile (Q3)**: The value that separates the lowest 75% of the data from the highest 25% of the data.
5. **Maximum value**: The largest value in the dataset.

The five-number summary is often represented visually using a box plot, which displays the range of the dataset, the median, and the quartiles.
The five-number summary is a useful way to quickly summarize the central tendency, variability, and distribution of a dataset.



Interquartile Range

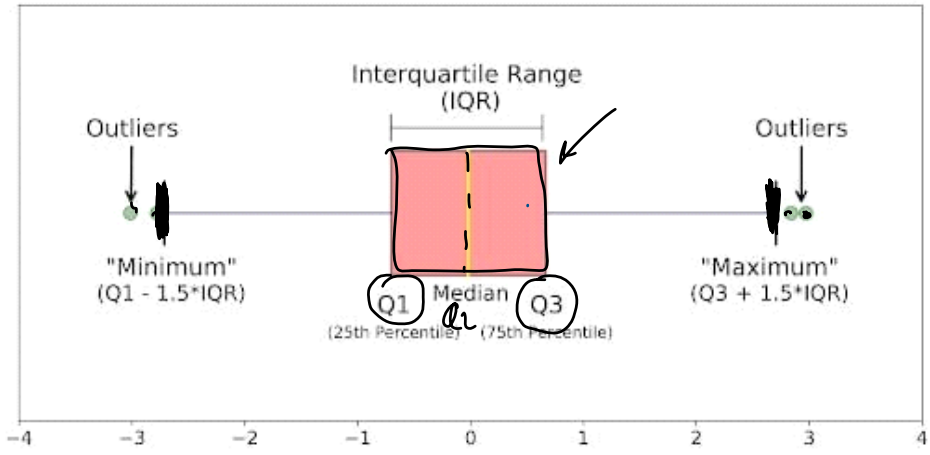
The interquartile range (IQR) is a measure of variability that is based on the five-number summary of a dataset. Specifically, the IQR is defined as the difference between the third quartile (Q3) and the first quartile (Q1) of a dataset.

Boxplots

13 March 2023 06:57

1. What is a boxplot

A box plot, also known as a box-and-whisker plot, is a graphical representation of a dataset that shows the distribution of the data. The box plot displays a summary of the data, including the minimum and maximum values, the first quartile (Q1), the median (Q2), and the third quartile (Q3).



2. How to create a boxplot with example

6 213 241 260 281 290 314 321 350 1500

1 2 3 4 5 6 7 8 9 10

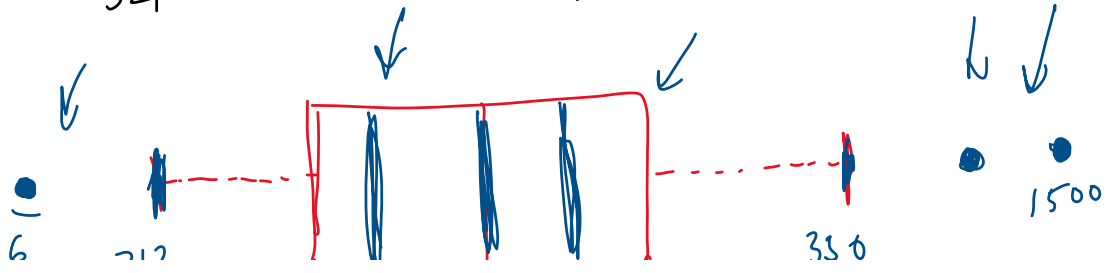
$Q_2 = \frac{50}{100}(11) = 5.5 = 285.5$

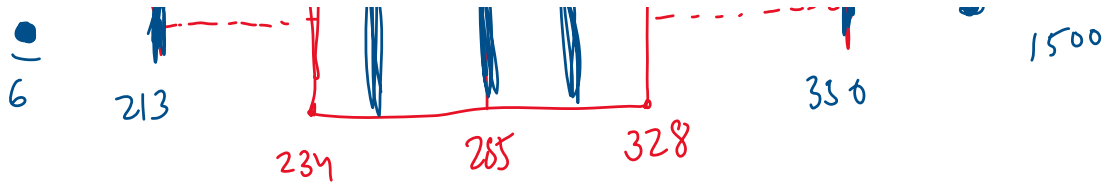
$Q_1 = \frac{25}{100} \times 11 = \frac{11}{4} = 2.75$

$213 + 0.75(241 - 213) = 234$

$Q_3 = \frac{75}{100} \times 11 = \frac{33}{4} = 8.25$

$321 + 0.25(350 - 321) = 328.25$





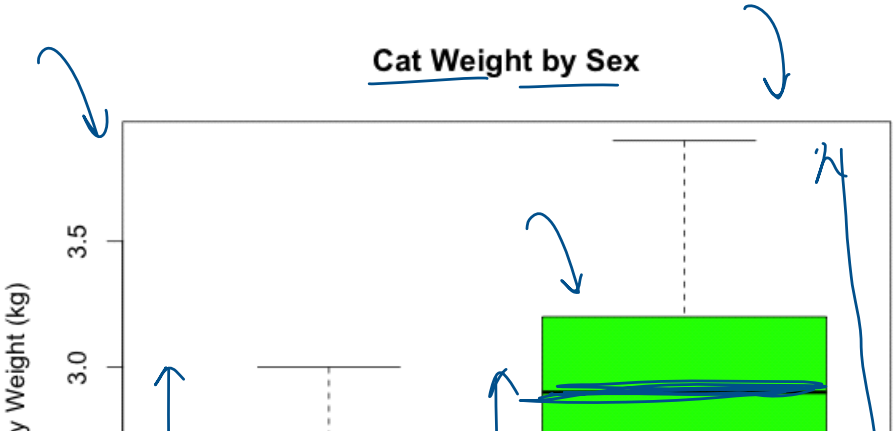
IQR = 94

min and max

$min = Q_1 - 1.5(IQR) = 93$

$max = Q_3 + 1.5(IQR) = 469$

1. Benefits of a Boxplot
   - Easy way to see the distribution of data
   - Tells about skewness of data
   - Can identify outliers
   - Compare 2 categories of data
2. Side by side boxplot

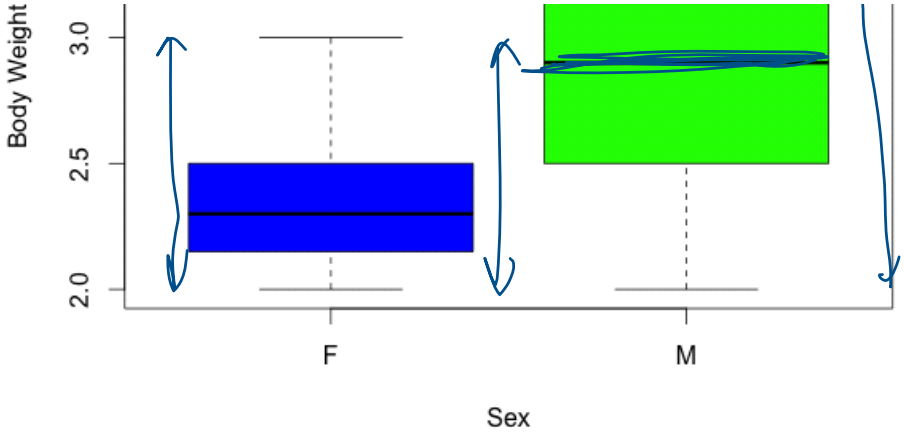


Weight | Age | gender

21 ↔ M

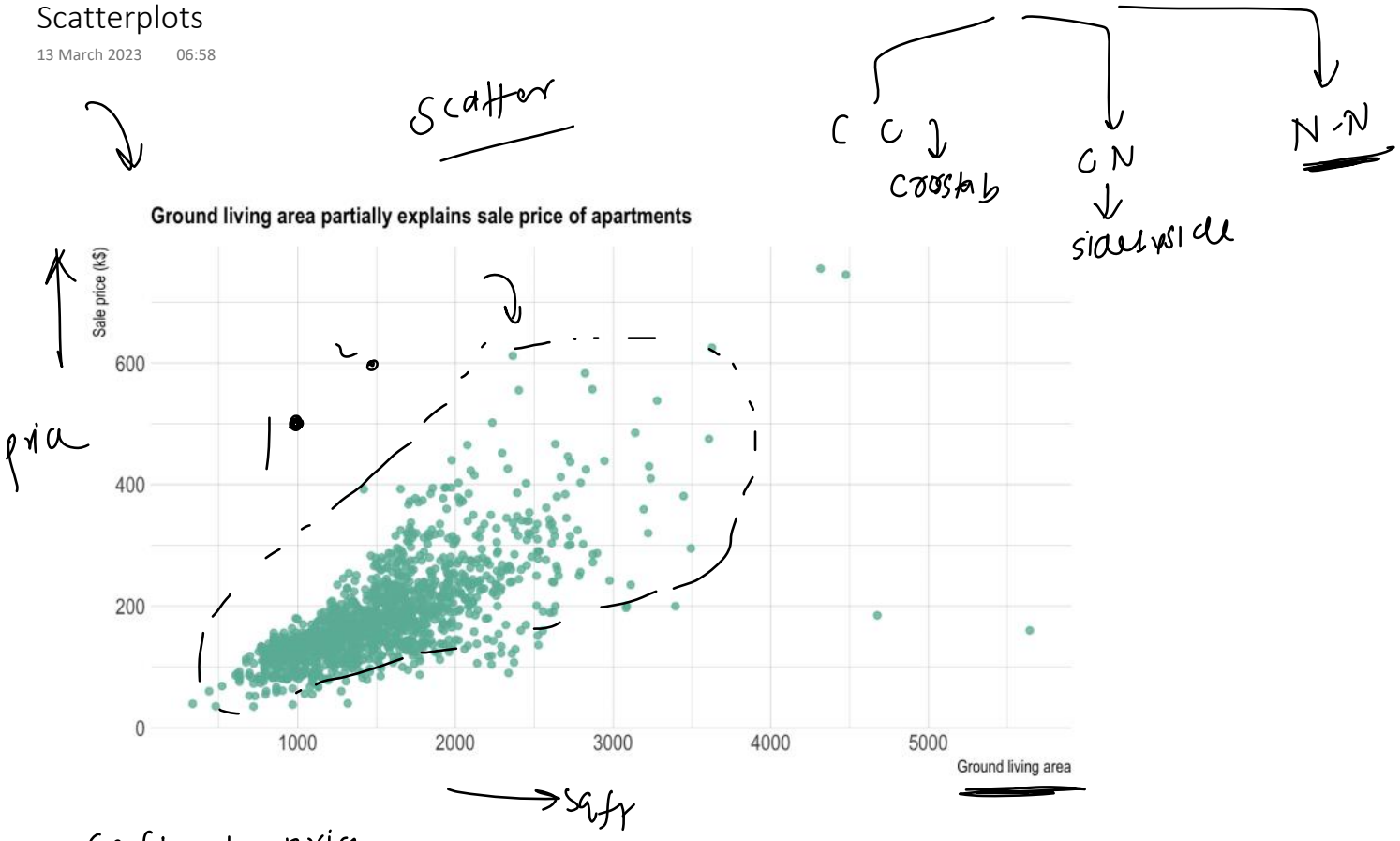
34 F

43 M


Body Weight
3.0
2.5
2.0
F
M
Sex

Scatterplots

13 March 2023 06:58

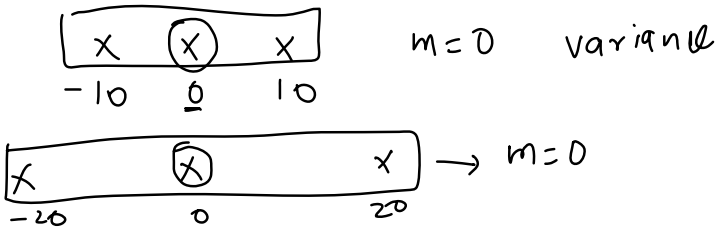


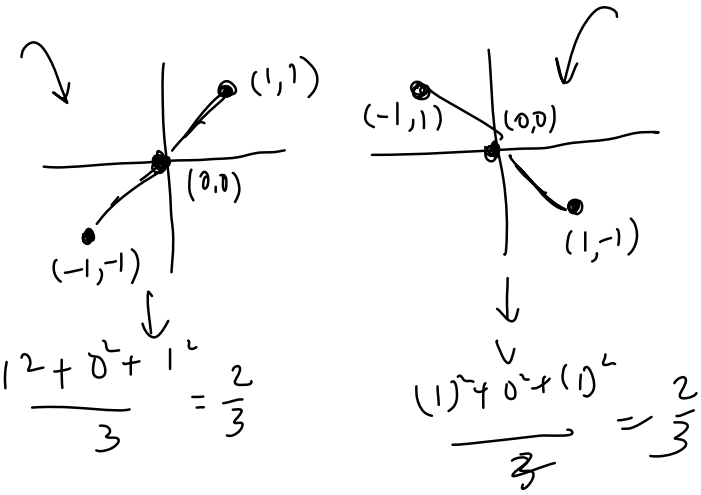
Sqft	price
1500	60L
1700	80L
1200	35L

Covariance

13 March 2023 06:57

- What problem does Covariance solve?

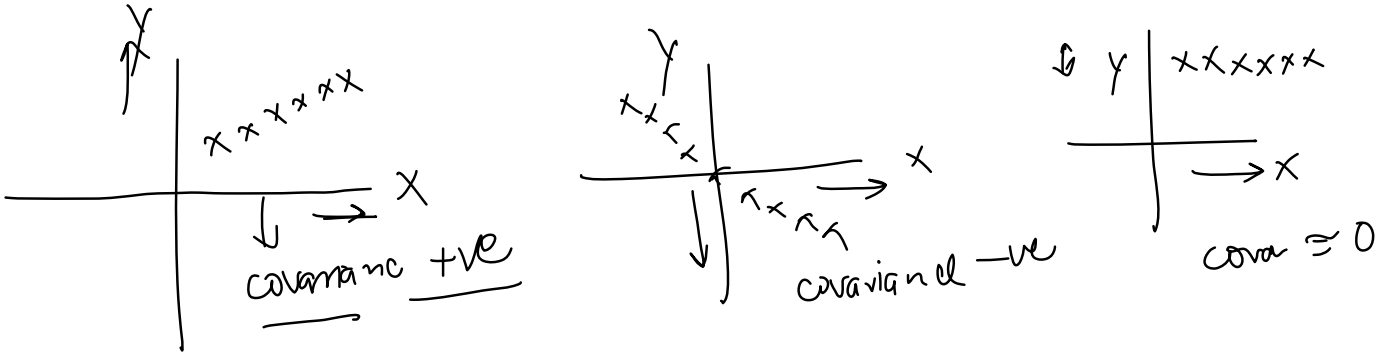




- What is covariance and how is it interpreted?

Covariance is a statistical measure that describes the degree to which two variables are linearly related. It measures how much two variables change together, such that when one variable increases, does the other variable also increase, or does it decrease?

If the covariance between two variables is positive, it means that the variables tend to move together in the same direction. If the covariance is negative, it means that the variables tend to move in opposite directions. A covariance of zero indicates that the variables are not linearly related.



- How is it calculated?

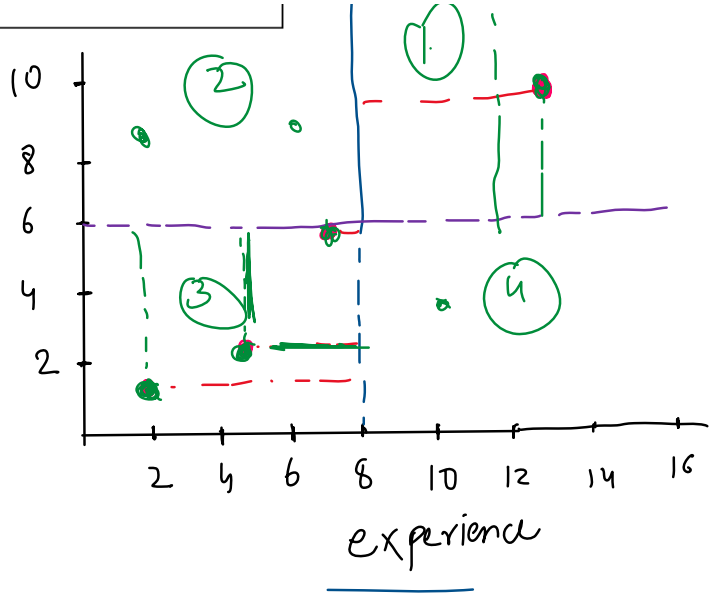
Covariance Formula	
Population	**Sample**
$\sigma_{xy} = \frac{\sum(X-\mu_x)(Y-\mu_y)}{N}$	$s_{xy} = \frac{\sum(X-\bar{x})(Y-\bar{y})}{n-1}$
X, Y – The Value of X and Y in the Population *μ_x, μ_y – The population Mean of X and Y* *N – Total Number of Observations*	*X, Y – The Value of X and Y in the Sample Data* *$\bar{x}, \bar{y}$ – The Sample Mean of X and Y* *n - Total Number of Observations*

Exp(x)	Salary(y)	X-Xmean	Y-Ymean	(X-Xmean)*(Y-Ymean)
2	1	-6	-5	30
5	2	-3	-4	12
8	5	0	-1	-1
12	12	4	6	24
13	10	5	4	20

$\bar{X} = 8 \quad \bar{Y} = 6$

$$\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \qquad n-1$$

$\frac{85}{4} =$ COV = 21.5

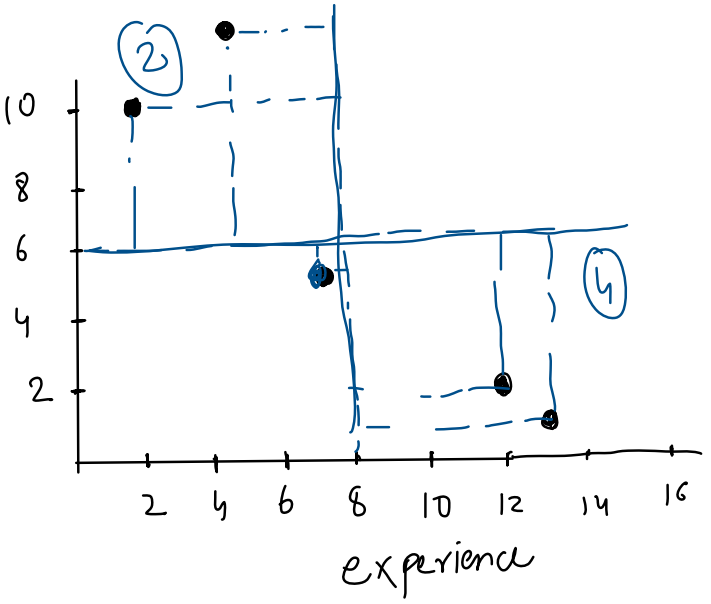


Backlogs(x)	package(y)	X-Xmean	Y-Ymean	(X-Xmean)*(Y-Ymean)
2	10	-6	4	-24
5	12	-3	6	-18
8	5	0	1	0
12	2	4	-4	-16
13	1	5	-5	-25

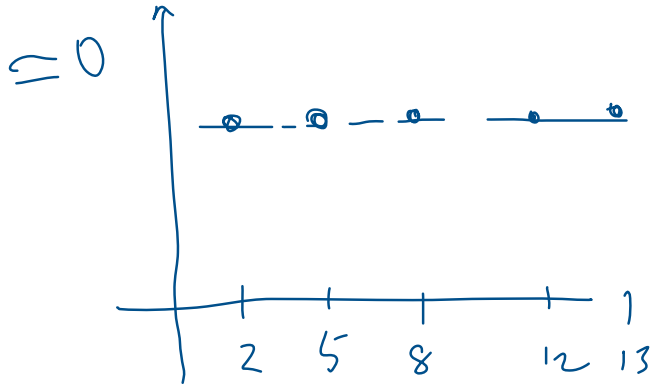
$\bar{X} = 8 \quad \bar{Y} = 6$

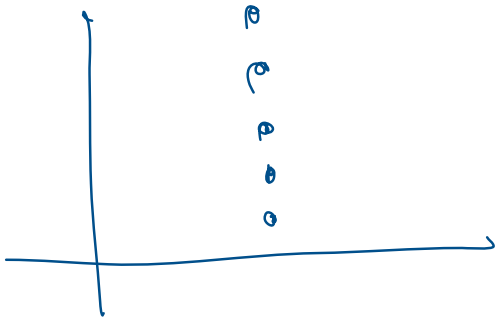
$\frac{-83}{4}$

-21.3



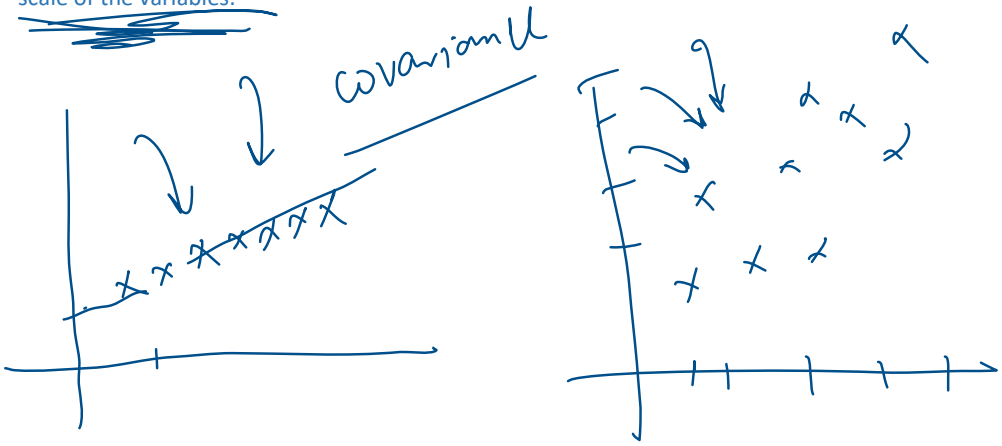
Backlogs(x)	package(y)	X-Xmean	Y-Ymean	(X-Xmean)*(Y-Ymean)
2	10			
5	10			
8	10			
12	10			
13	10			





- Disadvantages of using Covariance

  One limitation of covariance is that it does not tell us about the strength of the relationship between two variables, since the magnitude of covariance is affected by the scale of the variables.



- Covariance of a variable with itself

$$\sum (x-\bar{x})(y-\bar{y})$$

$$\sum_{i=1} \frac{(x-\bar{x})(y-\bar{y})}{n-1}$$

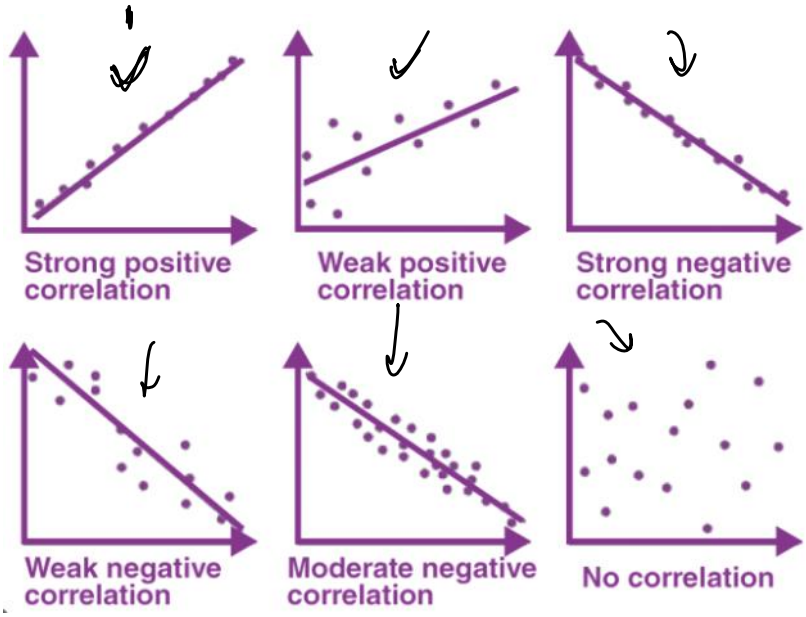
$$= \sum_{i=1}^{n} \frac{(x-\bar{x})(x-\bar{x})}{n-1}$$

$$= \sum_{i=1}^{n} \frac{(x-\bar{x})^2}{n-1}$$

Correlation

13 March 2023 06:58

1. What problem does Correlation solve?



Can we quantify this weak and strong relationship?

2. What is correlation?

Correlation refers to a statistical relationship between two or more variables. Specifically, it measures the degree to which two variables are related and how they tend to change together.

Correlation is often measured using a statistical tool called the correlation coefficient, which ranges from -1 to 1. A correlation coefficient of -1 indicates a perfect negative correlation, a correlation coefficient of 0 indicates no correlation, and a correlation coefficient of 1 indicates a perfect positive correlation.

$$Correlation = \frac{Cov\,(x,y)}{\sigma x * \sigma y}$$

Correlation and Causation

13 March 2023 18:31

The phrase "**correlation does not imply causation**" means that just because two variables are associated with each other, it does not necessarily mean that one causes the other. In other words, a correlation between two variables does not necessarily imply that one variable is the reason for the other variable's behaviour.

Suppose there is a positive correlation between the number of firefighters present at a fire and the amount of damage caused by the fire. One might be tempted to conclude that the presence of firefighters causes more damage. However, this correlation could be explained by a third variable - the severity of the fire. More severe fires might require more firefighters to be present, and also cause more damage.

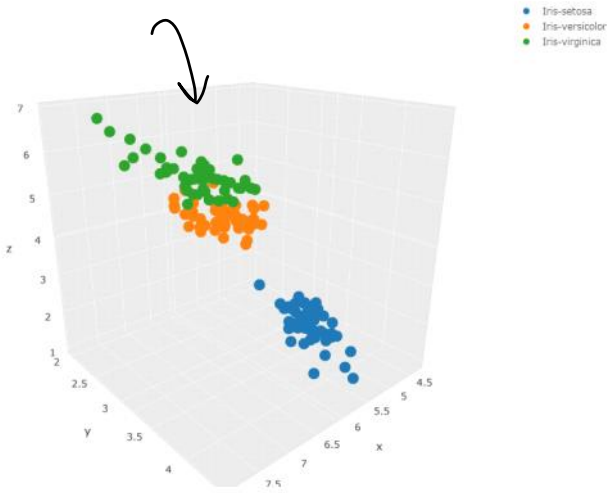
Thus, while correlations can provide valuable insights into how different variables are related, they cannot be used to establish causality. Establishing causality often requires additional evidence such as experiments, randomized controlled trials, or well-designed observational studies.

Visualizing Multiple Variables

13 March 2023 06:58

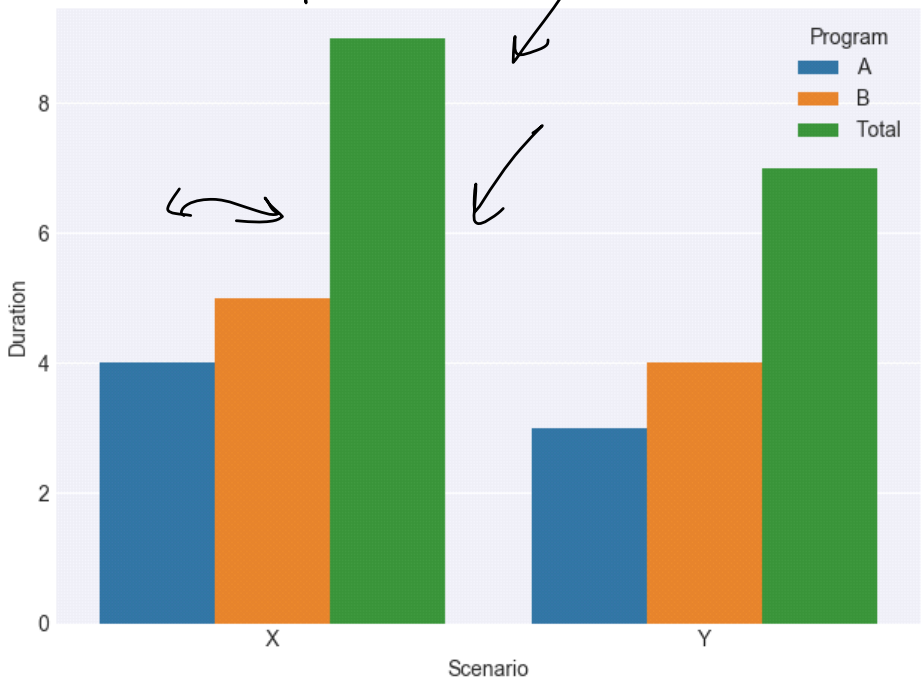
1. 3D Scatter Plots



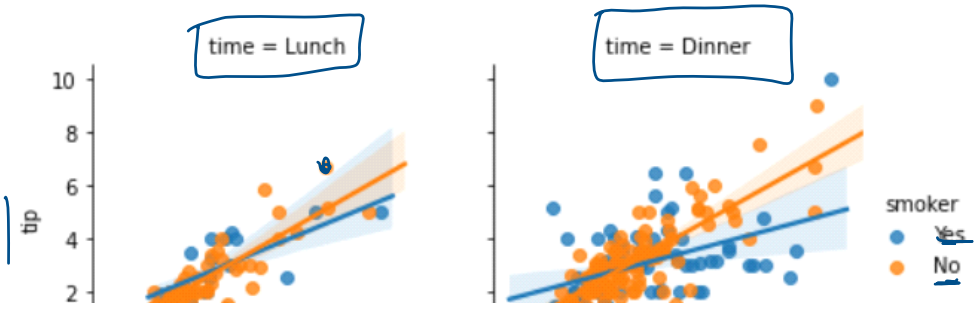


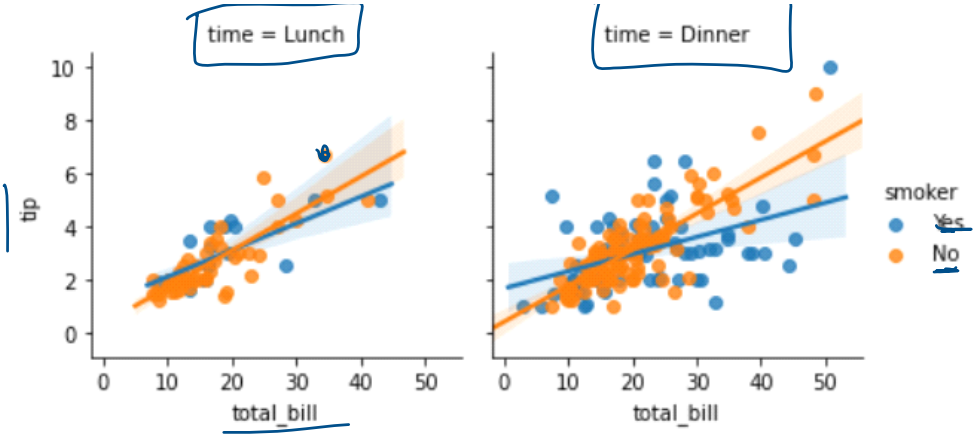
2. Hue Parameter



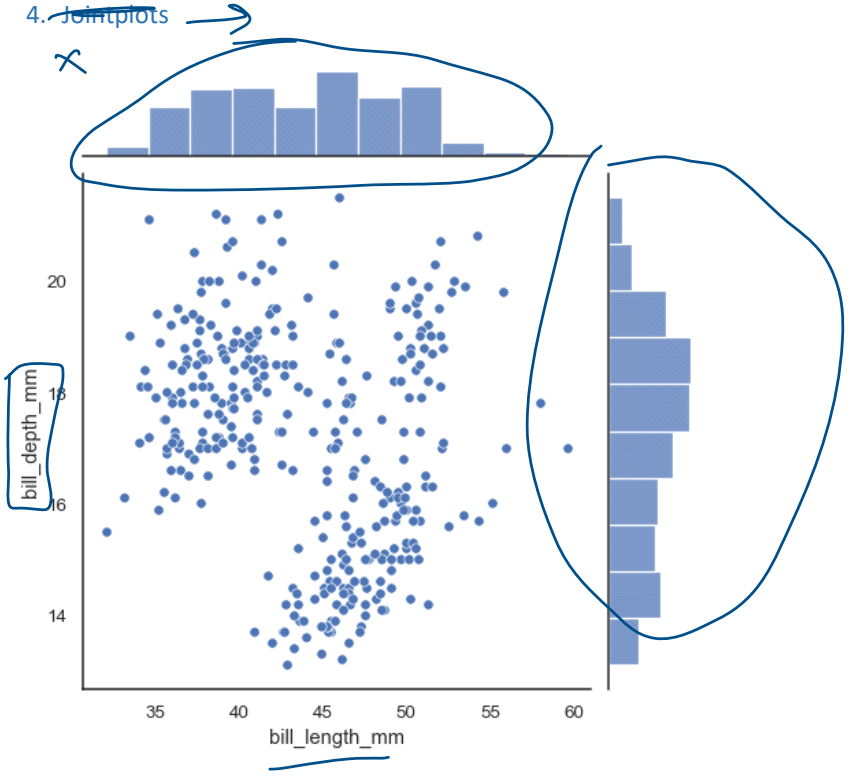


3. Facetgrids

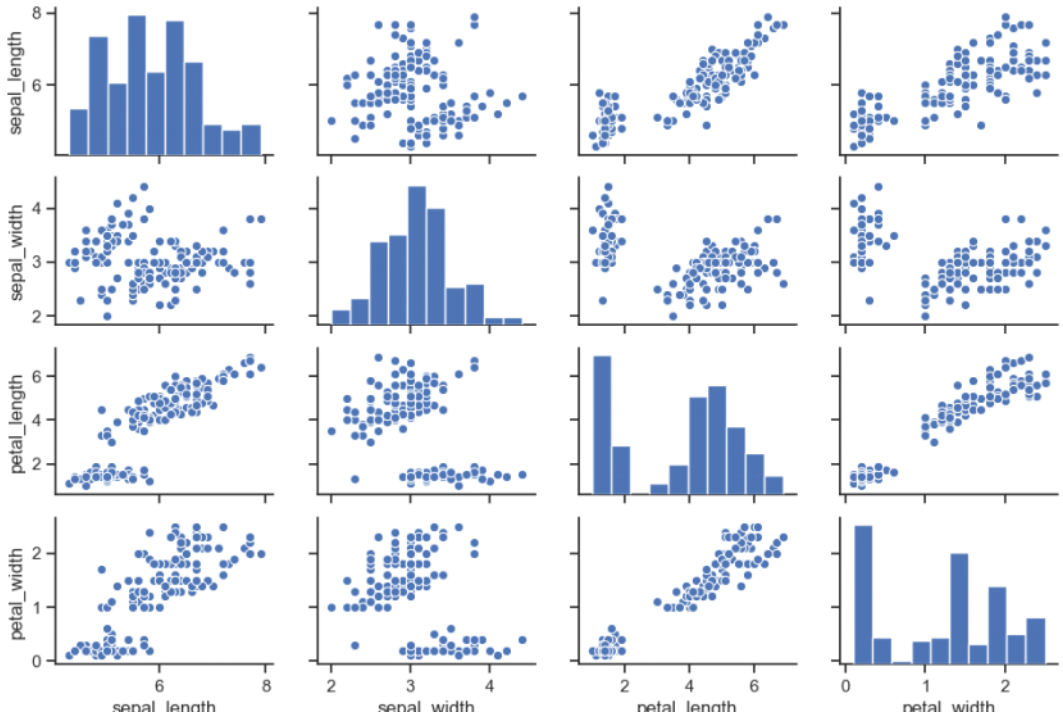




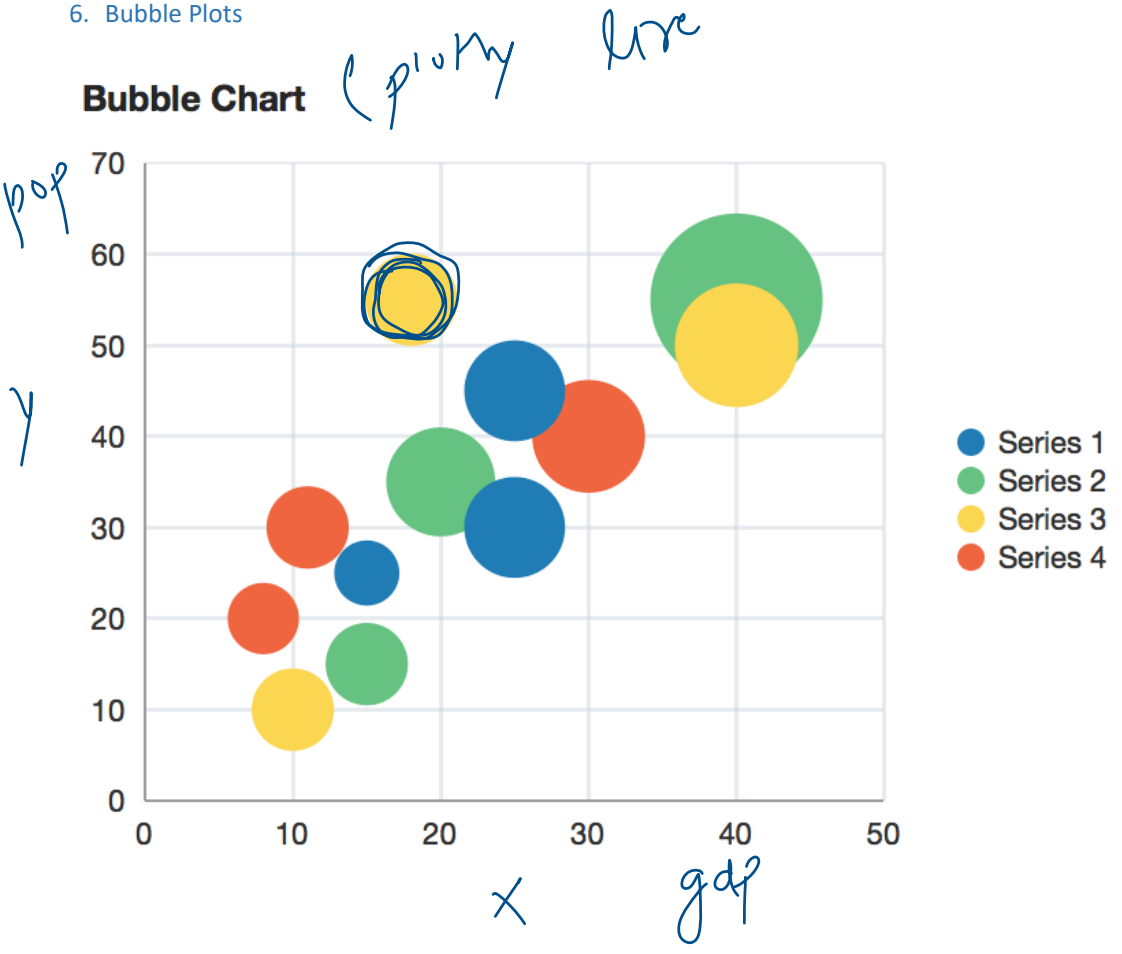
4. Jointplots



5. Pairplots



6. Bubble Plots



Flights Case Study

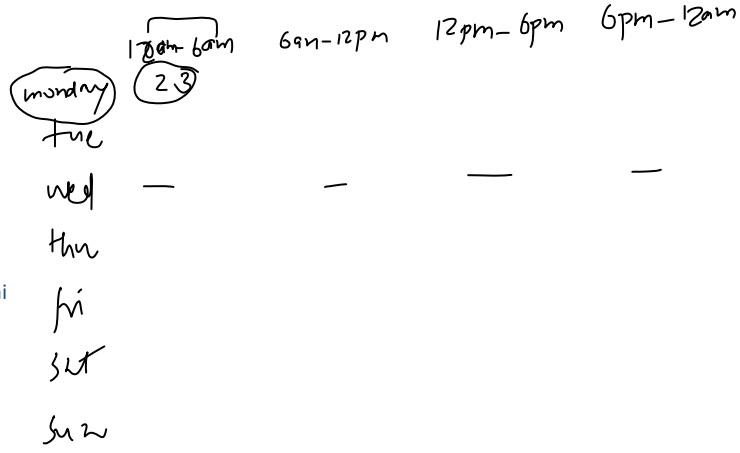
17 March 2023 14:52

1. Find the month with most number of flights
2. Which week day has most costly flights
3. Find number of indigo flights every month
4. Find list of all flights that depart between 10AM and 2PM from Delhi to Banglore
5. Find the number of flights departing on weekends from Bangalore
6. Calculate the arrival time for all flights by adding the duration to the departure time.

7. Calculate the arrival date for all the flights
8. Find the number of flights which travel on multiple dates.
9. Calculate the average duration of flights between all city pairs. The answer should In xh ym format

10. Find all flights which departed before midnight but arrived at their destination after midnight having only 0 stops.

11. Find quarter wise number of flights for each airline
12. Find the longest flight distance(between cities in terms of time) in India
13. Average time duration for flights that have 1 stop vs more than 1 stops
14. Find all Air India flights in a given date range originating from Delhi
15. Find the longest flight of each airline
16. Find all the pair of cities having average time duration > 3 hours
17. Make a weekday vs time grid showing frequency of flights from Banglore and Delhi
18. Make a weekday vs time grid showing avg flight price from Banglore and Delhi

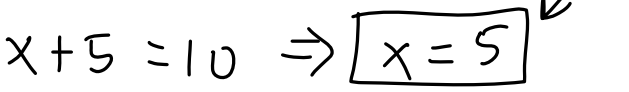


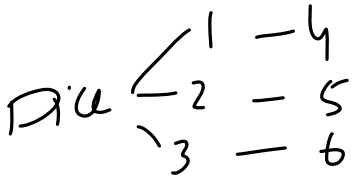
Random Variables

15 March 2023 11:43

- What are Algebraic Variables?

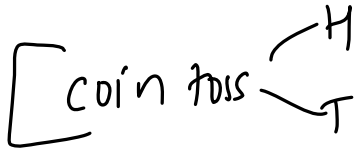
  In Algebra a variable, like x, is an unknown value

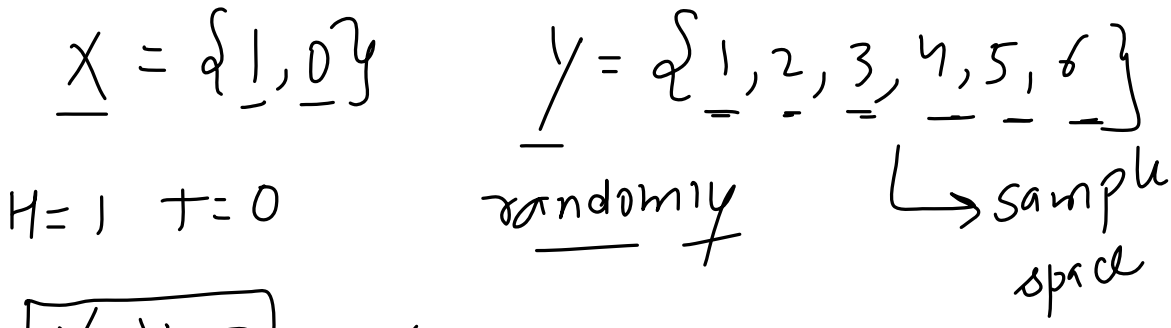




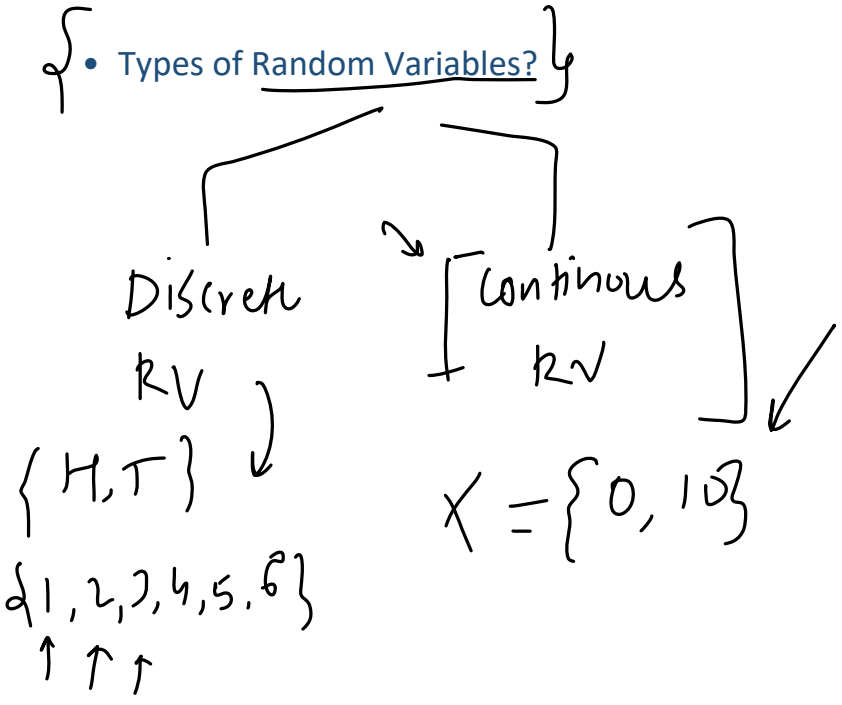
- What are Random Variables in Stats and Probability?

  A Random Variable is a set of possible values from a random experiment.





- Types of Random Variables?



Probability Distributions

15 March 2023 11:53

1. What are Probability Distributions?

A probability distribution is a list of all of the possible outcomes of a random variable along with their corresponding probability values.

coin toss	1 (H)	0 (T)
probab	$\frac{1}{2}$	$\frac{1}{2}$

2 dice →

2 3 4 5 6 7 8 9

dice

1	2	3	4	5	6
1/6	1/6	1/6	1/6	1/6	1/6

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

2 → 1/36
3 → 2/36
4 → 3/36
5 → 4/36
6 → 5/36
7 → 6/36
8 → 5/36
9 → 4/36
10 → 3/36
11 → 2/36
12 → 1/36

Problem with Distribution?

In many scenarios, the number of outcomes can be much larger and hence a table would be tedious to write down. Worse still, the number of possible outcomes could be infinite, in which case, good luck writing a table for that.

Example - Height of people, Rolling 10 dice together

→ Solution - Function?

→ What if we use a mathematical function to model the relationship between outcome and probability?

X → outcome
Y → prob

1	2	3	4	5	6
1/6	1/6	1/6	1/6	1/6	1/6

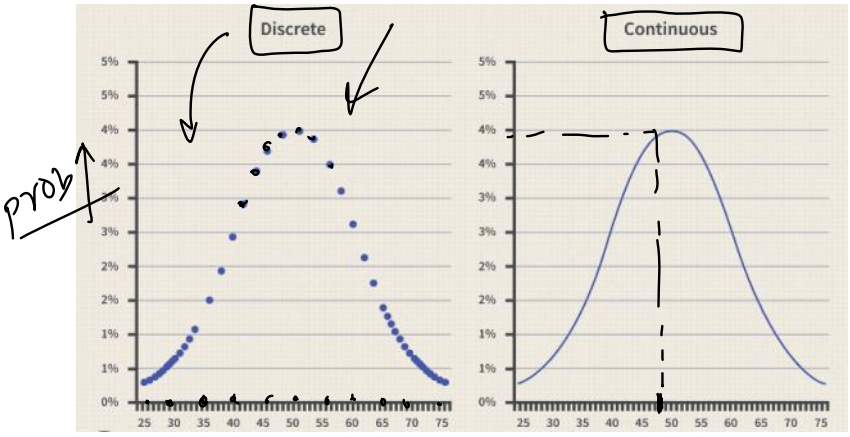
Y = f(X)

PDF → table → f(X)

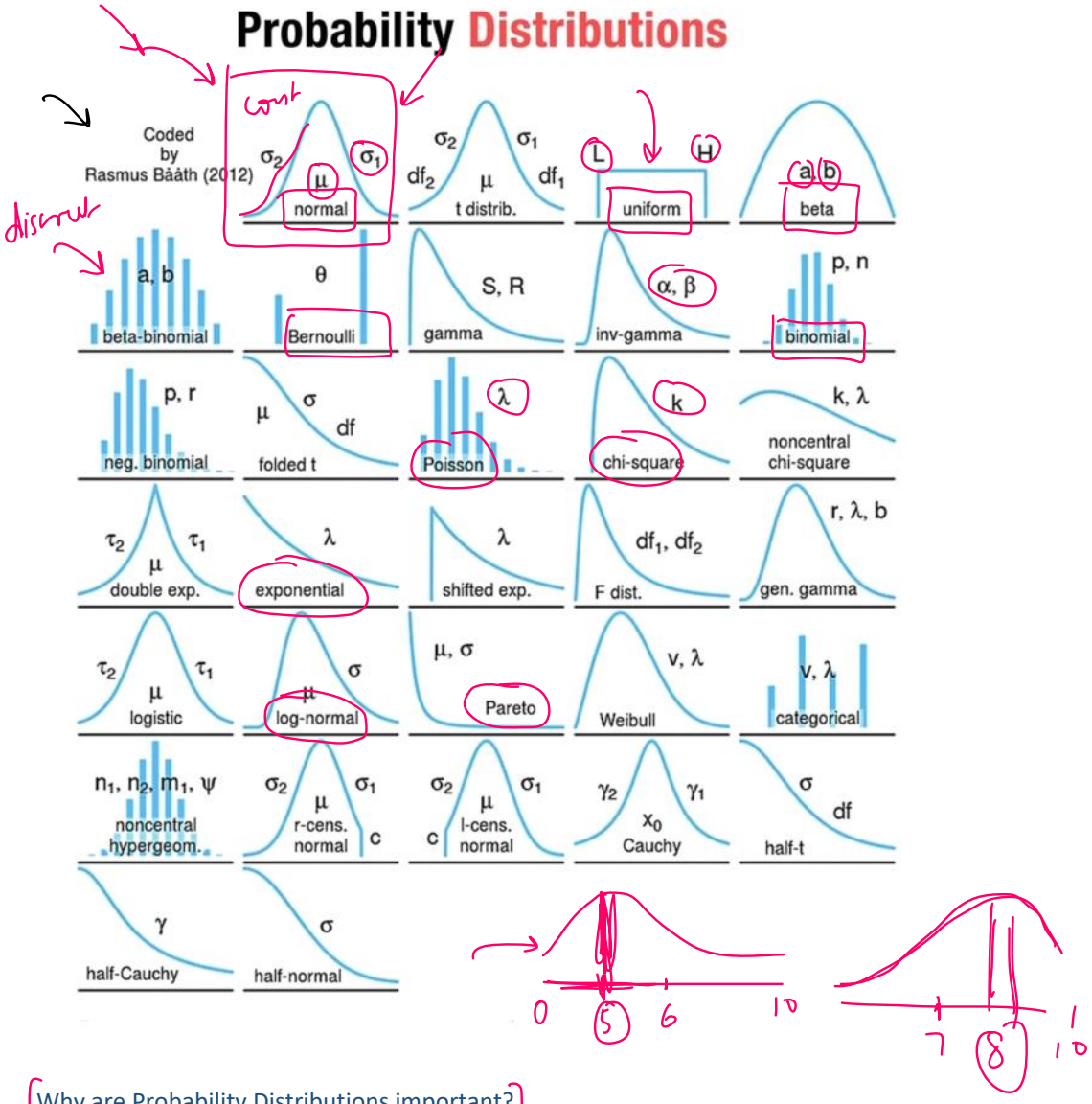
Note - A lot of time Probability Distribution and Probability Distribution Functions are

Note - A lot of time Probability Distribution and Probability Distribution Functions are used interchangeably.

1. Types of Probability Distributions



Famous Probability Distributions



Why are Probability Distributions important?

- Gives an idea about the shape/distribution of the data.
- And if our data follows a famous distribution then we automatically know a lot about the data.

A note on Parameters

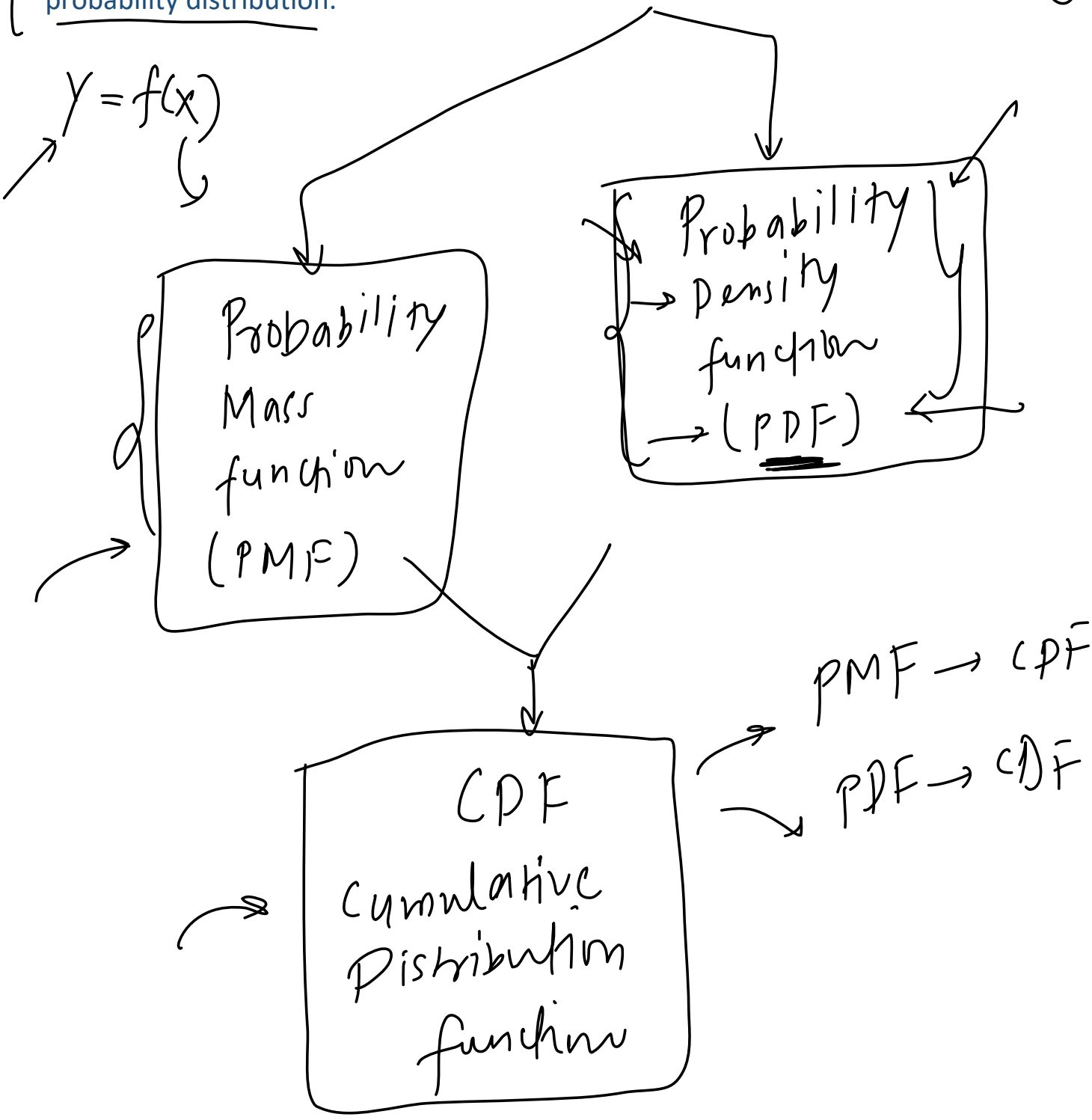
Parameters in probability distributions are numerical values that determine the shape, location, and scale of the distribution.

Different probability distributions have different sets of parameters that determine their shape and characteristics, and understanding these parameters is essential in statistical analysis and inference.

Probability Distribution Functions → PDF

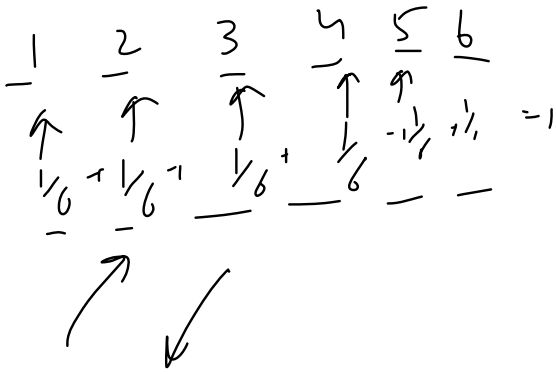
15 March 2023 20:08

A probability distribution function (~~PDF~~) is a mathematical function that describes the probability of obtaining different values of a random variable in a particular probability distribution.



Probability Mass Function (PMF)

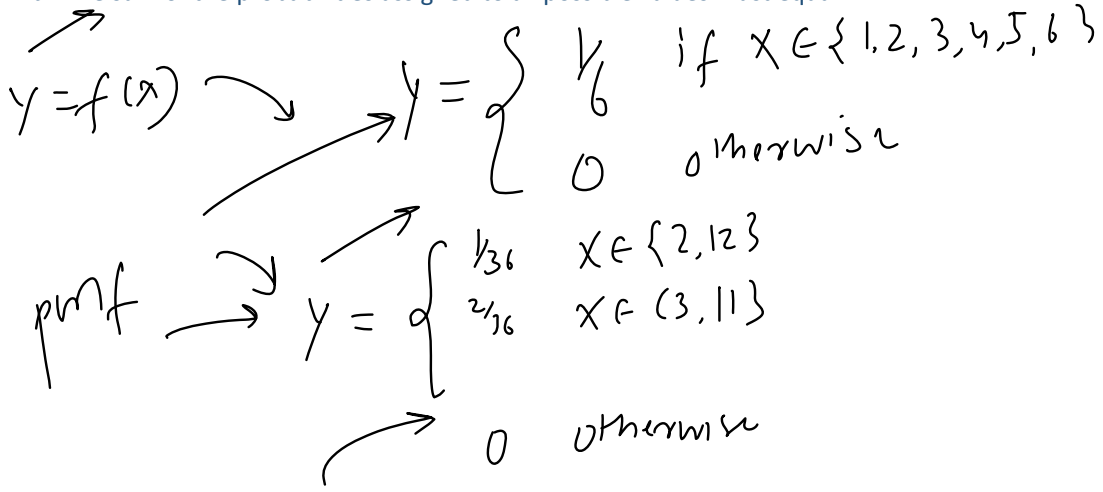
15 March 2023 15:25



PMF stands for Probability Mass Function. It is a mathematical function that describes the probability distribution of a **discrete random variable**.

The PMF of a discrete random variable assigns a probability to each possible value of the random variable. The probabilities assigned by the PMF must satisfy two conditions:

a. The probability assigned to each value must be non-negative (i.e., greater than or equal to zero).
b. The sum of the probabilities assigned to all possible values must equal 1.



Examples

https://en.wikipedia.org/wiki/Bernoulli_distribution

https://en.wikipedia.org/wiki/Binomial_distribution

Cumulative Distribution Function(CDF) of PMF

15 March 2023 20:09

The cumulative distribution function (CDF) F(x) describes the probability that a random variable X with a given probability distribution will be found at a value less than or equal to x

[F(x) = P(X <= x)]

f(x) =

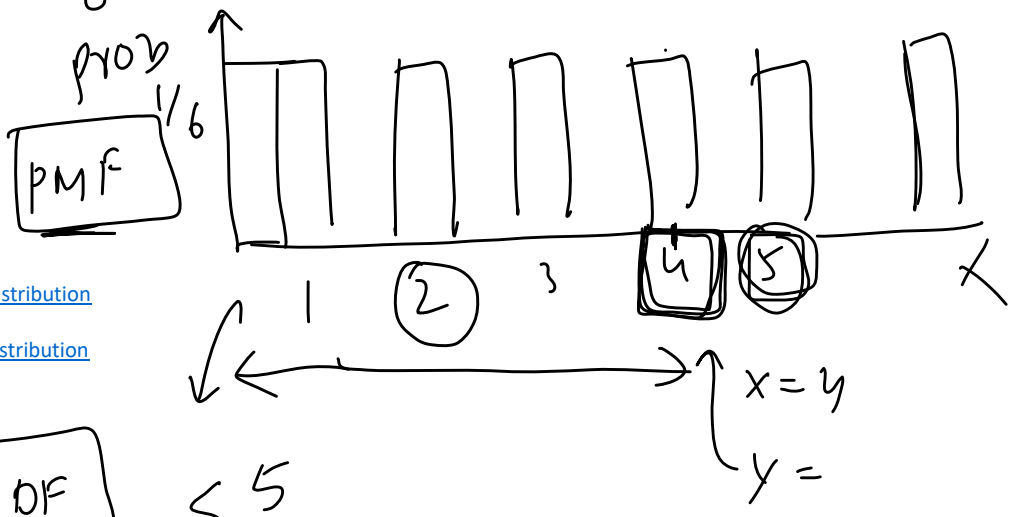
PMF f(x)

f(X=6) = 1/6

f(X=5) = 1/6

f(X≤4) = f(X=4) + f(X=3) + f(X=2) + f(X=1)

1/6 + 1/6 + 1/6 + 1/6



Examples:

https://en.wikipedia.org/wiki/Bernoulli_distribution

https://en.wikipedia.org/wiki/Binomial_distribution

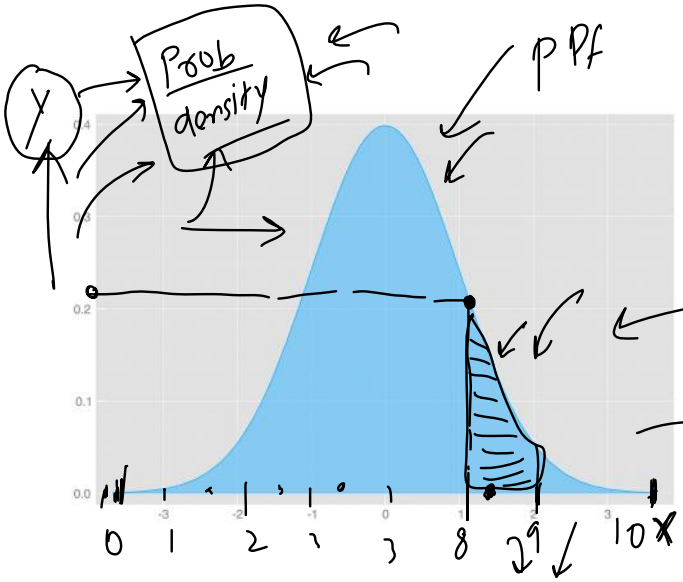
CDF ≤ 5

	PMF	CDF
1	1/6	1/6
2	1/6	2/6
3	1/6	3/6
4	1/6	4/6
5	1/6	5/6
6	1/6	6/6 → 1

Probability Density Function (PDF)

15 March 2023 15:25

PDF stands for Probability Density Function. It is a mathematical function that describes the probability distribution of a **continuous random variable.**



sample PDF

cgpa = 7.912

7.91234567 fg

$\int_8^9 f(x)\,dx$

$P(0 \le x \le 10) = 1$

area

8 → 9

probability between

8 → 8.1

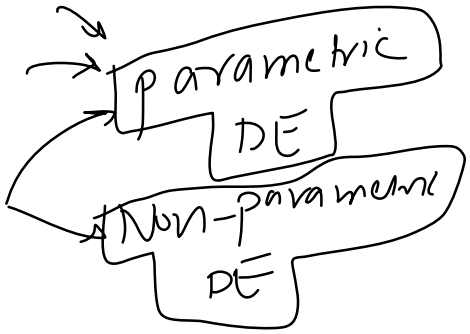
8 → 8.01

8 → 8.001

1. Why Probability Density and why not Probability?
2. What does the area of this graph represents?
3. How to calculate Probability then?
4. Examples of PDF
   a. https://en.wikipedia.org/wiki/Normal_distribution
   b. https://en.wikipedia.org/wiki/Log-normal_distribution
   c. https://en.wikipedia.org/wiki/Poisson_distribution
5. How is graph calculated?

Density Estimation

16 March 2023 06:54



Density estimation is a statistical technique used to estimate the probability density function (PDF) of a random variable based on a set of observations or data. In simpler terms, it involves estimating the underlying distribution of a set of data points.

Density estimation can be used for a variety of purposes, such as **hypothesis testing, data analysis, and data visualization**. It is particularly useful in areas such as **machine learning**, where it is often used to estimate the probability distribution of input data or to model the likelihood of certain events or outcomes.

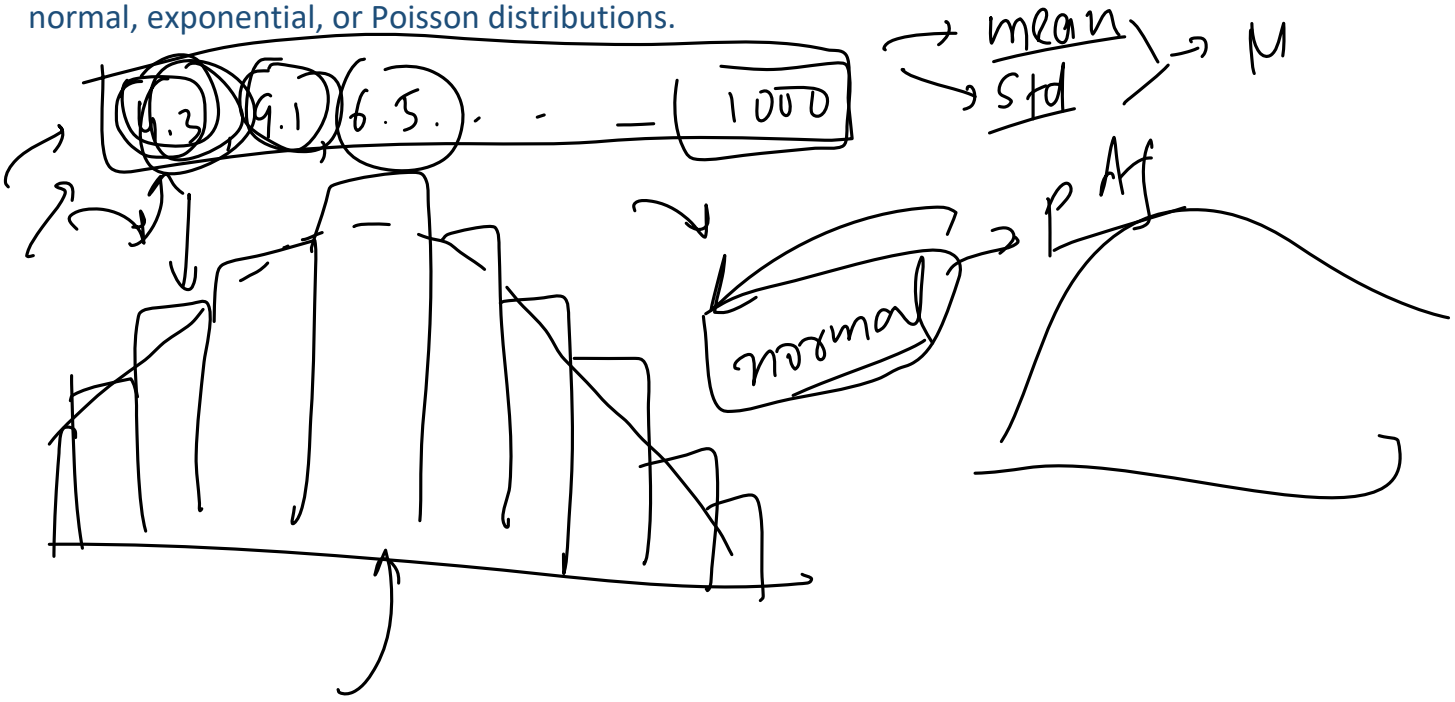
There are various methods for density estimation, including **parametric** and **non-parametric approaches**. Parametric methods assume that the data follows a specific probability distribution (such as a normal distribution), while non-parametric methods do not make any assumptions about the distribution and instead estimate it directly from the data.

Commonly used techniques for density estimation include **kernel density estimation (KDE)**, **histogram estimation**, and **Gaussian mixture models (GMMs)**. The choice of method depends on the specific characteristics of the data and the intended use of the density estimate.

Parametric Density Estimation

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Parametric density estimation is a method of estimating the probability density function (PDF) of a random variable by assuming that the underlying distribution belongs to a specific parametric family of probability distributions, such as the normal, exponential, or Poisson distributions.



Non-Parametric Density Estimation (KDE)

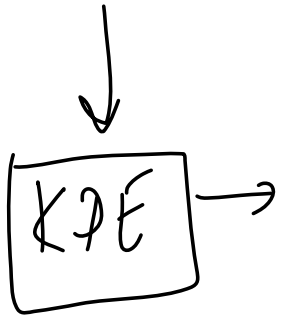
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But sometimes the distribution is not clear or it's not one of the famous distributions.

Non-parametric density estimation is a statistical technique used to estimate the probability density function of a random variable without making any assumptions about the underlying distribution. It is also referred to as non-parametric density estimation because it does not require the use of a predefined probability distribution function, as opposed to parametric methods such as the Gaussian distribution.

The non-parametric density estimation technique involves constructing an estimate of the probability density function using the available data. This is typically done by creating a **kernel density estimate**

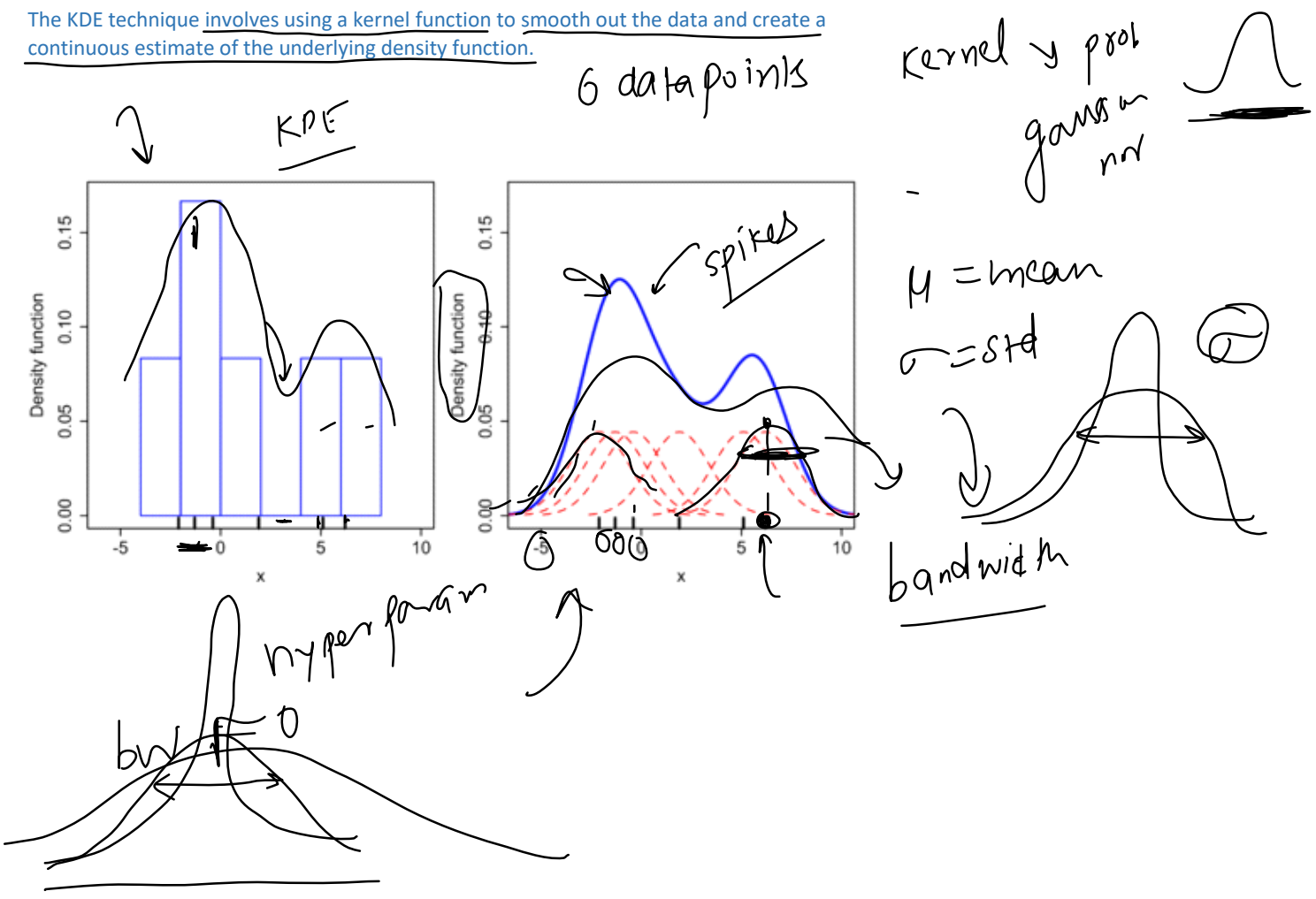
Non-parametric density estimation has several advantages over parametric density estimation. One of the main **advantages** is that it **does not require the assumption of a specific distribution**, which allows for more flexible and accurate estimation in situations where the underlying distribution is unknown or complex. However, non-parametric density estimation can be **computationally intensive and may require more data to achieve accurate estimates** compared to parametric methods.



Kernel Density Estimate(KDE)

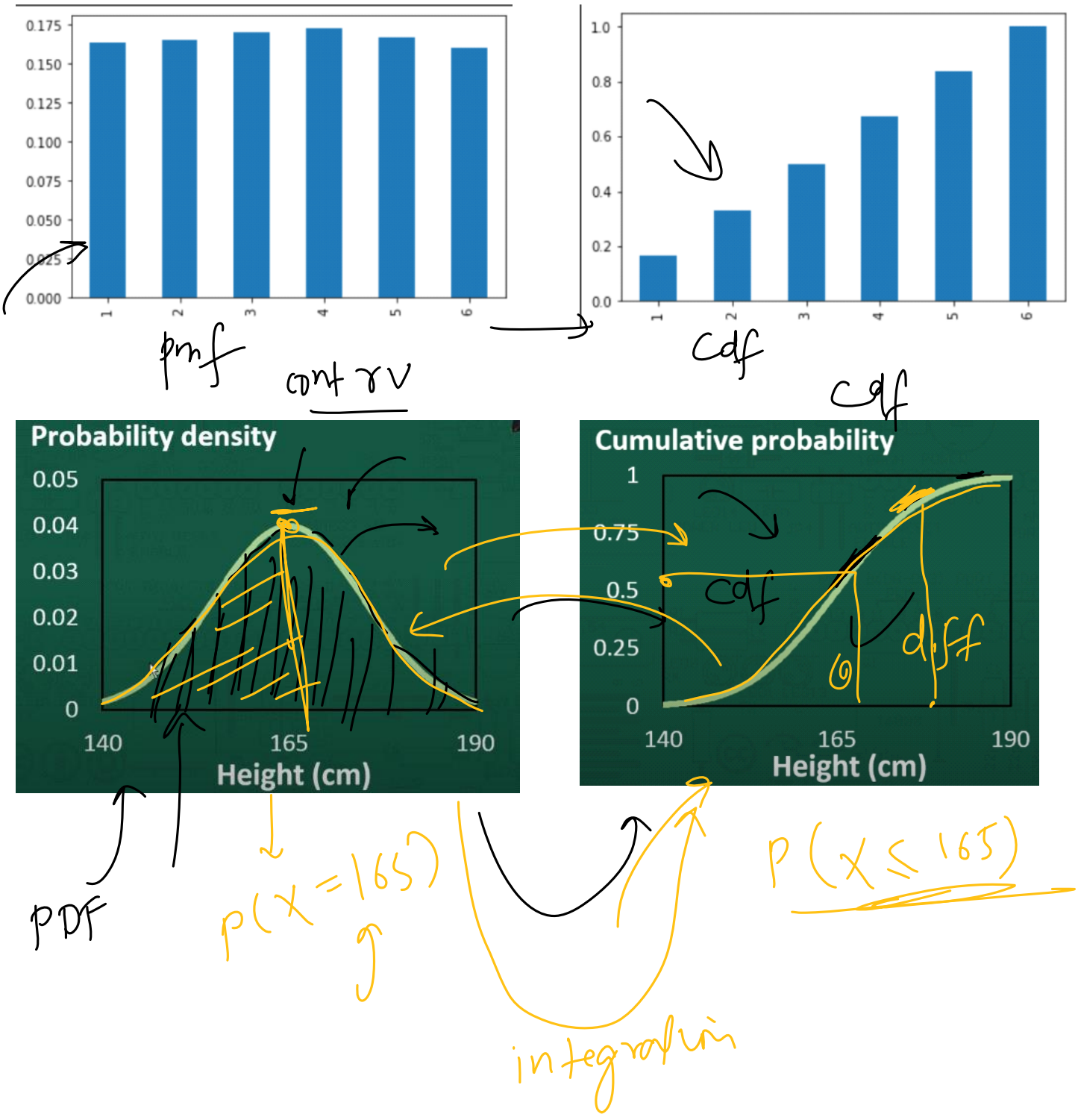
16 March 2023 16:08

The KDE technique involves using a kernel function to smooth out the data and create a continuous estimate of the underlying density function.



Cumulative Distribution Function(CDF) of PDF

15 March 2023 15:25



How to use PDF and CDF in Data Analysis

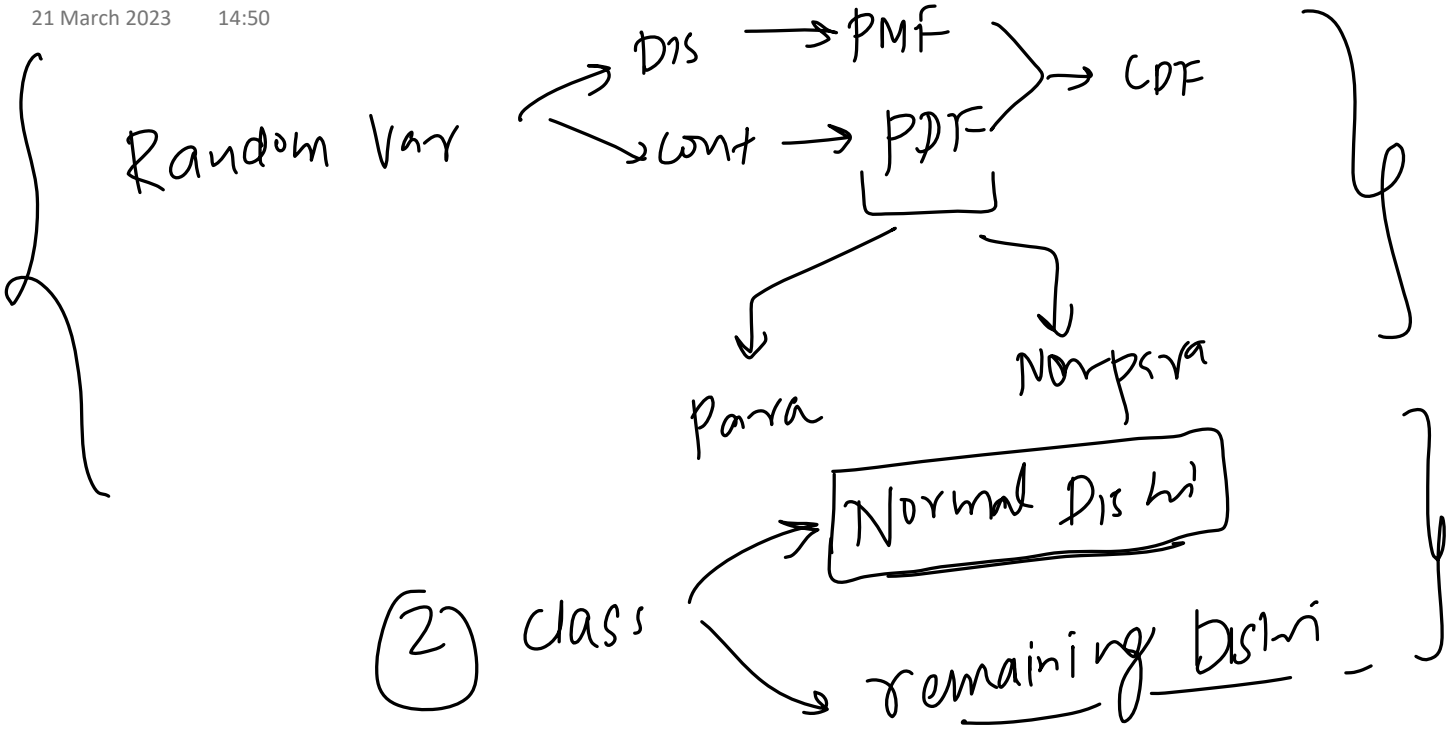
15 March 2023 20:10

2D Probability Density Plots

16 March 2023 06:50

Recap

21 March 2023 14:50



How to use PDF in Data Science

20 March 2023 18:11

PDF → CDF

PDF $P(X = x)$

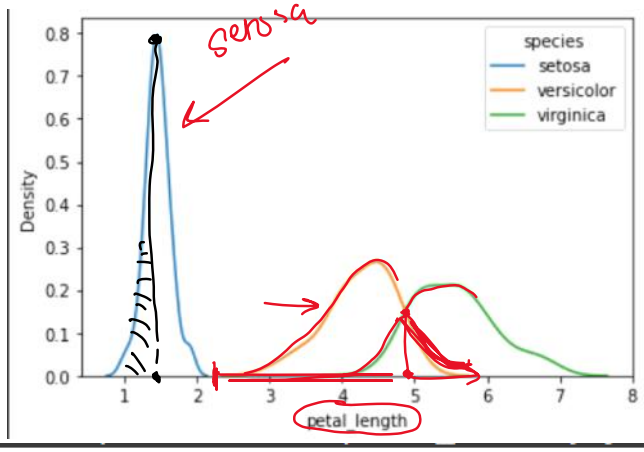
$2.3 < pl < 5$ → versicolor

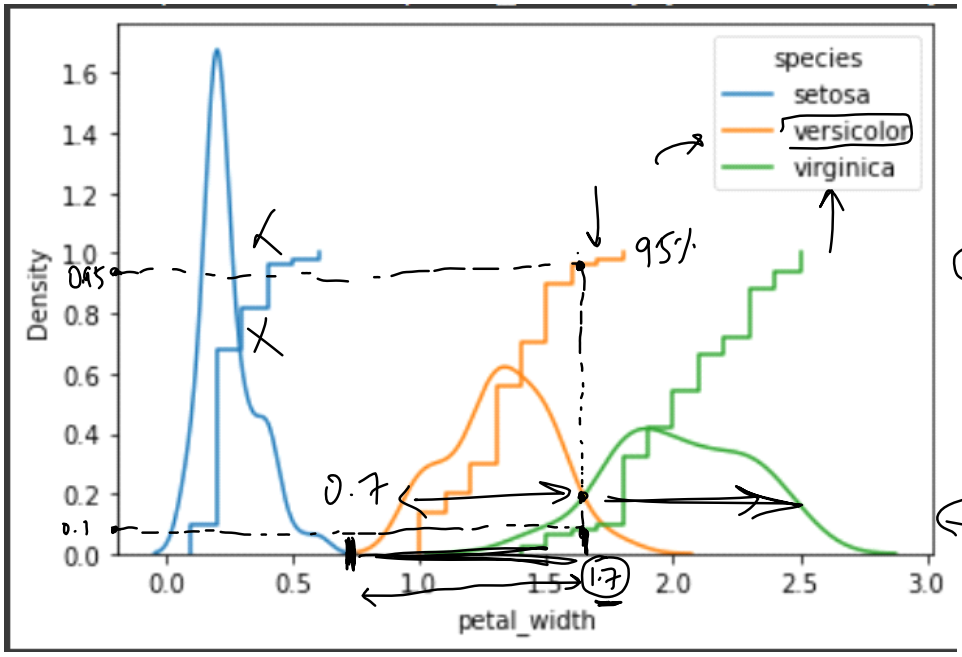
$P(X \leq x)$

$pl < 2.3$

> 2.3

$pl > 5$ virginica





95% of times

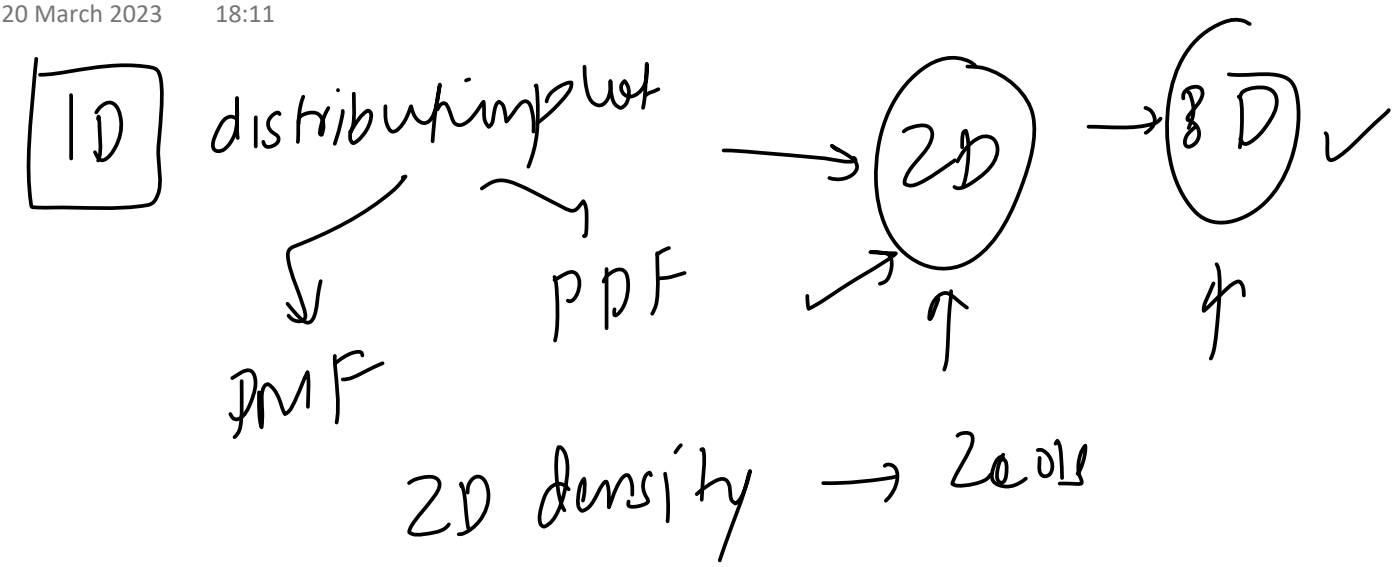
If $0.7 < pw < 1.7$ → versicolor

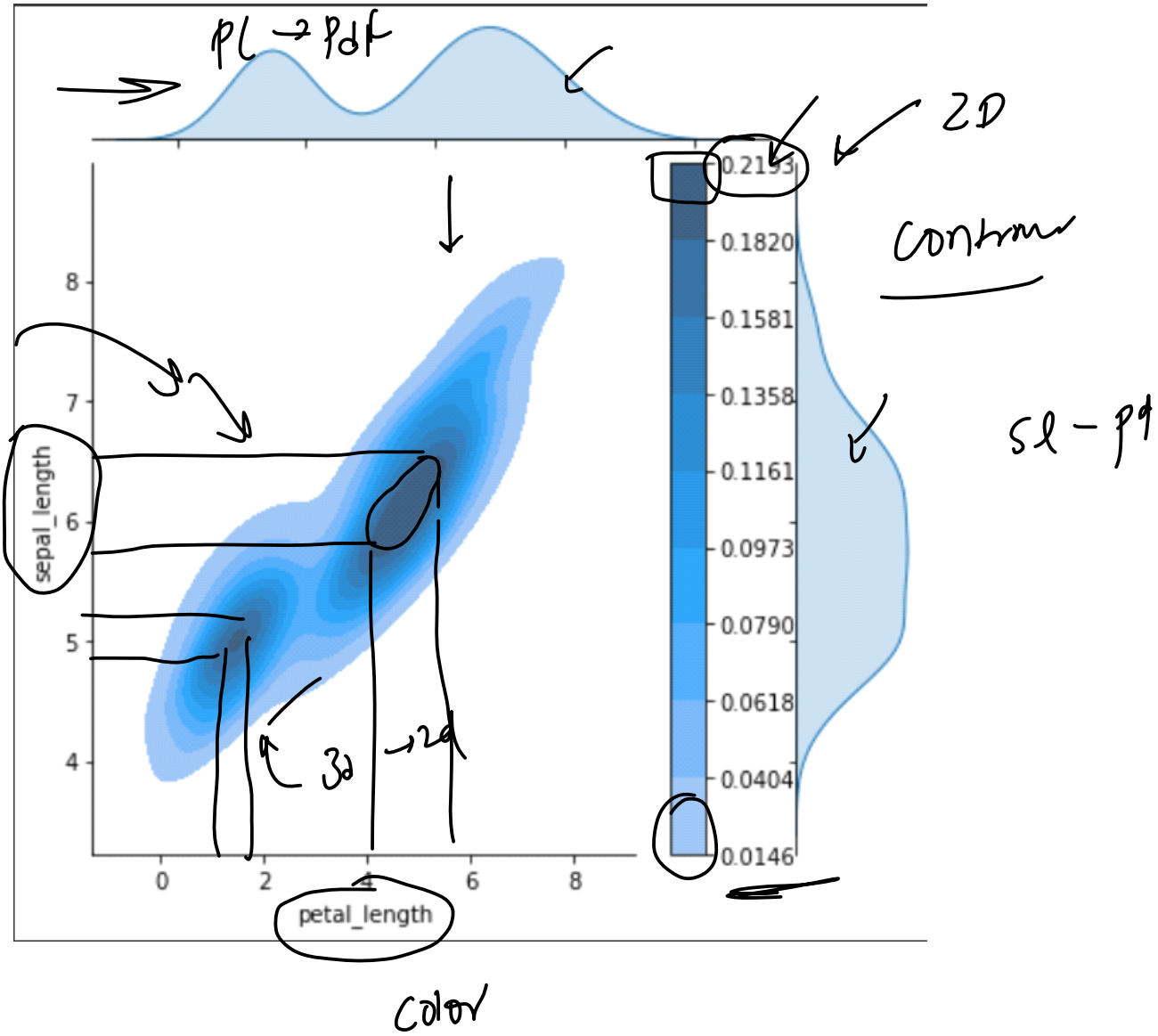
$pw > 1.7$ → virgin → 90%

10% virgin

2D Density Plots

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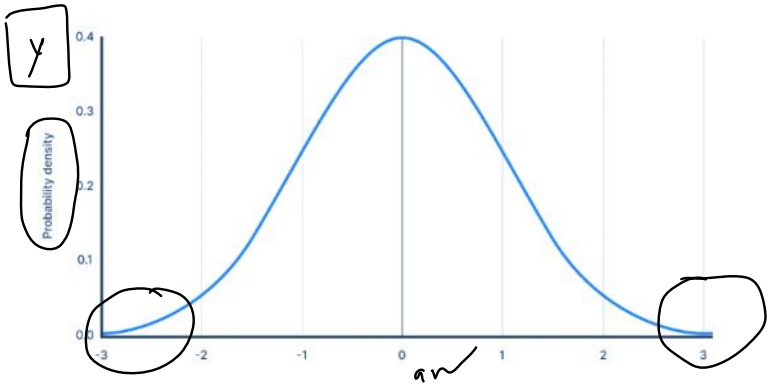


Normal Distribution

20 March 2023 18:06

1. What is normal distribution?

Normal distribution, also known as Gaussian distribution, is a probability distribution that is commonly used in statistical analysis. It is a continuous probability distribution that is symmetrical around the mean, with a bell-shaped curve.



-> Tail
-> Asymptotic in nature
-> Lots of points near the mean and very few far away

The normal distribution is characterized by two parameters: the mean (μ) and the standard deviation (σ). The mean represents the centre of the distribution, while the standard deviation represents the spread of the distribution.

Denoted as:

$$X \sim N(\mu, \sigma)$$

Why is it so important?

Commonality in Nature: Many natural phenomena follow a normal distribution, such as the heights of people, the weights of objects, the IQ scores of a population, and many more. Thus, the normal distribution provides a convenient way to model and analyse such data.

PDF Equation of Normal Distribution

$$y = f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

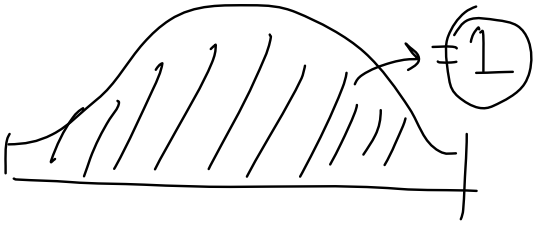
Parameters in Normal Distribution

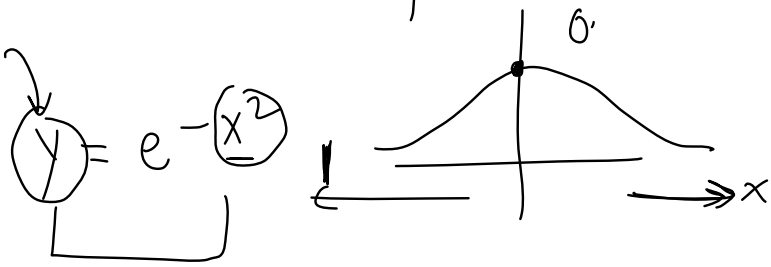
https://samp-suman-normal-dist-visualize-app-lkntug.streamlit.app/

Equation in detail:

$$y = e^{\frac{-(x-\mu)^2}{2\sigma^2}} = e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$\sigma\sqrt{2\pi}$





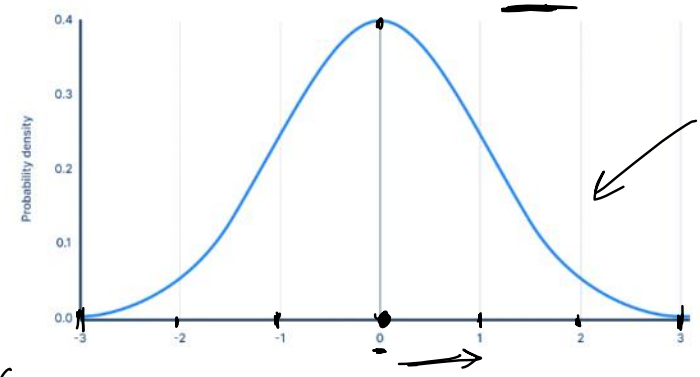
$$y = \frac{1}{e^{x^2}}$$

Standard Normal Variate

20 March 2023 18:08

- What is Standard Normal Variate

A Standard Normal Variate(Z) is a standardized form of the normal distribution with mean = 0 and standard deviation = 1.



Standardizing a normal distribution allows us to compare different distributions with each other, and to calculate probabilities using standardized tables or software.

Equation:

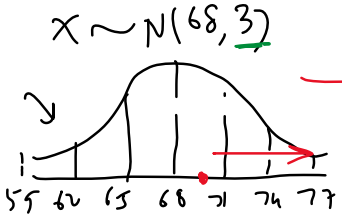
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

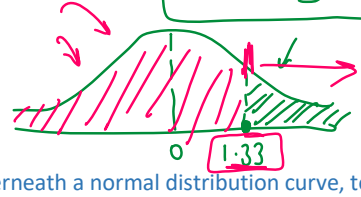
- How to transform a normal distribution to Standard Normal Variate

Refer Python code

Kya Fayda Standardize karne ka?

Suppose the heights of adult males in a certain population follow a normal distribution with a mean of 68 inches and a standard deviation of 3 inches. What is the probability that a randomly selected adult male from this population is taller than 72 inches?





- What are Z-tables

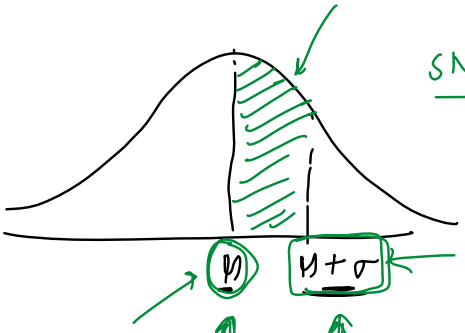
A z-table tells you the area underneath a normal distribution curve, to the left of the z-score

https://www.ztable.net/

For a Normal Distribution X~(u,std) what percent of population lie between mean and 1 standard deviation, 2 std and 3 std?

standard deviation, 2 std and 3 std.

$X \sim N(\mu, \sigma)$ %



SNV(z) →

$\mu \rightarrow$ ↓

$$z = \frac{x-\mu}{\sigma}$$

$$z = \frac{\mu-\mu}{\sigma} = 0$$

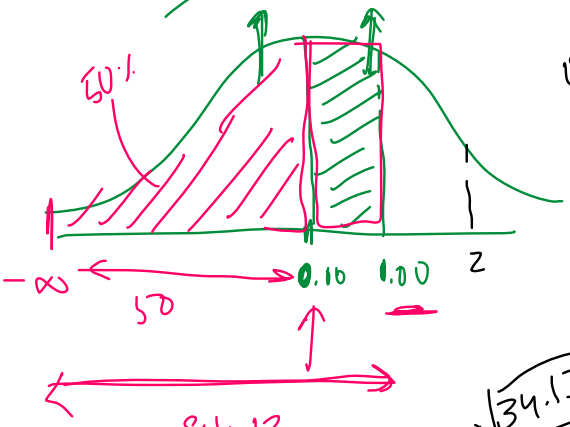
$$z = \frac{x-\mu}{\sigma} = \frac{\mu+\sigma-\mu}{\sigma}$$

$$\frac{\sigma}{\sigma} = 1$$

0.9772

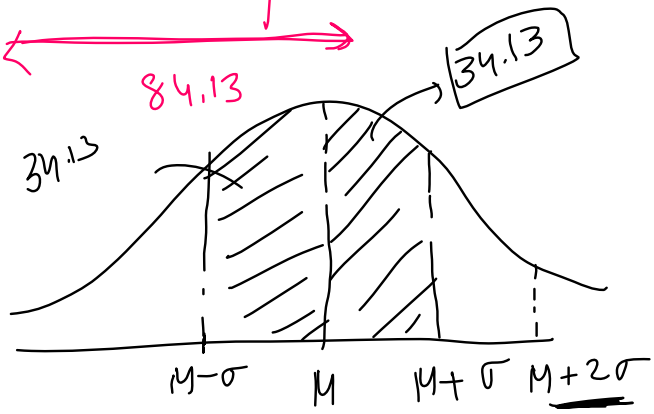
97%

84.13 − 50 = 34.13

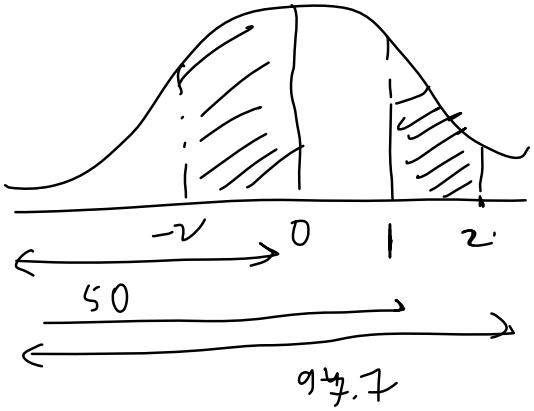


−1σ +1σ

64.28 %



$$z = \frac{\mu+2\sigma-\mu}{\sigma} = 2$$

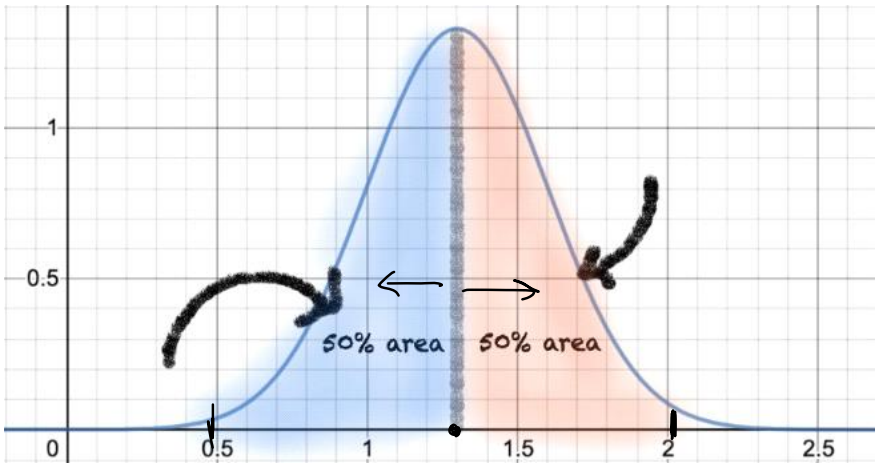


Properties of Normal Distribution

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1. Symmetricity

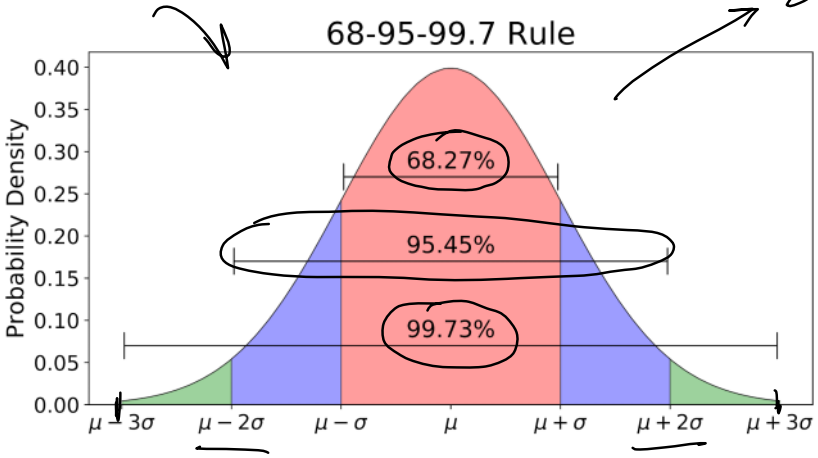
The normal distribution is symmetric about its mean which means that the probability of observing a value above the mean is the same as the probability of observing a value below the mean. The bell-shaped curve of the normal distribution reflects this symmetry.

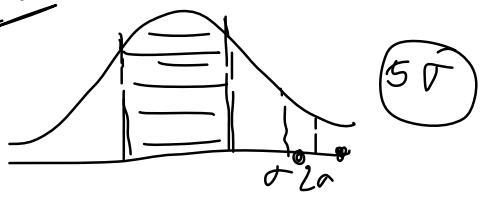


2. Measures of Central Tendencies are equal → mean → median → mode

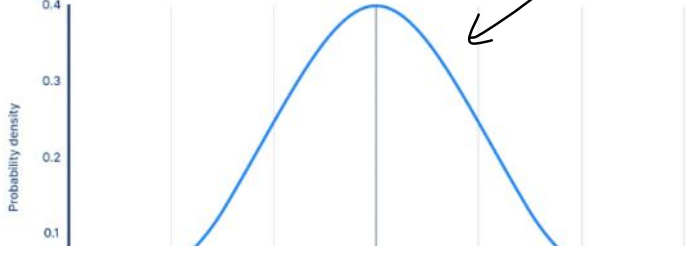
3. Empirical Rule

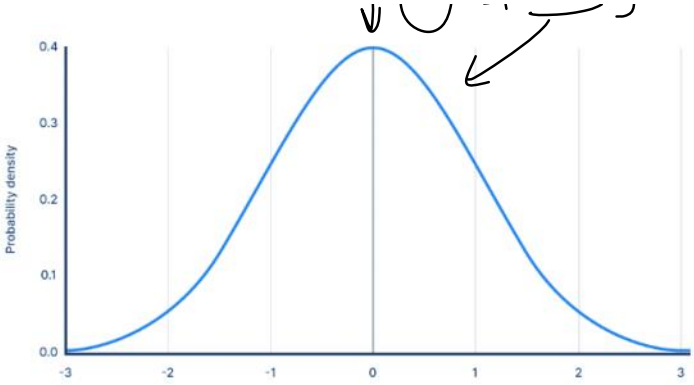
The normal distribution has a well-known empirical rule, also called the 68-95-99.7 rule, which states that approximately 68% of the data falls within one standard deviation of the mean, about 95% of the data falls within two standard deviations of the mean, and about 99.7% of the data falls within three standard deviations of the mean.





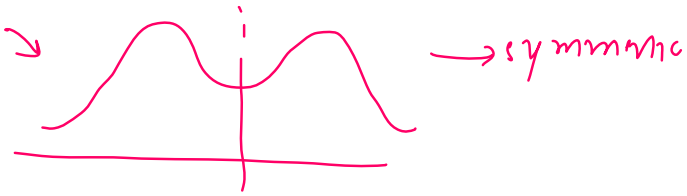
4. The area under the curve




Probability density
0.4
0.3
0.2
0.1
0.0
-3
-2
-1
0
1
2
3

Skewness

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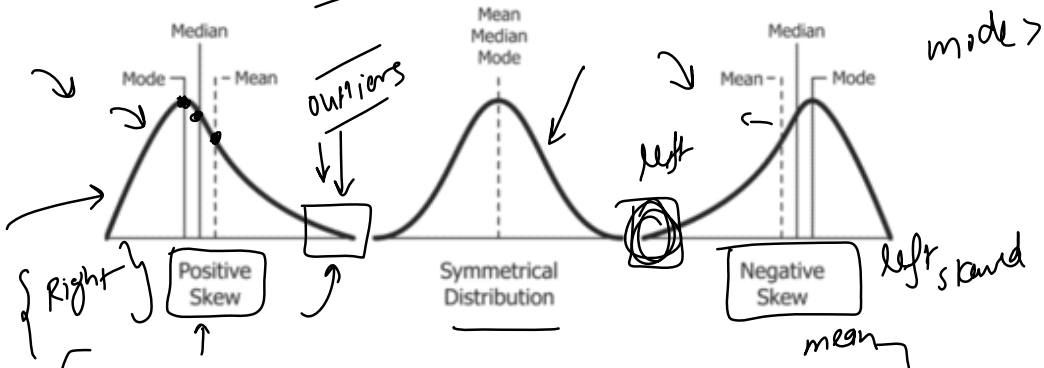
- What is skewness?

A normal distribution is a bell-shaped, symmetrical distribution with a specific mathematical formula that describes how the data is spread out. Skewness indicates that the data is not symmetrical, which means it is not normally distributed.

Skewness is a measure of the asymmetry of a probability distribution. It is a statistical measure that describes the degree to which a dataset deviates from the normal distribution.

In a symmetrical distribution, the mean, median, and mode are all equal. In contrast, in a skewed distribution, the mean, median, and mode are not equal, and the distribution tends to have a longer tail on one side than the other.

Skewness can be positive, negative, or zero. A positive skewness means that the tail of the distribution is longer on the right side, while a negative skewness means that the tail is longer on the left side. A zero skewness indicates a perfectly symmetrical distribution.

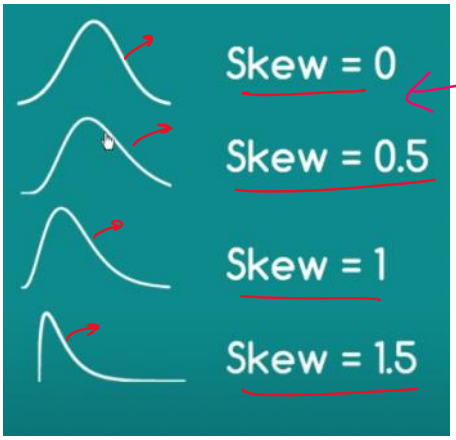


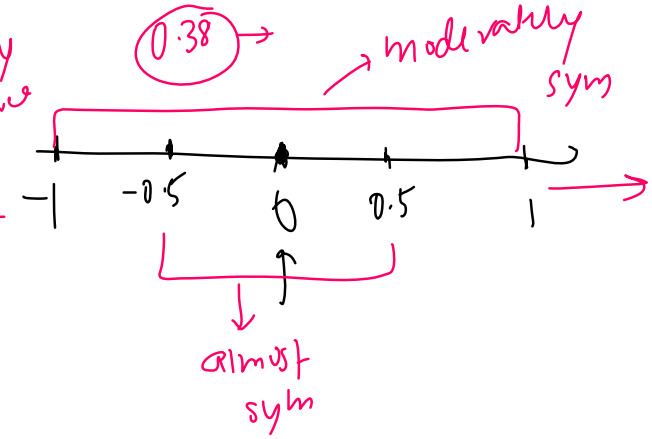
The greater the skew the greater the distance between mode, median and ~~mode~~ mean.

- How skewness is calculated?

$$\frac{n}{(n-1)(n-2)}\sum\left(\frac{(x-\bar{x})}{s}\right)^3$$

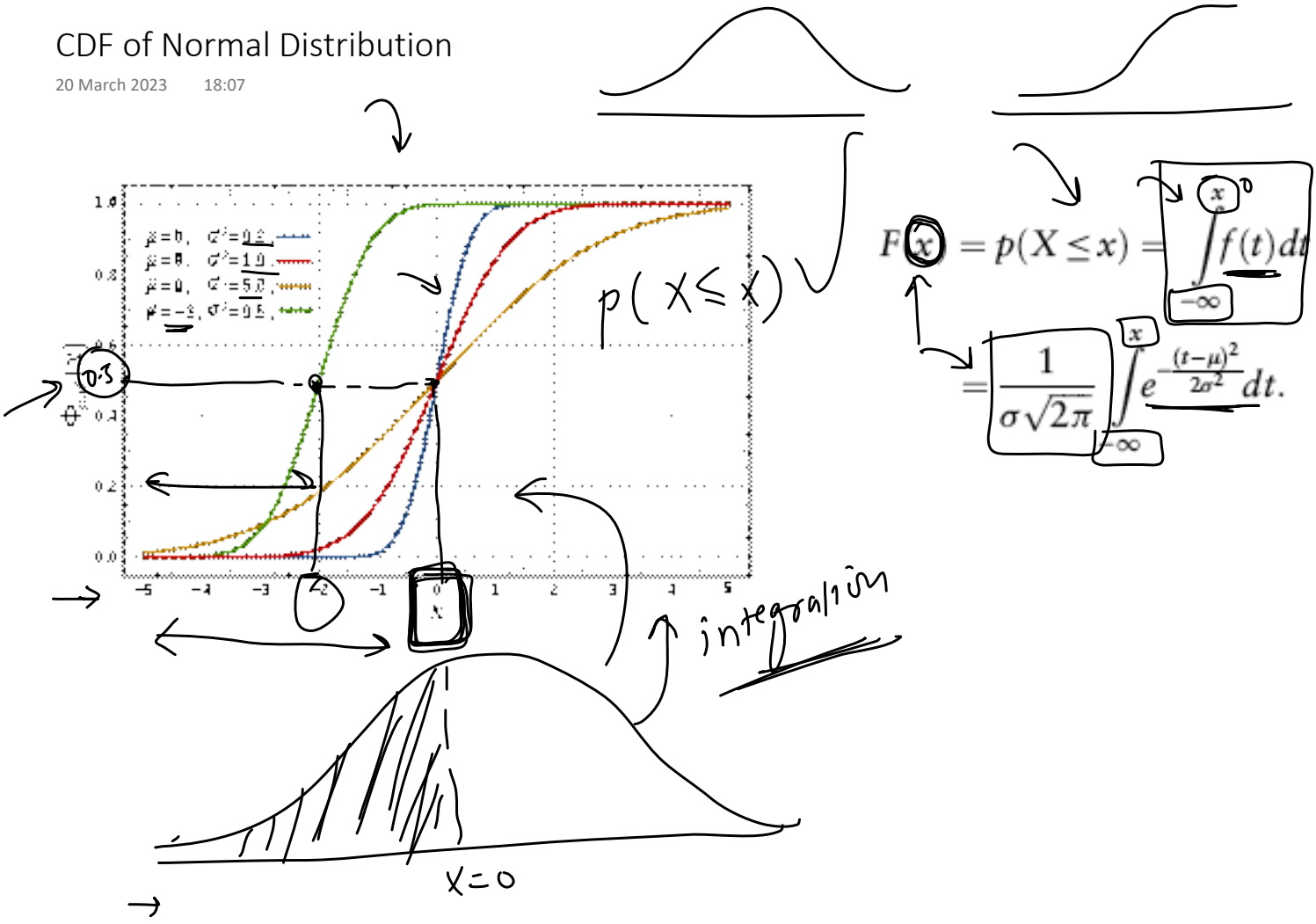
- Python Example
- Interpretation





CDF of Normal Distribution

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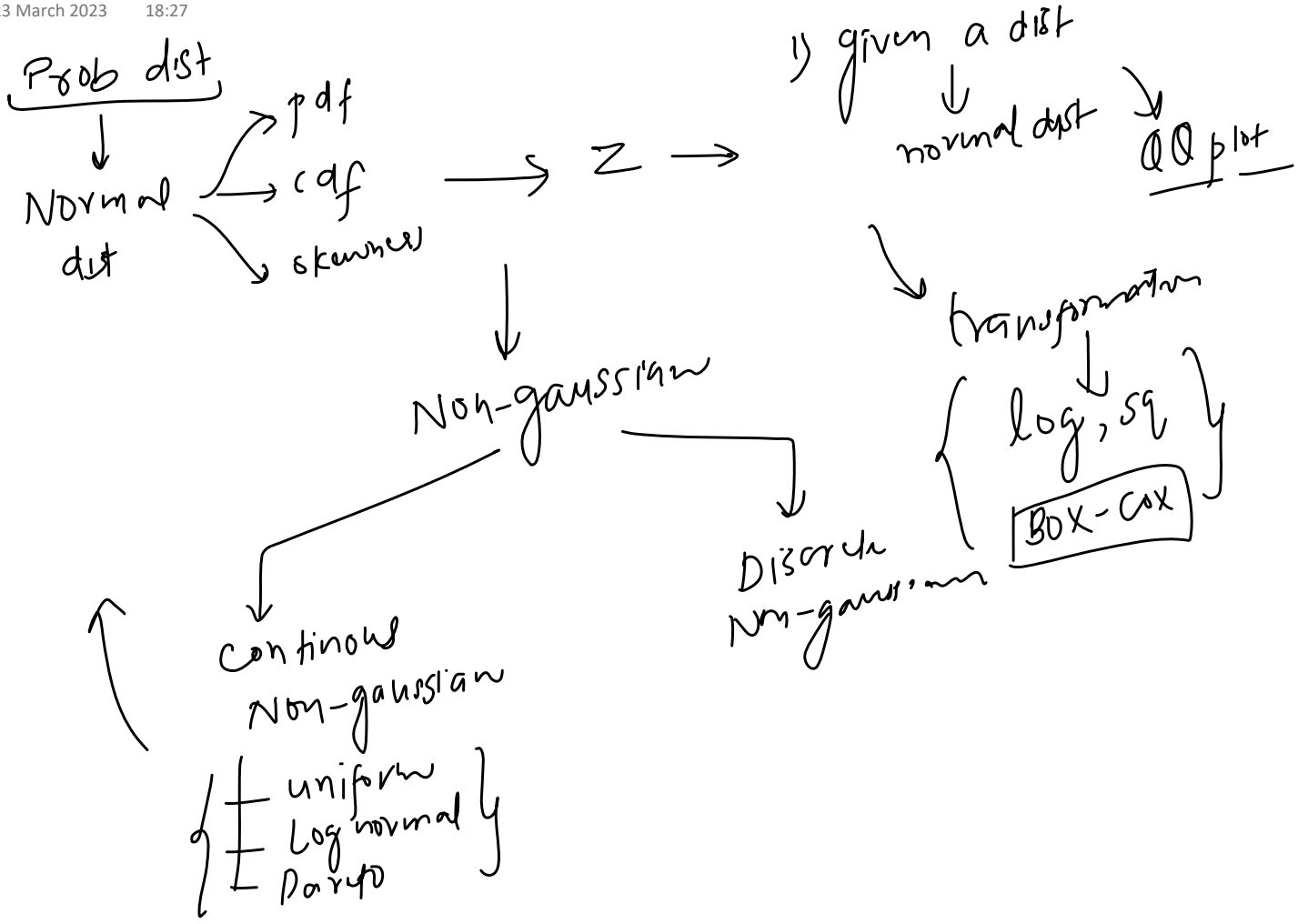
Use in Data Science

20 March 2023 18:08

- Outlier detection
- Assumptions on data for ML algorithms -> Linear Regression and GMM
- Hypothesis Testing
- Central Limit Theorem

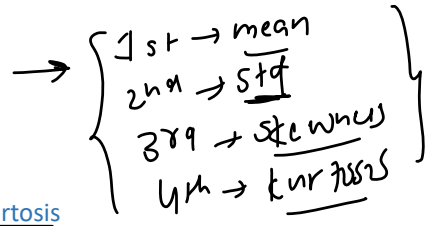
Recap

23 March 2023 18:27



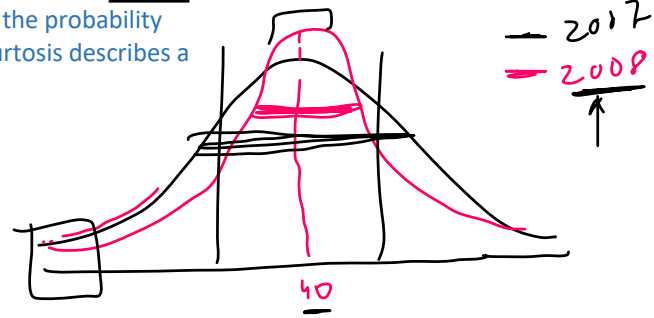
Kurtosis

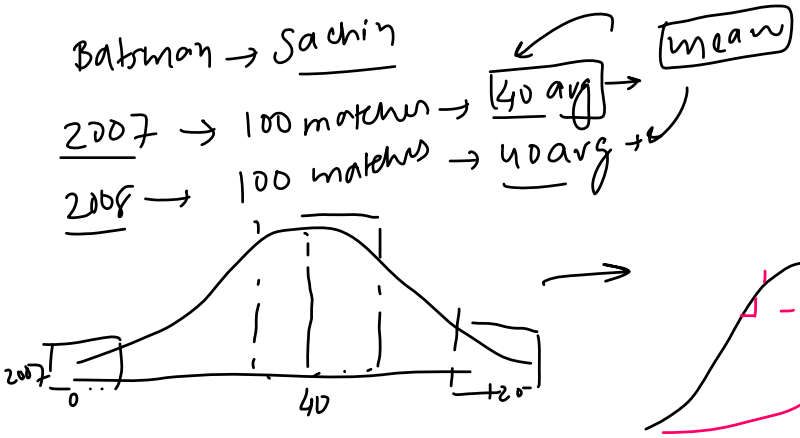
23 March 2023 13:18



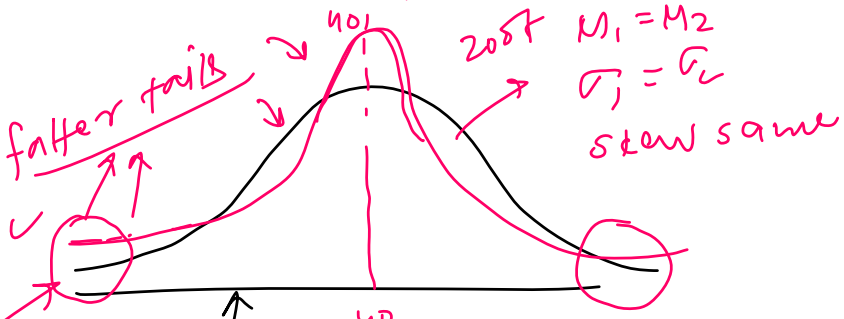
- What is Kurtosis?

Kurtosis is the 4th statistical moment. In probability theory and statistics, kurtosis (meaning "curved, arching") is a measure of the "tailedness" of the probability distribution of a real-valued random variable. Like skewness, kurtosis describes a particular aspect of a probability distribution.









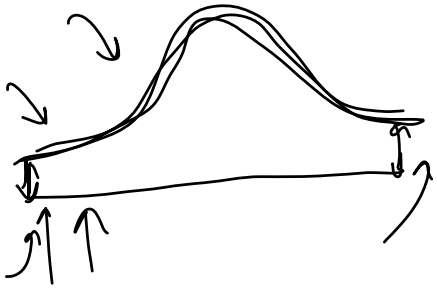
- False notation about Kurtosis

https://en.wikipedia.org/wiki/Kurtosis

- Formula

sample kurtosis

$$\left\{\frac{n*(n+1)}{(n-1)*(n-2)*(n-3)} * \sum_i^n \left(\frac{x_i - \bar{x}}{s}\right)^4\right\} - \frac{3*(n-1)^2}{(n-2)*(n-3)}$$



- Practical Use-case

In finance, kurtosis risk refers to the risk associated with the possibility of extreme outcomes or "fat tails" in the distribution of returns of a particular asset or portfolio.

If a distribution has high kurtosis, it means that there is a higher likelihood of extreme events occurring, either positive or negative, compared to a normal distribution.

In finance, kurtosis risk is important to consider because it indicates that there is a greater probability of large losses or gains occurring, which can have significant implications for investors. As a result, investors may want to adjust their investment strategies to account for kurtosis risk.

- Excess Kurtosis & Types

Excess kurtosis is a measure of how much more peaked or flat a distribution is

Excess Kurtosis & Types

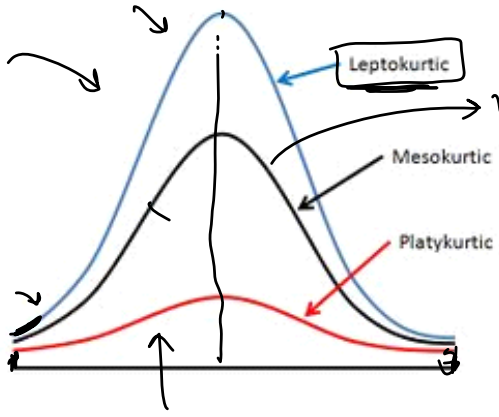
Excess kurtosis is a measure of how much more peaked or flat a distribution is compared to a normal distribution, which is considered to have a kurtosis of 0. It is calculated by subtracting 3 from the sample kurtosis coefficient.

Types of Kurtosis

Leptokurtic

A distribution with positive excess kurtosis is called leptokurtic. "Lepto-" means "slender". In terms of shape, a leptokurtic distribution has fatter tails. This indicates that there are more extreme values or outliers in the distribution.

Example - Assets with positive excess kurtosis are riskier and more volatile than those with a normal distribution, and they may experience sudden price movements that can result in significant gains or losses.



Platykurtic

A distribution with negative excess kurtosis is called platykurtic. "Platy-" means "broad". In terms of shape, a platykurtic distribution has thinner tails. This indicates that there are fewer extreme values or outliers in the distribution.

Assets with negative excess kurtosis are less risky and less volatile than those with a normal distribution, and they may experience more gradual price movements that are less likely to result in large gains or losses.

Mesokurtic

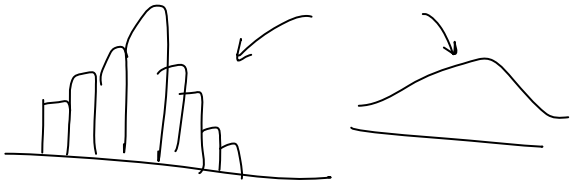
Distributions with zero excess kurtosis are called mesokurtic. The most prominent example of a mesokurtic distribution is the normal distribution family, regardless of the values of its parameters.

Mesokurtic is a term used to describe a distribution with a excess kurtosis of 0, indicating that it has the same degree of "peakedness" or "flatness" as a normal distribution.

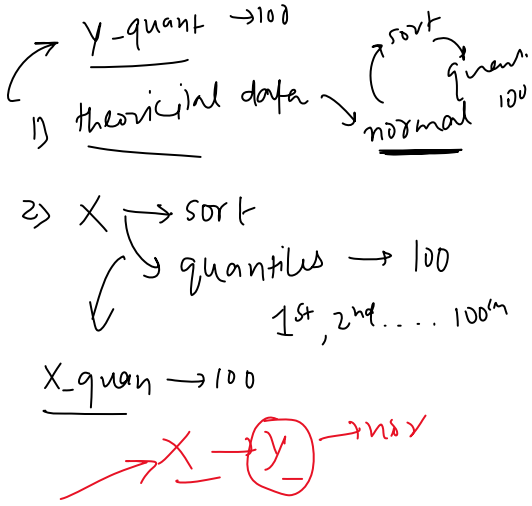
Example -
In finance, a mesokurtic distribution is considered to be the ideal distribution for assets or portfolios, as it represents a balance between risk and return.

QQ Plot

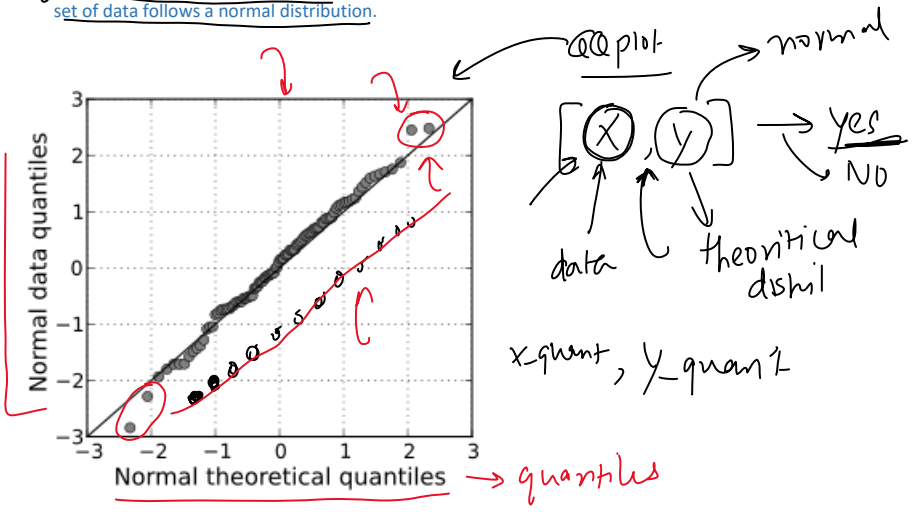
23 March 2023 13:19



- How to find if a given distribution is normal or not?
  - *Visual inspection* One of the easiest ways to check for normality is to visually inspect a histogram or a density plot of the data. A normal distribution has a bell-shaped curve, which means that the majority of the data falls in the middle, and the tails taper off symmetrically. If the distribution looks approximately bell-shaped, it is likely to be normal.
  - *QQ Plot* Another way to check for normality is to create a normal probability plot (also known as a Q-Q plot) of the data. A normal probability plot plots the observed data against the expected values of a normal distribution. If the data points fall along a straight line, the distribution is likely to be normal.
  - *Statistical tests*: There are several statistical tests that can be used to test for normality, such as the Shapiro-Wilk test, the Anderson-Darling test, and the Kolmogorov-Smirnov test. These tests compare the observed data to the expected values of a normal distribution and provide a p-value that indicates whether the data is likely to be normal or not. A p-value less than the significance level (usually 0.05) suggests that the data is not normal.



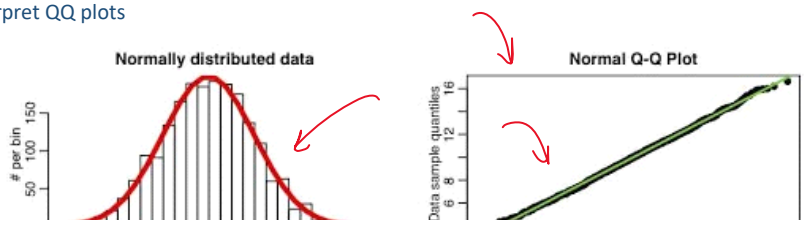
- What is a QQ Plot and how is it plotted?
  - A QQ plot (quantile-quantile plot) is a graphical tool used to assess the similarity of the distribution of two sets of data. It is particularly useful for determining whether a set of data follows a normal distribution.



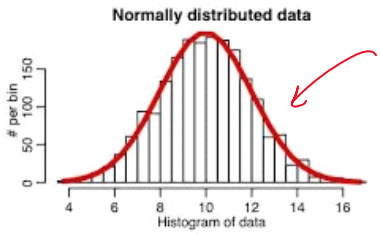
  - In a QQ plot, the quantiles of the two sets of data are plotted against each other. The quantiles of one set of data are plotted on the x-axis, while the quantiles of the other set of data are plotted on the y-axis. If the two sets of data have the same distribution, the points on the QQ plot will fall on a straight line. If the two sets of data do not have the same distribution, the points will deviate from the straight line.

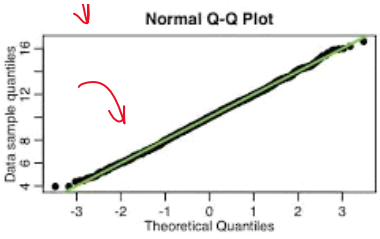
- Python example
- https://www.statsmodels.org/dev/generated/statsmodels.graphics.gofplots.qqplot.html

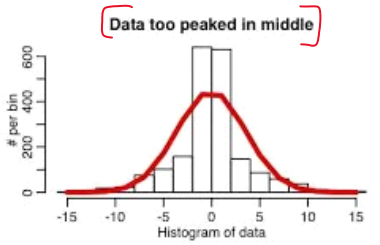
- How to interpret QQ plots

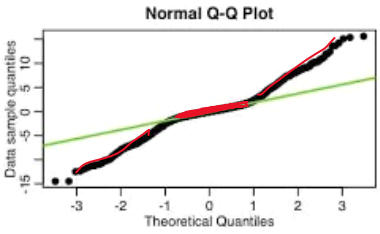


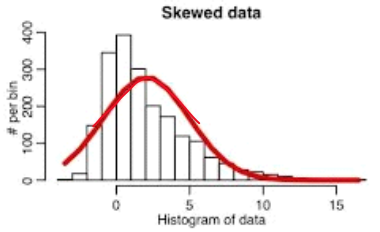
1.

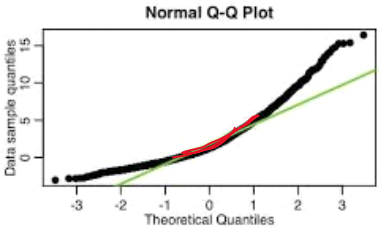

Normally distributed data
per bin
Histogram of data

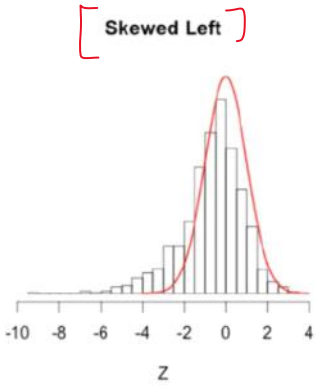

Normal Q-Q Plot
Data sample quantiles
Theoretical Quantiles

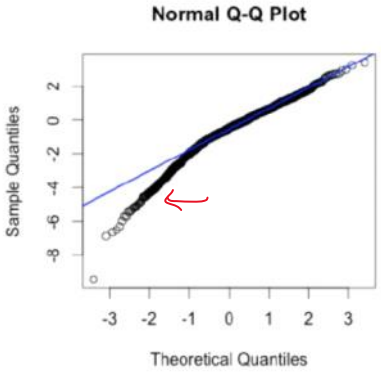

Data too peaked in middle
per bin
Histogram of data


Normal Q-Q Plot
Data sample quantiles
Theoretical Quantiles

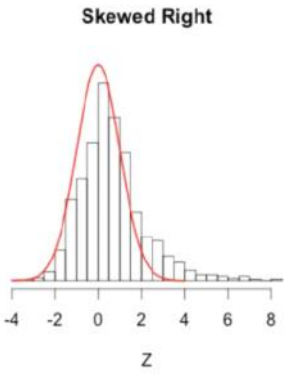

Skewed data
per bin
Histogram of data

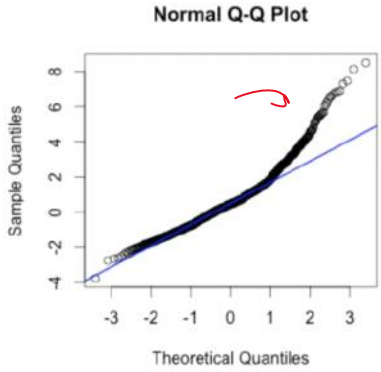

Normal Q-Q Plot
Data sample quantiles
Theoretical Quantiles


Skewed Left
Z

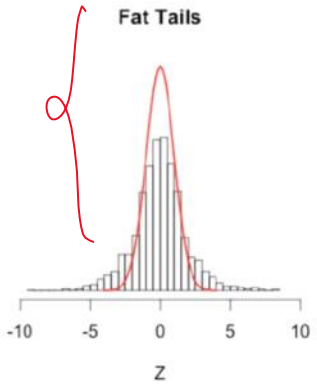

Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles

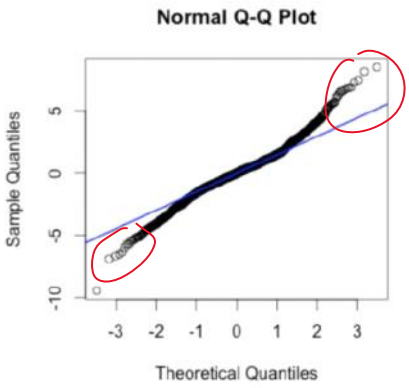
1.

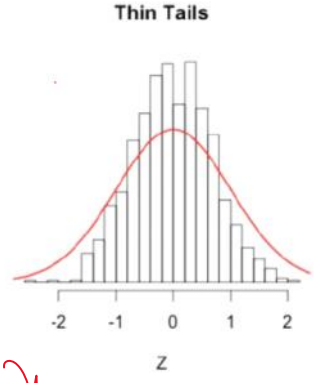

Skewed Right
Z

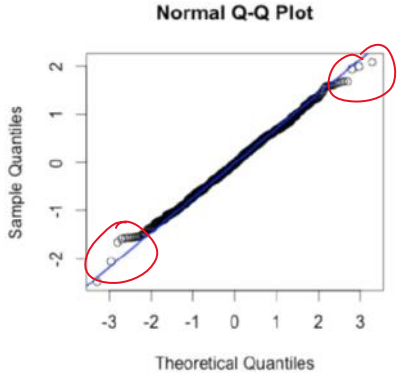

Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles

1.

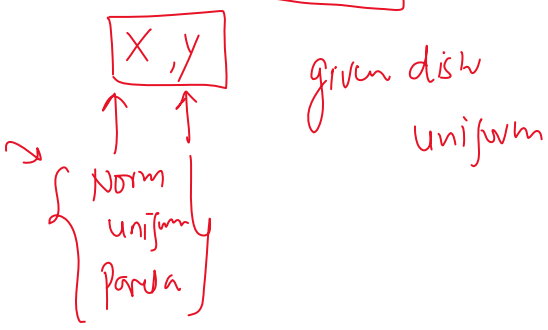


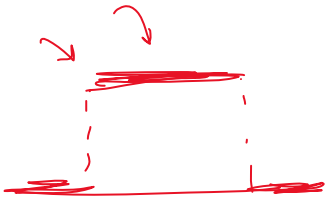






- Does QQ plot only detect normal distribution?





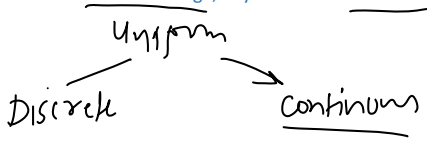
Uniform Distribution

23 March 2023 13:19

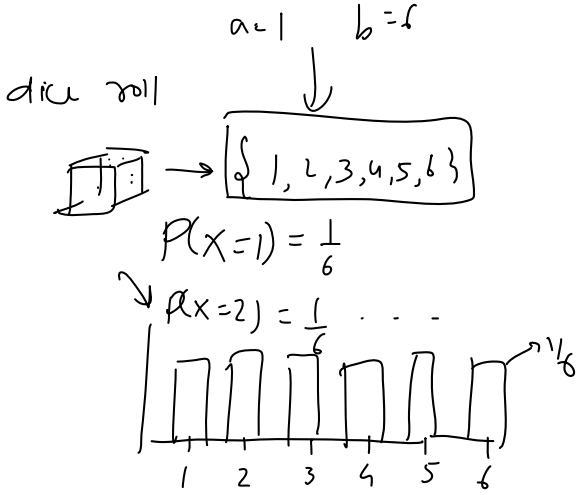
- What is Uniform Distribution and it's types

In probability theory and statistics, a uniform distribution is a probability distribution where all outcomes are equally likely within a given range. This means that if you were to select a random value from this range, any value would be as likely as any other value.

Types



Denoted as $X \sim U(a,b)$ → parameters (a ↔ lower, b ↔ high)

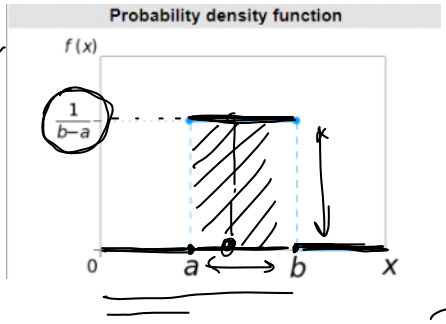


- Examples

a. The height of a person randomly selected from a group of individuals whose heights range from 5'6" to 6'0" would follow a continuous uniform distribution.
b. The time it takes for a machine to produce a product, where the production time ranges from 5 to 10 minutes, would follow a continuous uniform distribution.
c. The distance that a randomly selected car travels on a tank of gas, where the distance ranges from 300 to 400 miles, would follow a continuous uniform distribution.
d. The weight of a randomly selected apple from a basket of apples that weighs between 100 and 200 grams, would follow a continuous uniform distribution.

- PDF CDF and Graphs

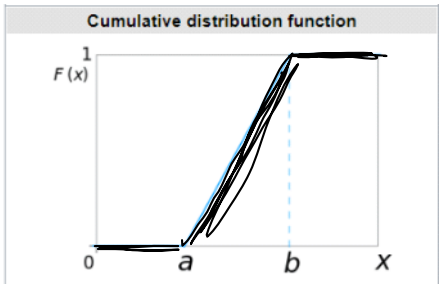
PDF



$\frac{1}{b-a}$ ← $f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \le x \le b \\ 0 & \text{for anywhere } x<a \text{ or } x>b \end{cases}$

$b-a$

CDF

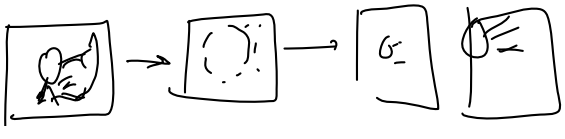


https://en.wikipedia.org/wiki/Continuous_uniform_distribution

- Skewness → 0 → Symmetric → Normal

- Application in Machine learning and Data Science

a. **Random initialization** In many machine learning algorithms, such as neural networks and k-means clustering, the initial values of the parameters can have a significant impact on the final result. Uniform distribution is often used to randomly initialize the parameters, as it ensures that all values in the range have an equal probability of being selected.

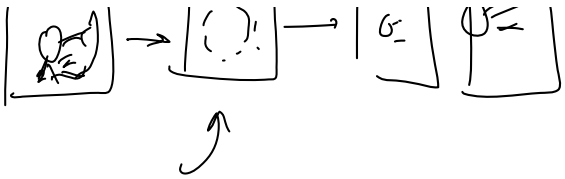


a. **Random initialization**: In many machine learning algorithms, such as neural networks and k-means clustering, the initial values of the parameters can have a significant impact on the final result. Uniform distribution is often used to randomly initialize the parameters, as it ensures that all values in the range have an equal probability of being selected.

b. **Sampling**: Uniform distribution can also be used for sampling. For example, if you have a dataset with an equal number of samples from each class, you can use uniform distribution to randomly select a subset of the data that is representative of all the classes.

c. **Data augmentation**: In some cases, you may want to artificially increase the size of your dataset by generating new examples that are similar to the original data. Uniform distribution can be used to generate new data points that are within a specified range of the original data.

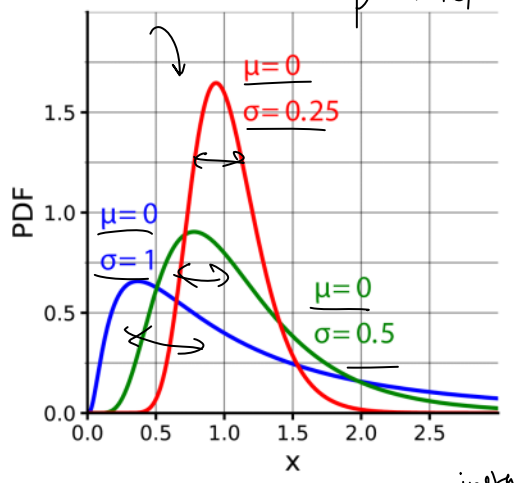
d. **Hyperparameter tuning**: Uniform distribution can also be used in hyperparameter tuning, where you need to search for the best combination of hyperparameters for a machine learning model. By defining a uniform prior distribution for each hyperparameter, you can sample from the distribution to explore the hyperparameter space.



Log Normal Distribution

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In probability theory and statistics, a lognormal distribution is a heavy tailed continuous probability distribution of a random variable whose logarithm is normally distributed.



Examples

- The length of comments posted in Internet discussion forums follows a log-normal distribution.
- Users' dwell time on online articles (jokes, news etc.) follows a log-normal distribution.
- The length of chess games tends to follow a log-normal distribution.
- In economics, there is evidence that the income of 97%–99% of the population is distributed log-normally.

Denoted as

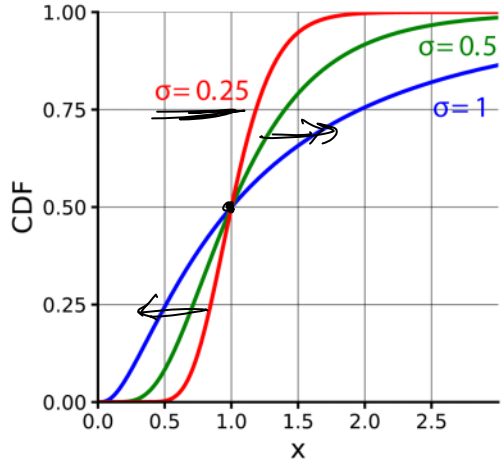
$X \sim \text{lognormal}(\mu, \sigma) \rightarrow \ln(x) \sim N(\mu', \sigma')$

PDF Equation

$$\frac{1}{x\sigma\sqrt{2\pi}} e^{\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)}$$

← similar Normal dist.

CDF



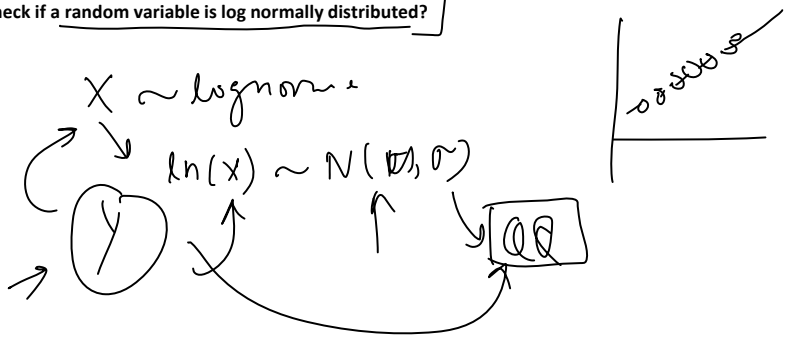
Skewness

Skewed

How to check if a random variable is log normally distributed?

Skewness

How to check if a random variable is log normally distributed?



Pareto Distribution

23 March 2023 13:19

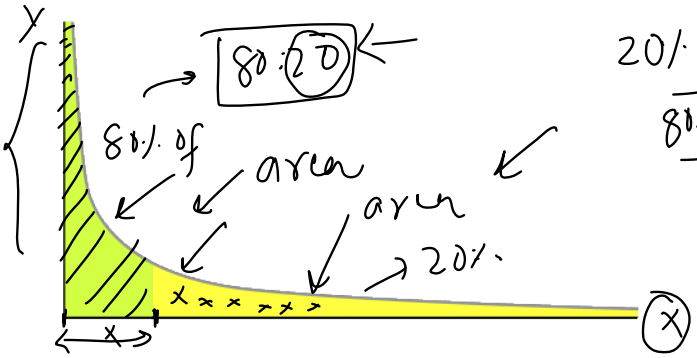
Pareto Distribution

The Pareto distribution is a type of probability distribution that is commonly used to model the distribution of wealth, income, and other quantities that exhibit a similar power-law behaviour

What is Power Law

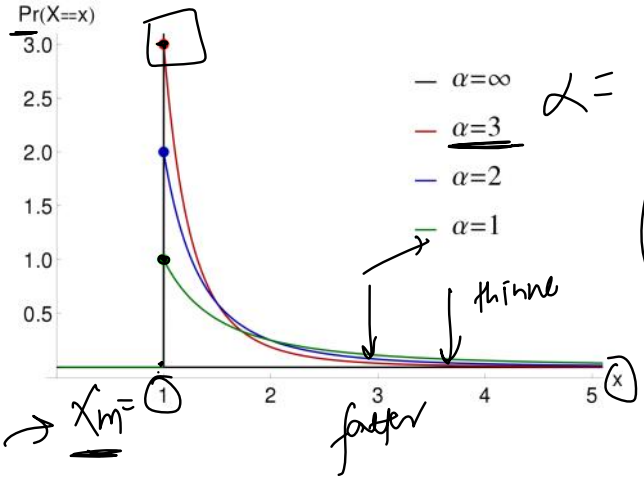
In mathematics, a power law is a functional relationship between two variables, where one variable is proportional to a power of the other. Specifically, if y and x are two variables related by a power law, then the relationship can be written as:

y = k * x^a



Vilfredo Pareto originally used this distribution to describe the allocation of wealth among individuals since it seemed to show rather well the way that a larger portion of the wealth of any society is owned by a smaller percentage of the people in that society. He also used it to describe distribution of income. This idea is sometimes expressed more simply as the Pareto principle or the "80-20 rule" which says that 20% of the population controls 80% of the wealth

Graph & Parameters

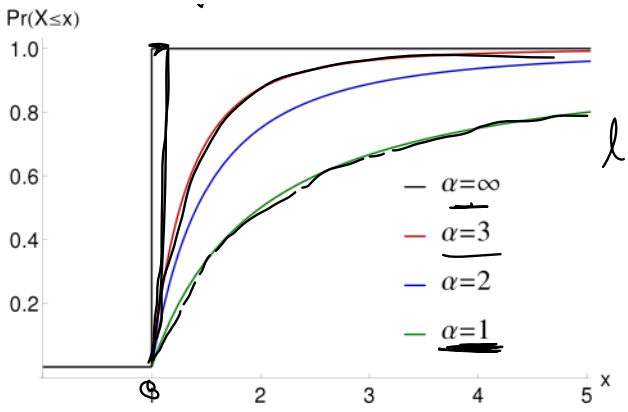


Examples

- The sizes of human settlements (few cities, many hamlets/villages)
- File size distribution of Internet traffic which uses the TCP protocol (many smaller files, few larger ones)

CDF

Pr(X<x)



log (y)

pareto

x log

Skewness → skewed

log-log plot → log

log x

[How to detect if a distribution is Pareto Distribution?]

$$y = \frac{\alpha x_m^{\alpha}}{x^{\alpha+1}}$$

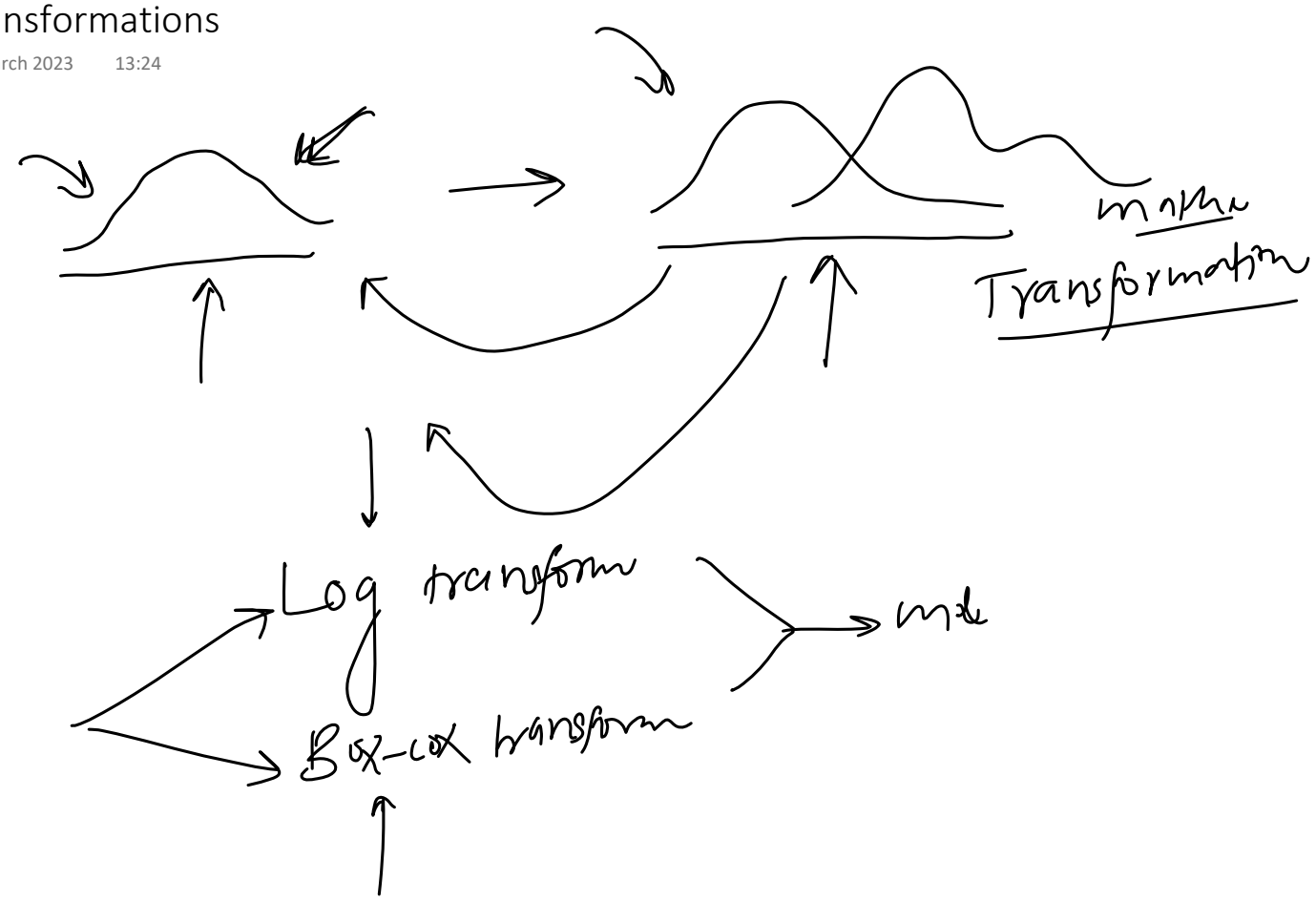
QQ plot

x, y

pareto

Transformations

23 March 2023 13:24



Views

24 March 2023 17:37

- What are Views?

In SQL, a view is a virtual table that does not store any data on its own but presents a customized view of one or more tables in a database. A view can be thought of as a pre-defined SELECT statement that retrieves data from one or more tables and returns a specific subset of data to the user.

So basically it is a logical table instead of a physical table

Once a view is created, it can be used in the same way as a table in SQL queries, and any changes made to the underlying tables will be reflected in the view.(Show)

Simple Views - Created from 1 single table

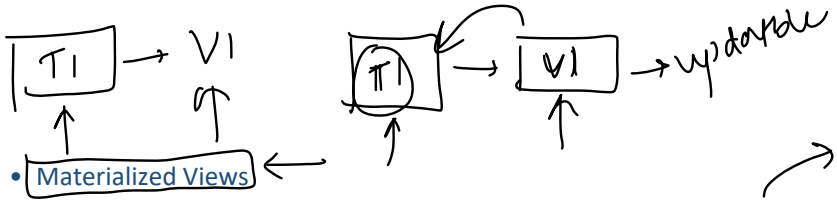
Complex Views - Created from multiple tables with the help of joins, subquery etc.

- Read only Vs Updatable Views

a. **Read-only views**: As the name suggests, read-only views are views that cannot be updated. They are used to simplify the process of querying data, but they cannot be used to modify or delete data in the underlying tables.

b. **Updatable views**: Updatable views are views that allow you to modify, insert or delete data in the underlying tables through the view. They behave like normal tables, but with restrictions.

To make a view updatable, certain conditions must be met. For example, the view must not contain any derived columns, subqueries, or aggregate functions. Additionally, the view must be based on a single table or a join of tables with a unique one-to-one relationship.

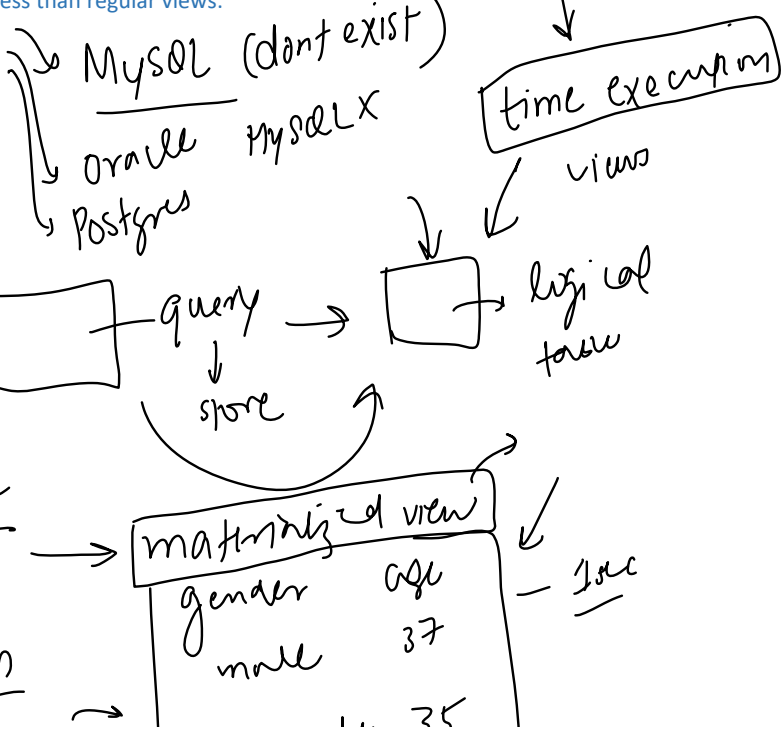


- Materialized Views

A materialized view is a database object in SQL that contains the results of a query. Unlike regular views, which are just virtual tables that store SQL queries, materialized views are physical tables that store the results of a query. Materialized views are precomputed and stored on disk, which makes them much faster to access than regular views.

Benefit - Faster queries

Disadvantage - Need to manually update the view

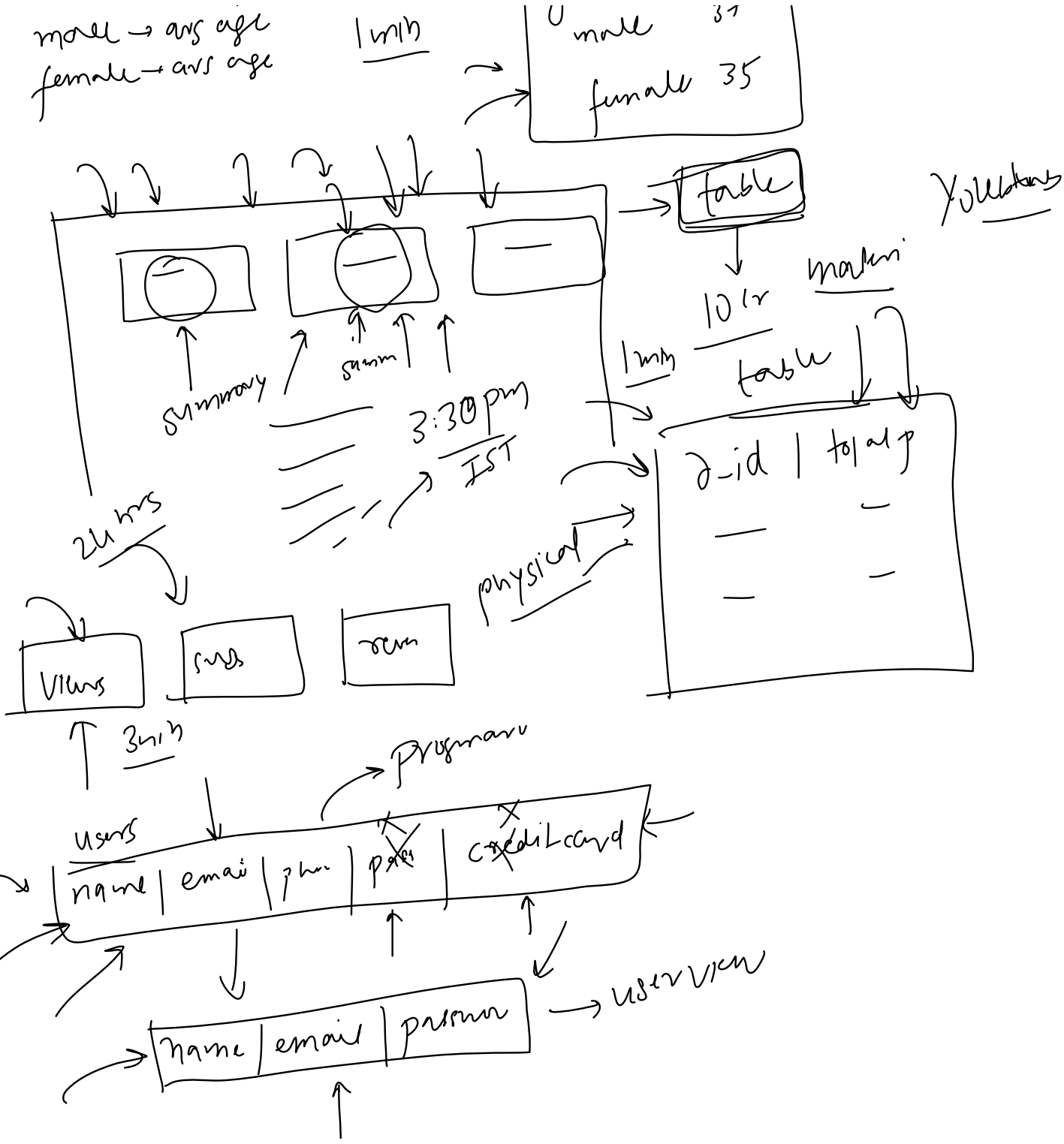


- Advantages of using a View

No Physical Storage

Security

Make complex queries simple

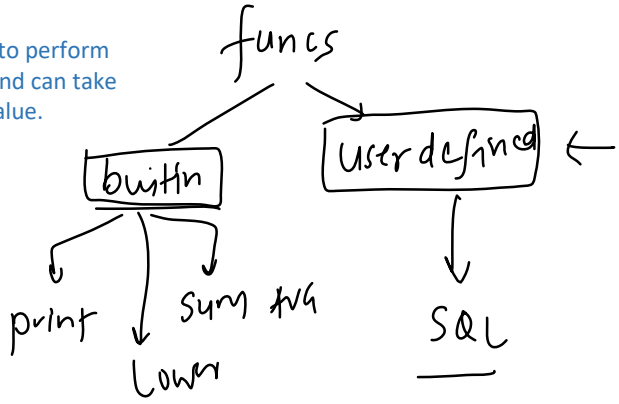

male → avg age
female → avg age
1 min
male
female 35
table
summary
3:30 pm
IST
24 hrs
1 min
10 hr
table
physical
Views
3 min
primary
users
name
email
password
credit card
name
email
password
user view

User Defined Functions

24 March 2023 18:58

- What are function and their advantages

User-defined functions (UDFs) in SQL are functions that are created by users to perform specific tasks. These functions can be used just like built-in functions in SQL and can take parameters as input, perform some operations on them, and then return a value.

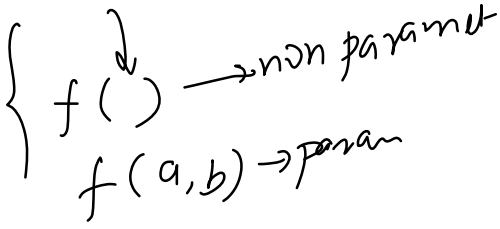


- Syntax

```
DELIMITER $$

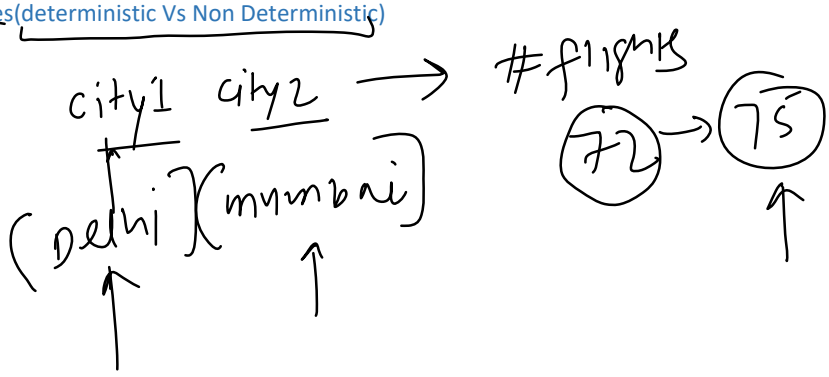
CREATE FUNCTION Function_Name(
    Parameter_1 DataType,
    Parameter_2 DataType,
    Parameter_n DataType,
)
RETURNS Return_Datatype
[NOT] DETERMINISTIC
BEGIN
      Function Body
      Return Return_Value
END $$

DELIMITER ;
```



Examples

- hello world
- parameterized vs non parameterized
- Calculate age in years(for col)
- greet with name -> conditional title
- Date formatting and flights between 2 cities(deterministic Vs Non Deterministic)
- show all functions of a database
- Drop function



Benefits

1. Simplifies SQL queries
2. Reusability
3. Enhances readability

debugging

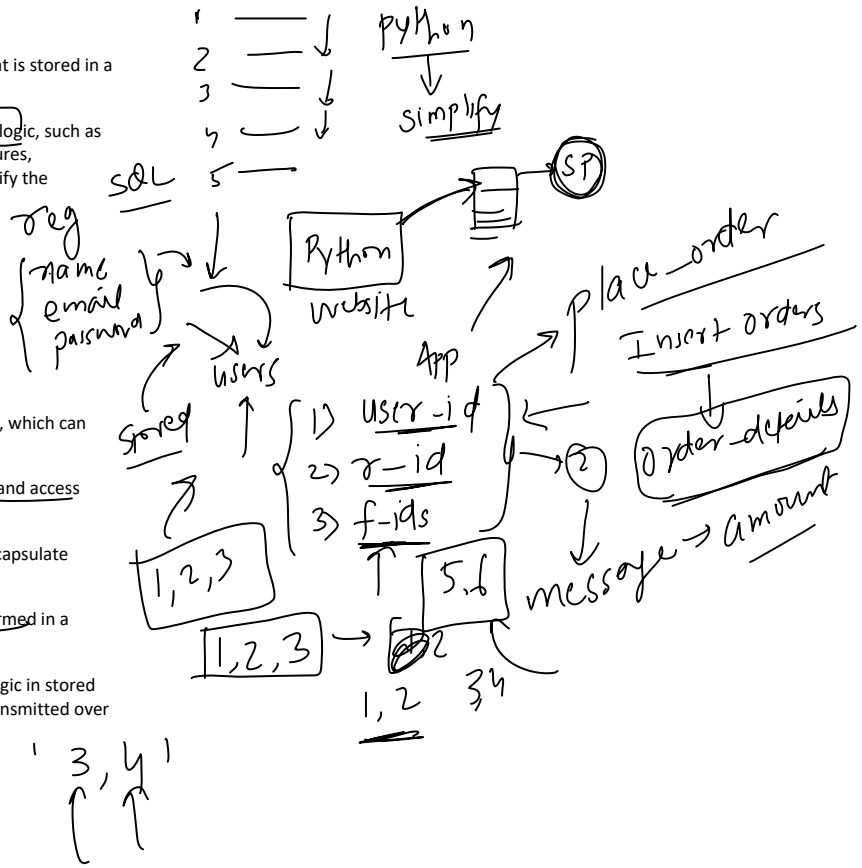
Stored Procedures

25 March 2023 16:39

A stored procedure is a named block of SQL statements and procedural logic that is stored in a database and can be executed by a user or application.

Stored procedures are often used to encapsulate business logic and application logic, such as data validation, data processing, and database updates. By using stored procedures, developers can separate application logic from the presentation layer and simplify the application code.

- Create hello world stored procedure
- Create stored procedure to create a new user
- Show error message
- Create stored procedure to show orders placed by 1 single user
- Create a stored procedure to place an order



Some of the benefits of using stored procedures include:

1. **Improved performance**: Stored procedures are precompiled and optimized, which can improve performance and reduce network traffic.
2. **Enhanced security**: Stored procedures can be granted specific permissions and access rights, which can improve security and limit access to sensitive data.
3. **Encapsulation of business logic**: Stored procedures allow developers to encapsulate complex business logic and make it easier to maintain and update.
4. **Consistency**: Stored procedures ensure that database operations are performed in a consistent manner, which can help to maintain data integrity.
5. **Reduced network traffic**: By encapsulating data access and manipulation logic in stored procedures, developers can reduce the amount of data that needs to be transmitted over the network.



Transactions

25 March 2023 08:19

What are Transactions?

A database transaction is a sequence of operations that are performed as a single logical unit of work in a database management system (DBMS). A transaction may consist of one or more database operations, such as inserts, updates, or deletes, which are treated as a single atomic operation by the DBMS.

It follows the principle of all or none.

What is Commit, Rollback and Savepoint?

In a database transaction, there are three main commands that are used to manage the transaction:

1. **Commit**: A commit command is used to permanently save the changes made by a transaction to the database. When a transaction is committed, all changes made by the transaction are made permanent and cannot be rolled back.
2. **Rollback**: A rollback command is used to undo the changes made by a transaction and return the database to its state before the transaction began. When a transaction is rolled back, all changes made by the transaction are discarded and the database is returned to its previous state.
3. **Savepoint**: A savepoint command is used to mark a specific point within a transaction where a rollback can be performed. This allows for partial rollbacks of a transaction, where only changes made after the savepoint are undone, while changes made before the savepoint are still committed to the database.

What is Autocommit?

Autocommit is a feature of database management systems (DBMS) that automatically commits each individual database transaction as soon as it is completed, rather than requiring an explicit commit command to be issued.

When Autocommit is enabled, each individual SQL statement issued against the database is treated as a separate transaction and is committed immediately after it is executed. This means that each SQL statement becomes a separate, independent transaction, and its effects are immediately visible to other users.

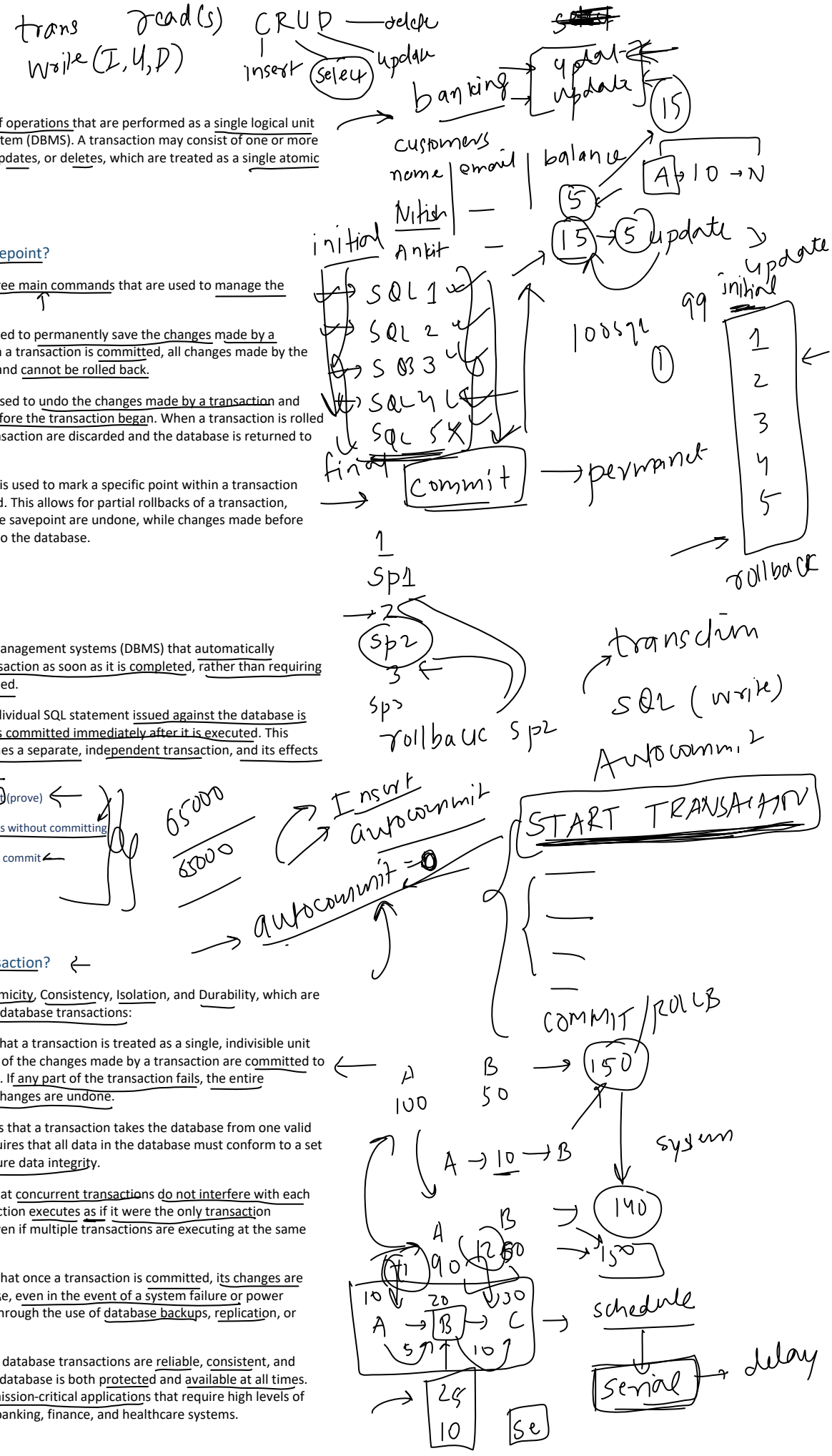
- Each SQL write statement is Autocommit (prove)
- set Autocommit = 0 and (show)
- START TRANSACTION -> show operations without committing
- START TRANSACTION -> with commit
- START TRANSACTION -> all or none with commit
- rollback
- rollback with savepoint
- rollback and commit together

What is ACID properties of a Transaction?

ACID is an acronym that stands for Atomicity, Consistency, Isolation, and Durability, which are a set of properties that ensure reliable database transactions:

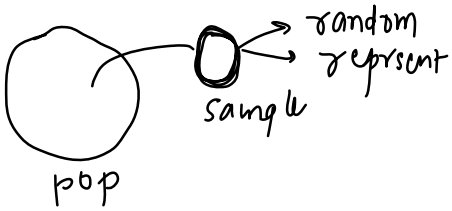
1. **Atomicity**: This property ensures that a transaction is treated as a single, indivisible unit of work. This means that either all of the changes made by a transaction are committed to the database, or none of them are. If any part of the transaction fails, the entire transaction is rolled back, and all changes are undone.
2. **Consistency**: This property ensures that a transaction takes the database from one valid state to another valid state. It requires that all data in the database must conform to a set of rules, or constraints, which ensure data integrity.
3. **Isolation**: This property ensures that concurrent transactions do not interfere with each other. It requires that each transaction executes as if it were the only transaction executing against the database, even if multiple transactions are executing at the same time.
4. **Durability**: This property ensures that once a transaction is committed, its changes are permanently stored in the database, even in the event of a system failure or power outage. This is typically achieved through the use of database backups, replication, or other forms of data redundancy.

Together, these properties ensure that database transactions are reliable, consistent, and accurate, and that the data stored in a database is both protected and available at all times. The ACID properties are essential for mission-critical applications that require high levels of data integrity and availability, such as banking, finance, and healthcare systems.



Some Terms

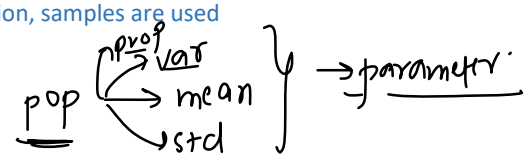
30 March 2023 07:09



Population Vs Sample

Population: A population is the entire group or set of individuals, objects, or events that a researcher wants to study or draw conclusions about. It can be people, animals, plants, or even inanimate objects, depending on the context of the study. The population usually represents the complete set of possible data points or observations.

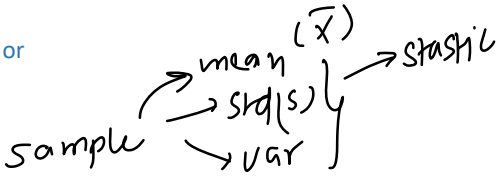
Sample: A sample is a subset of the population that is selected for study. It is a smaller group that is intended to be representative of the larger population. Researchers collect data from the sample and use it to make inferences about the population as a whole. Since it is often impractical or impossible to collect data from every member of a population, samples are used as an efficient and cost-effective way to gather information.



Parameter Vs ~~Estima~~te

statistics

Parameter: A parameter is a numerical value that describes a characteristic of a population. Parameters are usually denoted using Greek letters, such as μ (mu) for the population mean or σ (sigma) for the population standard deviation. Since it is often difficult or impossible to obtain data from an entire population, parameters are usually unknown and must be estimated based on available sample data.



$\bar{x} \rightarrow \mu$

Statistic: A statistic is a numerical value that describes a characteristic of a sample, which is a subset of the population. By using statistics calculated from a representative sample, researchers can make inferences about the unknown respective parameter of the population. Common statistics include the sample mean (denoted by x̄, pronounced "x-bar"), the sample median, and the sample standard deviation (denoted by s).

sample → population

Inferential Statistics

Inferential statistics is a branch of statistics that focuses on making predictions, estimations, or generalizations about a larger population based on a sample of data taken from that population. It involves the use of probability theory to make inferences and draw conclusions about the characteristics of a population by analysing a smaller subset or sample.

The key idea behind inferential statistics is that it is often impractical or impossible to collect data from every member of a population, so instead, we use a representative sample to make inferences about the entire group. Inferential statistical techniques include hypothesis testing, confidence intervals, and regression analysis, among others.

These methods help researchers answer questions like:

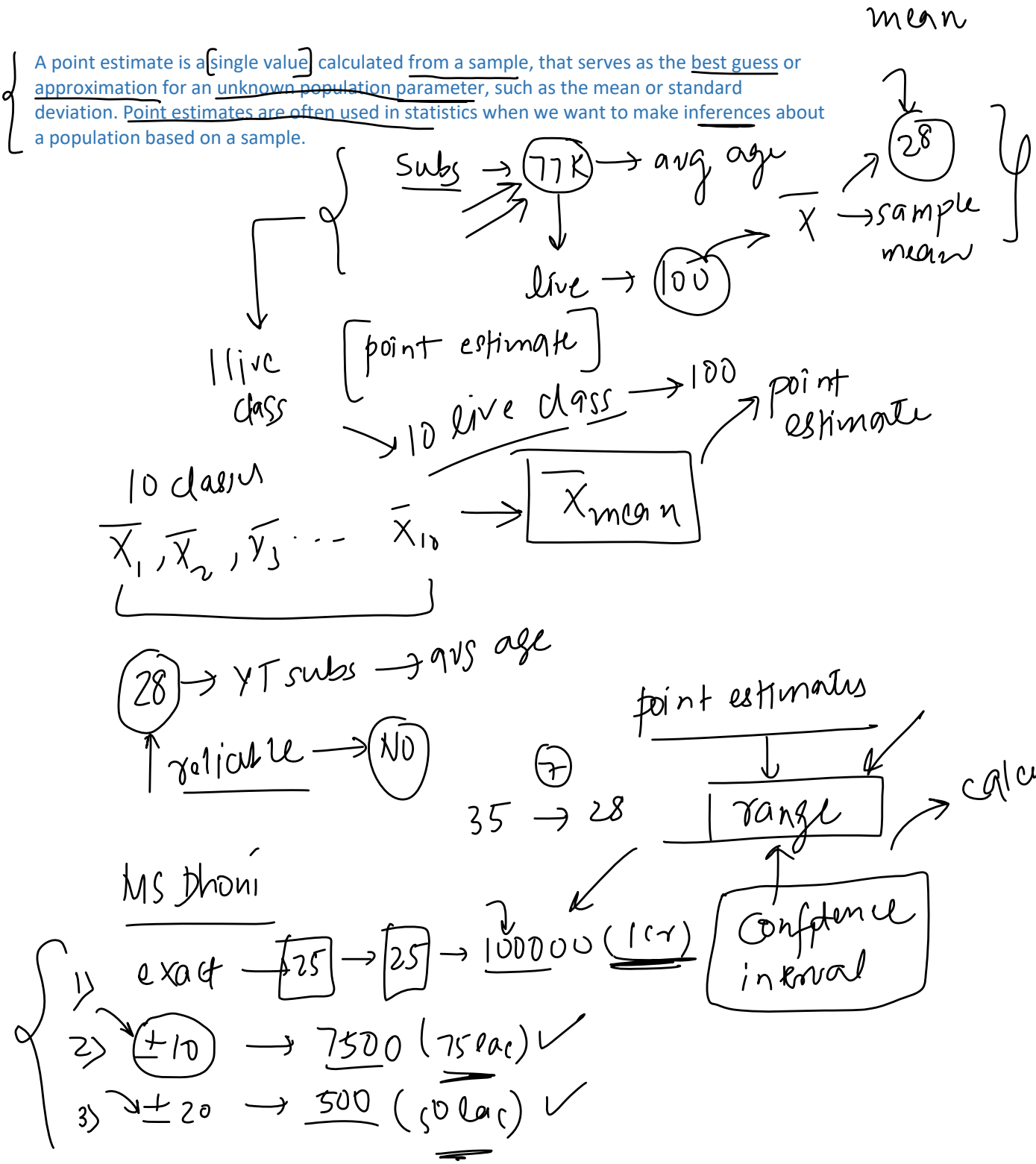
a. Is there a significant difference between two groups?
b. Can we predict the outcome of a variable based on the values of other variables?
c. What is the relationship between two or more variables?

Inferential statistics are widely used in various fields, such as economics, social sciences, medicine, and natural sciences, to make informed decisions and guide policy based on limited data.

Point Estimate

30 March 2023 07:19

A point estimate is a single value calculated from a sample, that serves as the best guess or approximation for an unknown population parameter, such as the mean or standard deviation. Point estimates are often used in statistics when we want to make inferences about a population based on a sample.



Confidence Interval

30 March 2023 07:18

Confidence interval, in simple words, is a range of values within which we expect a particular population parameter, like a mean, to fall. It's a way to express the uncertainty around an estimate obtained from a sample of data.

Confidence level, usually expressed as a percentage like 95%, indicates how sure we are that the true value lies within the interval.

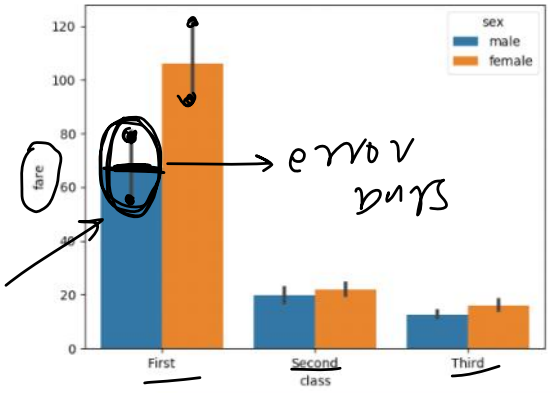
Confidence Interval = [Point Estimate] ± [Margin of Error]

Ways to calculate CI:

Confidence Interval is created for Parameters and not statistics. Statistics help us get the confidence interval for a parameter.

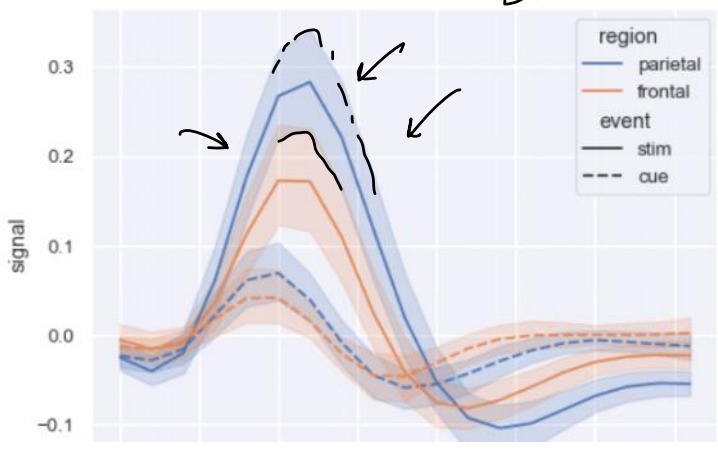
Examples of CT usage

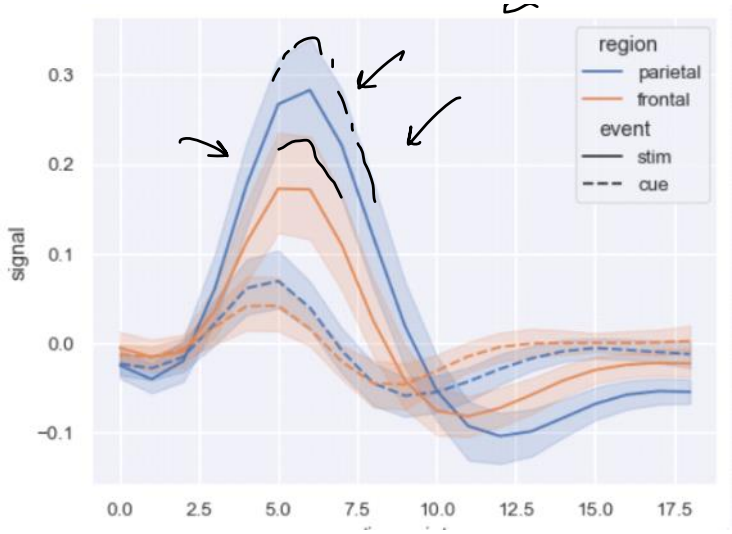
seaborn → bar plot



error bars

(CI)



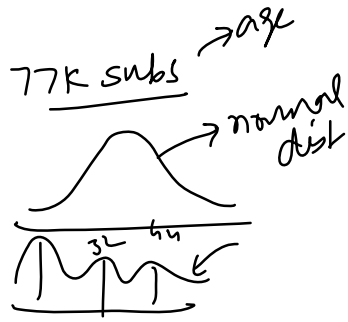

region
parietal
frontal
event
stim
cue
signal
0.3
0.2
0.1
0.0
−0.1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5

Confidence Interval (Sigma Known)

30 March 2023 07:13

Assumptions

1. Random sampling: The data must be collected using a random sampling method to ensure that the sample is representative of the population. This helps to minimize biases and ensures that the results can be generalized to the entire population.
2. Known population standard deviation: The population standard deviation (σ) must be known or accurately estimated. In practice, the population standard deviation is often unknown, and the sample standard deviation (s) is used as an estimate. However, if the sample size is large enough, the sample standard deviation can provide a reasonably accurate approximation.
3. Normal distribution or large sample size: The Z-procedure assumes that the underlying population is normally distributed. However, if the population distribution is not normal, the Central Limit Theorem can be applied when the sample size is large (usually, sample size $n \geq 30$ is considered large enough). According to the Central Limit Theorem, the sampling distribution of the sample mean will approach a normal distribution as the sample size increases, regardless of the shape of the population distribution.



A (1 - alpha)*100% Confidence Interval for mu:

$$CI = \bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

$Z \rightarrow ?$

$(1-\alpha) \rightarrow$ confidence level

$(1-\alpha) \rightarrow 95\%$

$\sigma \rightarrow$ std pop

$n \rightarrow$ sample size $\rightarrow 100$

Intuition

Point estimate $(\bar{X}) \rightarrow$ CLT

point estimate $(\bar{x})$ → CI

10 live class → 50 → avg class

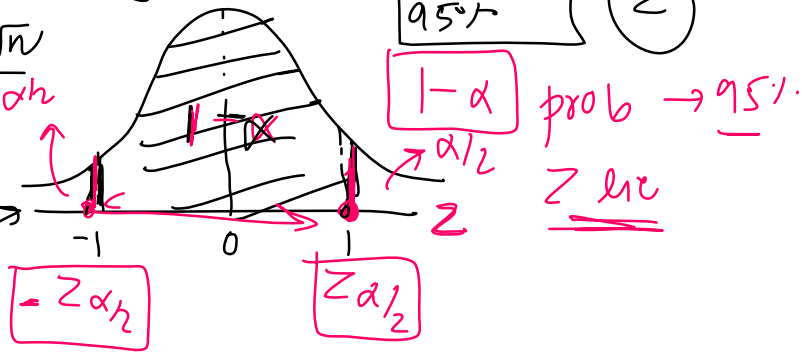
$[\bar{x}_1, \bar{x}_2 \ldots \bar{x}_{10}]$ → sampling dist of sample mean

normally dist → (z) → $N(0,1)$

$\bar{x}$ μ $\sigma/\sqrt{n}$

$z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

95% = $(1-\alpha)$ sure



$-z_{2.5} \longleftrightarrow z_{2.5}$

95%

$\alpha = 5$

$CI = \bar{x} \pm (z_{\alpha/2}) \dfrac{\sigma}{\sqrt{n}}$ (95)

$$P\left(-z_{\alpha/2} < z < z_{\alpha/2}\right) = 1-\alpha$$

$$P\left(-z_{\alpha/2} < \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) = 1-\alpha$$

μ → CI

$$P\left(-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \bar{x} - \mu < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha$$

$$P\left(-\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < -\mu < -\bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1-\alpha$$

$P\left(-X - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < -\mu < \quad + z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \right)$

$P\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \right) = 1-\alpha$

μ → fixed

95%

$1-\alpha$ → 0.95

$\alpha = 0.50$

$\frac{\alpha}{2} = 0.250$

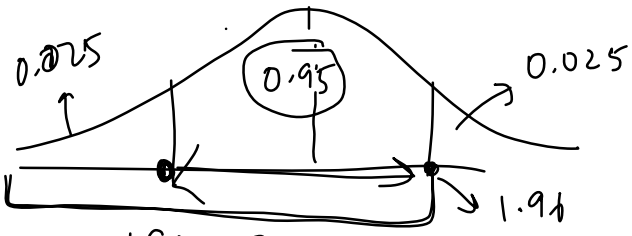
$\bar{X}$ → sample

CI $\mu = \bar{X} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$

confidence $(1-\alpha)$ → 95%

$\mu = \bar{X} \pm z_{0.250}\frac{\sigma}{\sqrt{n}}$ (z)

950
025
975



range → confidence interval

$\mu = \bar{X} \pm 1.96\frac{\sigma}{\sqrt{n}}$ → CI with a confidence of 95%.

$z_{\alpha/2} \rightarrow 1.96$

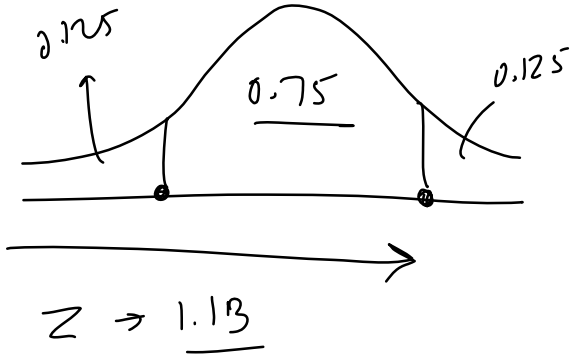
50%

75%

99%

0.87

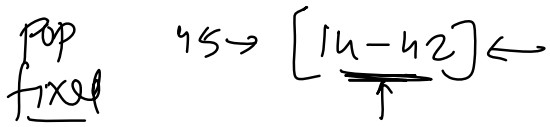
$z_{\alpha/2}$



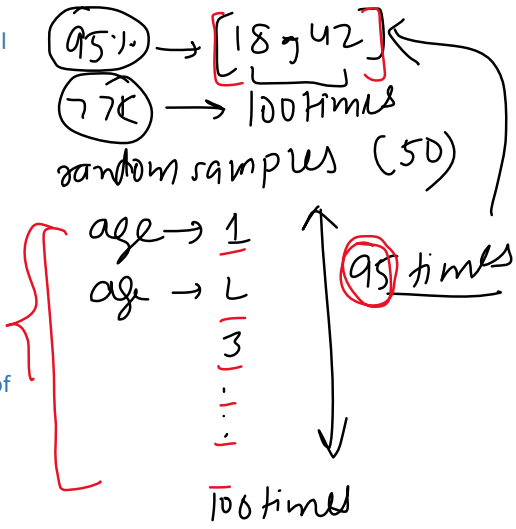
$z \rightarrow 1.15$

Interpreting Confidence Interval

30 March 2023 08:33

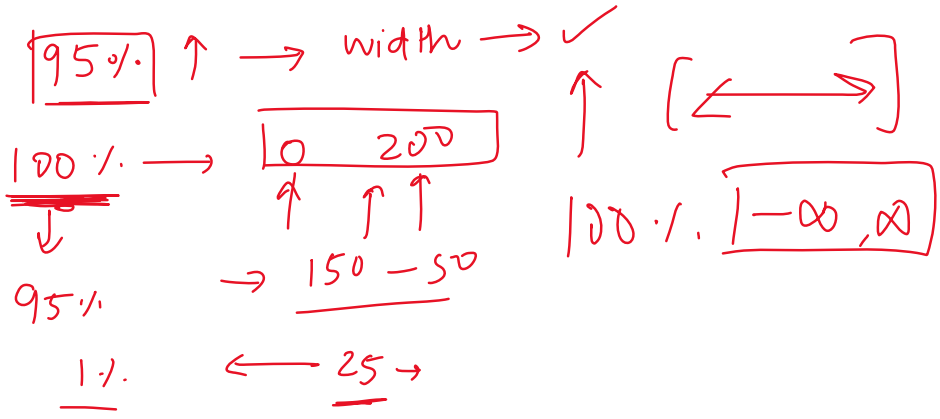


A confidence interval is a range of values within which a population parameter, such as the population mean, is estimated to lie with a certain level of confidence. The confidence interval provides an indication of the precision and uncertainty associated with the estimate. To interpret the confidence interval values, consider the following points:



1. **Confidence level**: The confidence level (commonly set at 90%, 95%, or 99%) represents the probability that the confidence interval will contain the true population parameter if the sampling and estimation process were repeated multiple times. For example, a 95% confidence interval means that if you were to draw 100 different samples from the population and calculate the confidence interval for each, approximately 95 of those intervals would contain the true population parameter.
2. **Interval range**: The width of the confidence interval gives an indication of the precision of the estimate. A narrower confidence interval suggests a more precise estimate of the population parameter, while a wider interval indicates greater uncertainty. The width of the interval depends on the sample size, variability in the data, and the desired level of confidence.
3. **Interpretation**: To interpret the confidence interval values, you can say that you are "X% confident that the true population parameter lies within the range (lower limit, upper limit)." Keep in mind that this statement is about the interval, not the specific point estimate, and it refers to the confidence level you chose when constructing the interval.

What is the trade-off



Factors Affecting Margin of Error

30 March 2023 07:15

1. Confidence Level (1-alpha)
2. Sample Size
3. Population Standard Deviation

upper − lower / 2 → margin

CI = point estimate ± margin of error

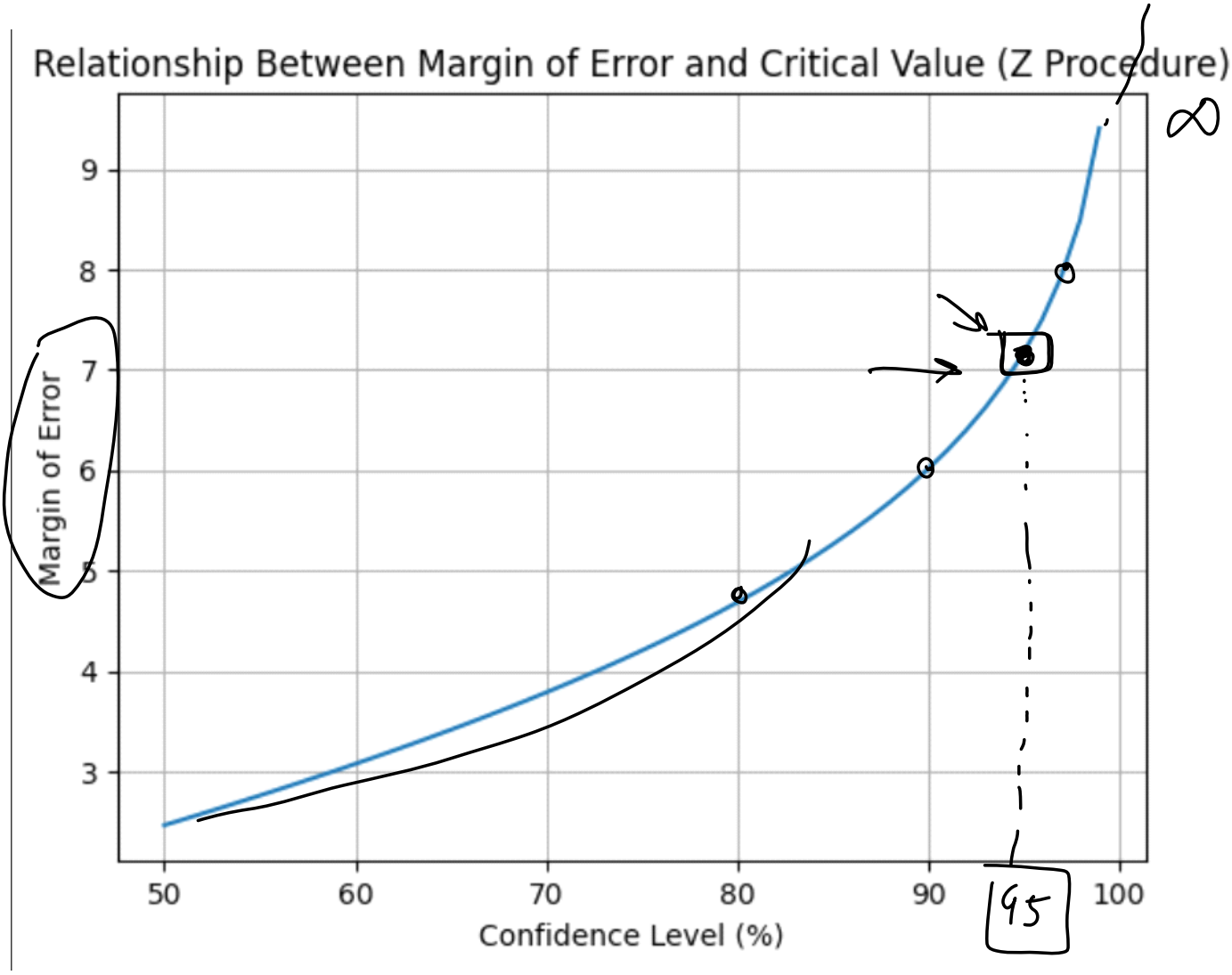
$= \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$\bar{x}$: sample mean (1)

$Z_{\alpha/2}$: critical value (1)

σ: pop std (2)

$\sqrt{n}$: sample size (3)



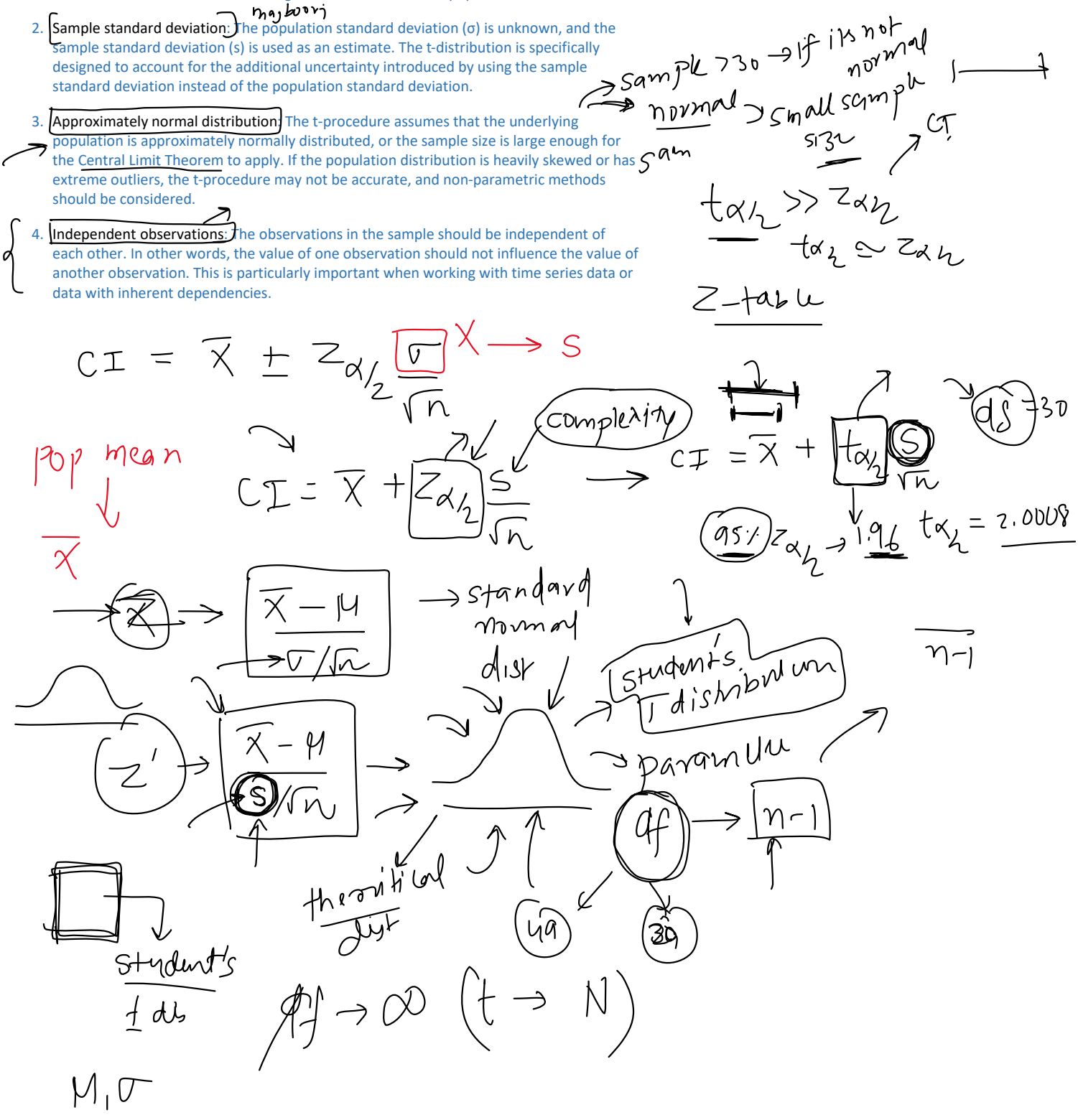
Confidence Interval (Sigma not known)

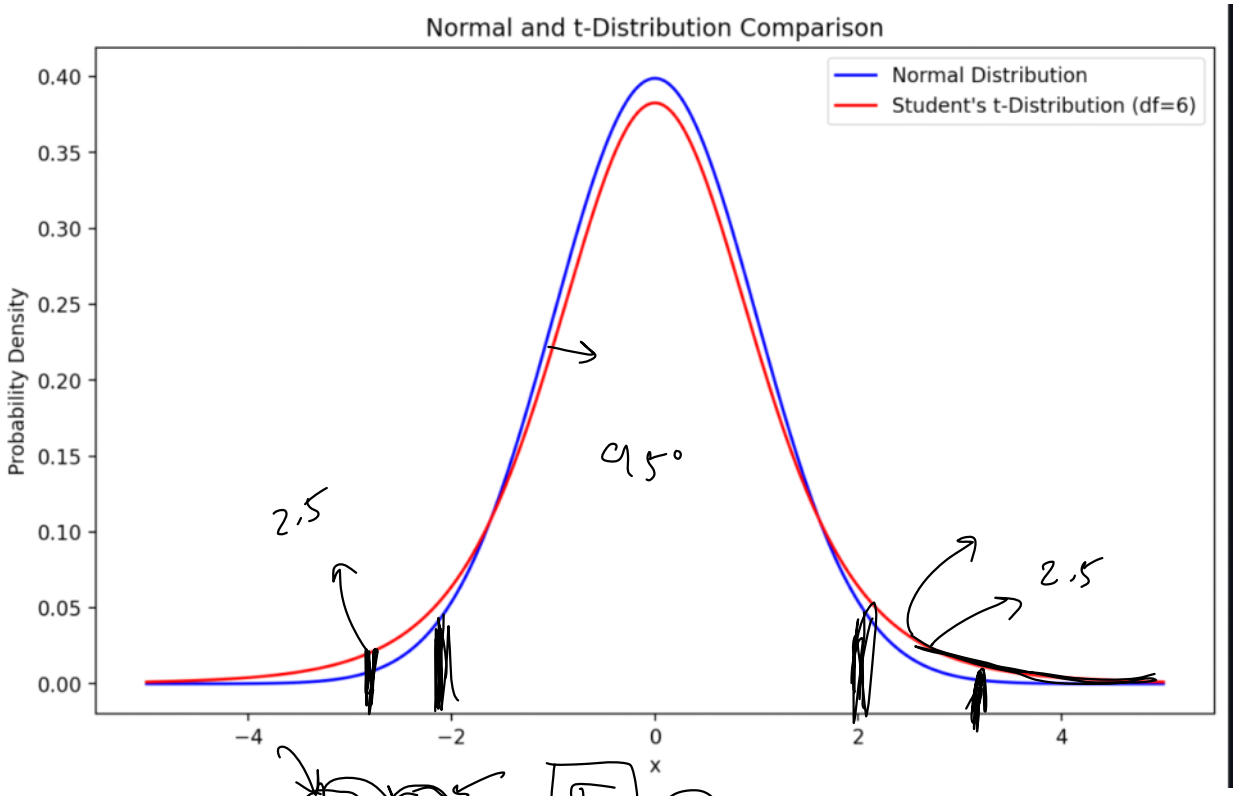
30 March 2023 07:15

Using the t procedure

Assumptions

1. **Random sampling:** The data must be collected using a random sampling method to ensure that the sample is representative of the population. This helps to minimize biases and ensures that the results can be generalized to the entire population.
2. **Sample standard deviation:** The population standard deviation (σ) is unknown, and the sample standard deviation (s) is used as an estimate. The t-distribution is specifically designed to account for the additional uncertainty introduced by using the sample standard deviation instead of the population standard deviation.
3. **Approximately normal distribution:** The t-procedure assumes that the underlying population is approximately normally distributed, or the sample size is large enough for the Central Limit Theorem to apply. If the population distribution is heavily skewed or has extreme outliers, the t-procedure may not be accurate, and non-parametric methods should be considered.
4. **Independent observations:** The observations in the sample should be independent of each other. In other words, the value of one observation should not influence the value of another observation. This is particularly important when working with time series data or data with inherent dependencies.





$$CI = \bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$$

σ

$$= \bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$CI = \bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \longleftrightarrow CI = \bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

1 Nitish 2 CLT

2 Ankit. Count

$\bar{x}_1$ s_1

$\bar{x}_2$ s_2

s_3

s_{10}

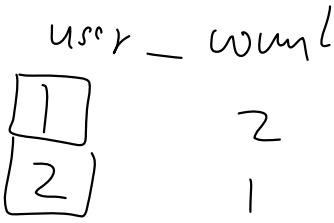
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user_id | name | event | dm

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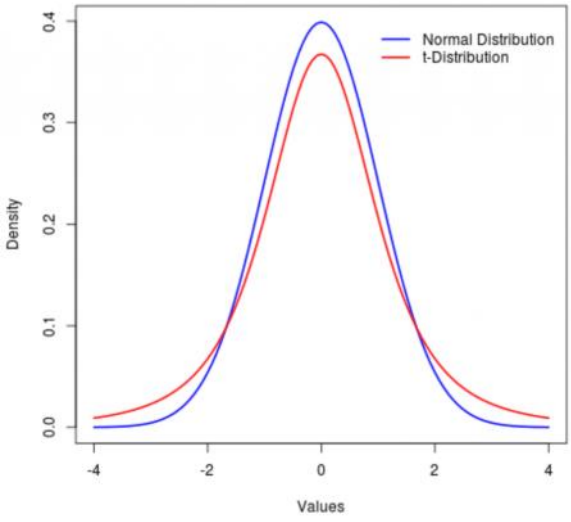
NIL comm

Ankit like


user_id
1
2
2
1

Student's T Distribution

30 March 2023 07:16



Student's t-distribution, or simply the t-distribution, is a probability distribution that arises when estimating the mean of a normally distributed population when the sample size is small and the population standard deviation is unknown. It was introduced by William Sealy Gosset, who published under the pseudonym "Student."

The t-distribution is similar to the normal distribution (also known as the Gaussian distribution or the bell curve) but has heavier tails. The shape of the t-distribution is determined by the degrees of freedom, which is closely related to the sample size (degrees of freedom = sample size - 1). As the degrees of freedom increase (i.e., as the sample size increases), the t-distribution approaches the normal distribution.

In hypothesis testing and confidence interval estimation, the t-distribution is used in place of the normal distribution when the sample size is small (usually less than 30) and the population standard deviation is unknown. The t-distribution accounts for the additional uncertainty that arises from estimating the population standard deviation using the sample standard deviation.

To use the t-distribution in practice, you look up critical t-values from a t-distribution table, which provides values corresponding to specific degrees of freedom and confidence levels (e.g., 95% confidence). These critical t-values are then used to calculate confidence intervals or perform hypothesis tests.

Titanic Case Study

31 March 2023 18:00

Pop → 1360

μ → x

σ → x

CLT → 10 times → 30 size

95% confidence level

inference

1. Why Probability Distributions are important?

01 April 2023 16:42

2. Why divide by n-1?

01 April 2023 16:43

1. Demo of this happening
2. Mathematical Derivation
3. Geometric Intuition

   The "sample standard deviation" is often called "the standard deviation of the sample," but that's not technically correct. Really, it's an *estimate*, derived from the sample, of the standard deviation *of the population*.

   $\bar{x}$ is estimated from the values xi, the points xi are a little closer to $\bar{x}$, on average than to μ. Therefore, the above estimate would underestimate the true population standard deviation!

   When we calculate the sample variance, we use the sample mean as an approximation for the population mean. This approximation usually results in an underestimation of the actual population variance because the sample mean is affected by the sample itself, and it's closer to the sample data points than the true population mean.

4. Why divide by n-1 (Bessel's correction)
5. But what is n-1 and why n-1 only

https://colab.research.google.com/drive/1BXNWprKm3jg6AwXP3JHjkdNpKmuUROxj?usp=sharing

3. Sampling Techniques

01 April 2023 16:43

Probability sampling is a method of selecting samples from a population in a way that each member of the population has a known, non-zero chance of being selected. This approach ensures that the sample is representative of the population and reduces sampling bias. There are several types of probability sampling techniques, including:

1. Simple Random Sampling: Every member of the population has an equal chance of being selected. This can be done with or without replacement.
2. Systematic Sampling: Members of the population are chosen at regular intervals, usually using a random starting point. For example, selecting every 10th individual in a list.
3. Stratified Sampling: The population is divided into homogeneous subgroups (strata) based on specific characteristics, such as age or income. Then, simple random sampling is used to select a proportional number of individuals from each stratum.
4. Cluster Sampling: The population is divided into clusters, usually based on geographic regions or natural groupings. A random sample of clusters is selected, and then all individuals within the chosen clusters are included in the sample.
5. Multistage Sampling: This method combines two or more sampling techniques, often using a hierarchical approach. For example, one might first use cluster sampling to select specific regions, and then use stratified sampling within those regions.

When to use Systematic sampling over SRS

Systematic sampling can be advantageous over simple random sampling (SRS) in certain situations. Here are some cases where you might prefer to use systematic sampling over SRS:

a. When you need a more evenly distributed sample: Systematic sampling ensures that the sample is spread out evenly across the entire population. This can be particularly useful when there is a pattern or trend in the population, and you want to make sure that the sample covers the entire range of the population.

b. When the population is ordered or sequential: In cases where the population is ordered or has a natural sequence (e.g., items on a production line, patients in a waiting list, or data points in time series), systematic sampling can be more practical and efficient than SRS. It helps to capture periodic patterns or trends that might be present in the population.

c. When data collection is easier with systematic sampling: In some situations, the logistics of data collection may be more manageable with systematic sampling. For example, if you are surveying households in a neighbourhood, it may be more practical to interview every nth household rather than randomly selecting households from a list. This can reduce travel time and effort for the survey team.

d. When simple random sampling is difficult or impossible: In cases where the population list is not readily available or it is difficult to generate random numbers (e.g., in remote areas with limited resources), systematic sampling can be a more feasible alternative to SRS.

Likelihood Vs Probability

01 April 2023 16:43

Expected Value Vs Expected Variance

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Statistical Moments

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Spearman Correlation Coefficient

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Discrete Uniform Distribution

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Poisson Distribution

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What is Degree of Freedom?

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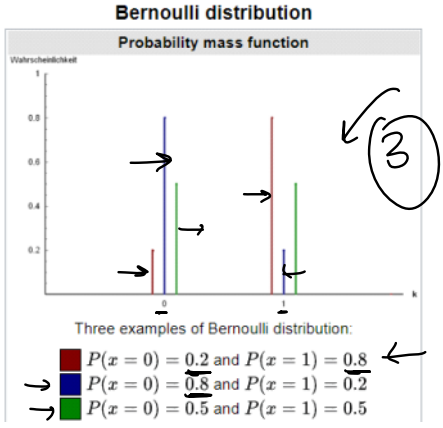
Bernoulli Distribution

27 March 2023 16:06

Bernoulli distribution is a probability distribution that models a binary outcome, where the outcome can be either success (represented by the value 1) or failure (represented by the value 0). The Bernoulli distribution is named after the Swiss mathematician Jacob Bernoulli, who first introduced it in the late 1600s.

The Bernoulli distribution is characterized by a single parameter, which is the probability of success, denoted by p. The probability mass function (PMF) of the Bernoulli distribution is:

$$\text{pmf} = P(X=x) = p^{x}(1-p)^{1-x}$$



The Bernoulli distribution is commonly used in machine learning for modelling binary outcomes, such as whether a customer will make a purchase or not, whether an email is spam or not, or whether a patient will have a certain disease or not.

Binomial Distribution

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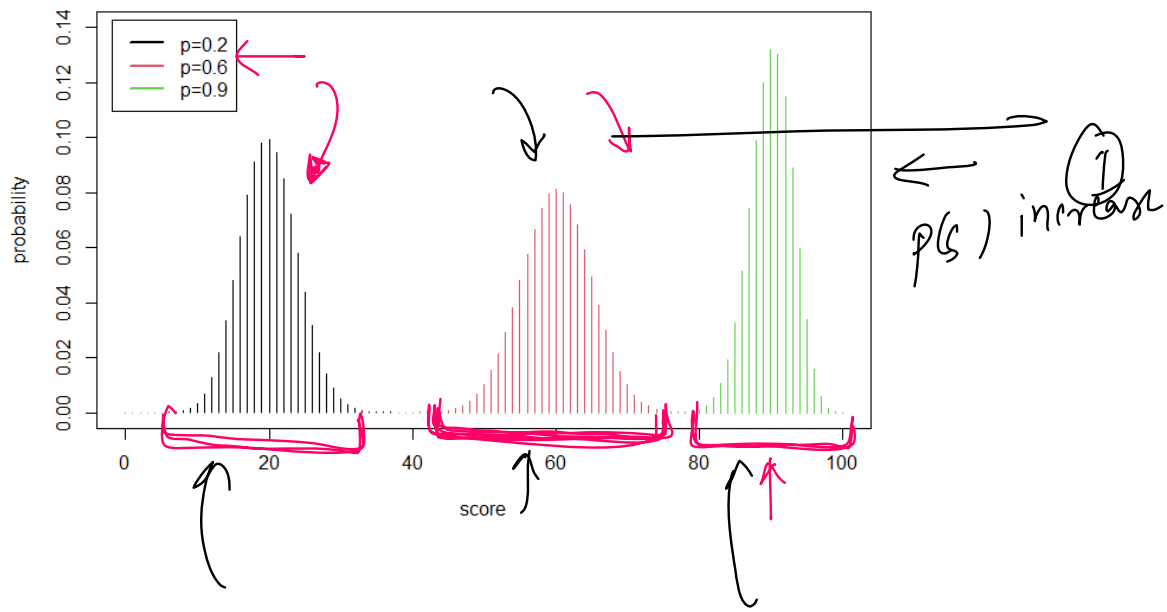
Binomial distribution is a probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials with two possible outcomes (often called "success" and "failure"), where the probability of success is constant for each trial. The binomial distribution is characterized by two parameters: the number of trials n and the probability of success p.

The Probability of anyone watching this lecture in the future and then liking it is 0.5. What is the probability that:

1. No-one out of 3 people will like it
1. 1 out of 3 people will like it
1. 2 out of 3 people will like it
1. 3 out of 3 people will like it

PDF Formula:

Graph of PDF:



Criteria:

Criteria:

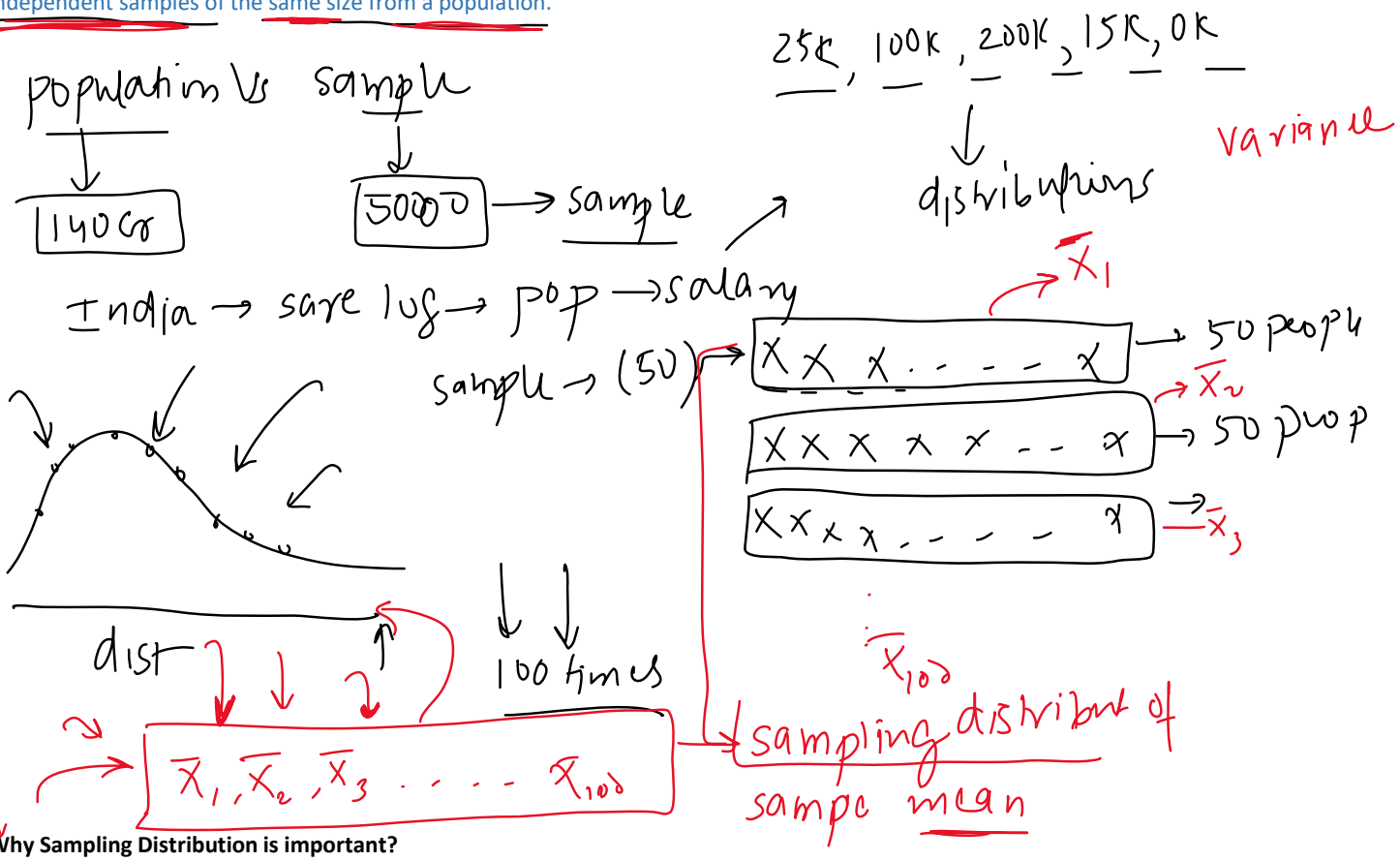
1. The process consists of n trials
2. Only 2 exclusive outcomes are possible, a success and a failure.
3. P(success) = p and P(failure) = 1-p and it is fixed from trial to trial
4. The trials are independent.

1. **Binary classification problems**: In binary classification problems, we often model the probability of an event happening as a binomial distribution. For example, in a spam detection system, we may model the probability of an email being spam or not spam using a binomial distribution.
2. **Hypothesis testing**: In statistical hypothesis testing, we use the binomial distribution to calculate the probability of observing a certain number of successes in a given number of trials, assuming a null hypothesis is true. This can be used to make decisions about whether a certain hypothesis is supported by the data or not.
3. **Logistic regression**: Logistic regression is a popular machine learning algorithm used for classification problems. It models the probability of an event happening as a logistic function of the input variables. Since the logistic function can be viewed as a transformation of a linear combination of inputs, the output of logistic regression can be thought of as a binomial distribution.
4. **A/B testing**: A/B testing is a common technique used to compare two different versions of a product, web page, or marketing campaign. In A/B testing, we randomly assign individuals to one of two groups and compare the outcomes of interest between the groups. Since the outcomes are often binary (e.g., click-through rate or conversion rate), the binomial distribution can be used to model the distribution of outcomes and test for differences between the groups.

Sampling Distribution

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Sampling distribution is a probability distribution that describes the statistical properties of a sample statistic (such as the sample mean or sample proportion) computed from multiple independent samples of the same size from a population.



Why Sampling Distribution is important?

Sampling distribution is important in statistics and machine learning because it allows us to estimate the variability of a sample statistic, which is useful for making inferences about the population. By analysing the properties of the sampling distribution, we can compute confidence intervals, perform hypothesis tests, and make predictions about the population based on the sample data.

central limit theorem

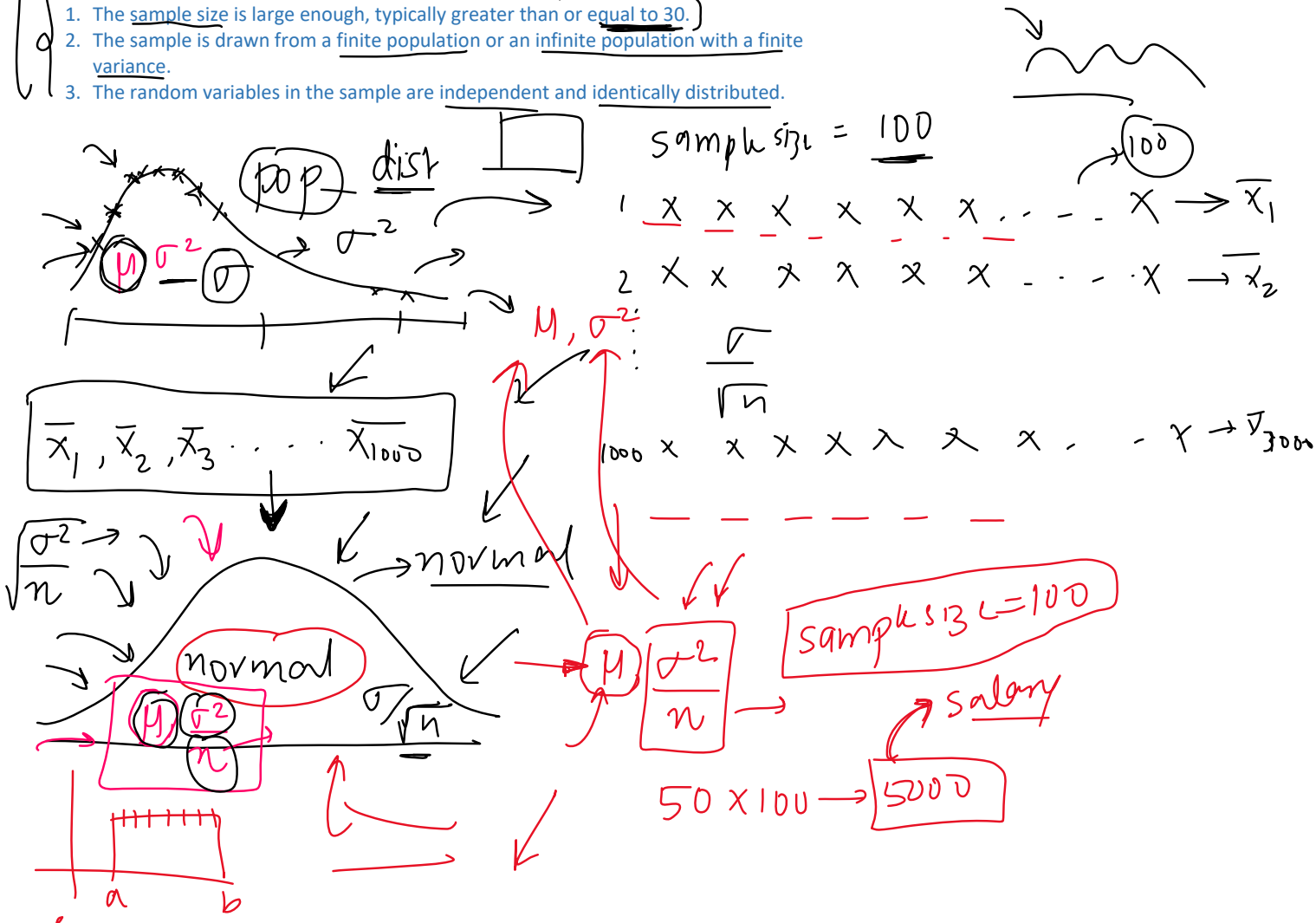
Central Limit Theorem

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The Central Limit Theorem (CLT) states that the distribution of the sample means of a large number of independent and identically distributed random variables will approach a normal distribution, regardless of the underlying distribution of the variables.

The conditions required for the CLT to hold are:

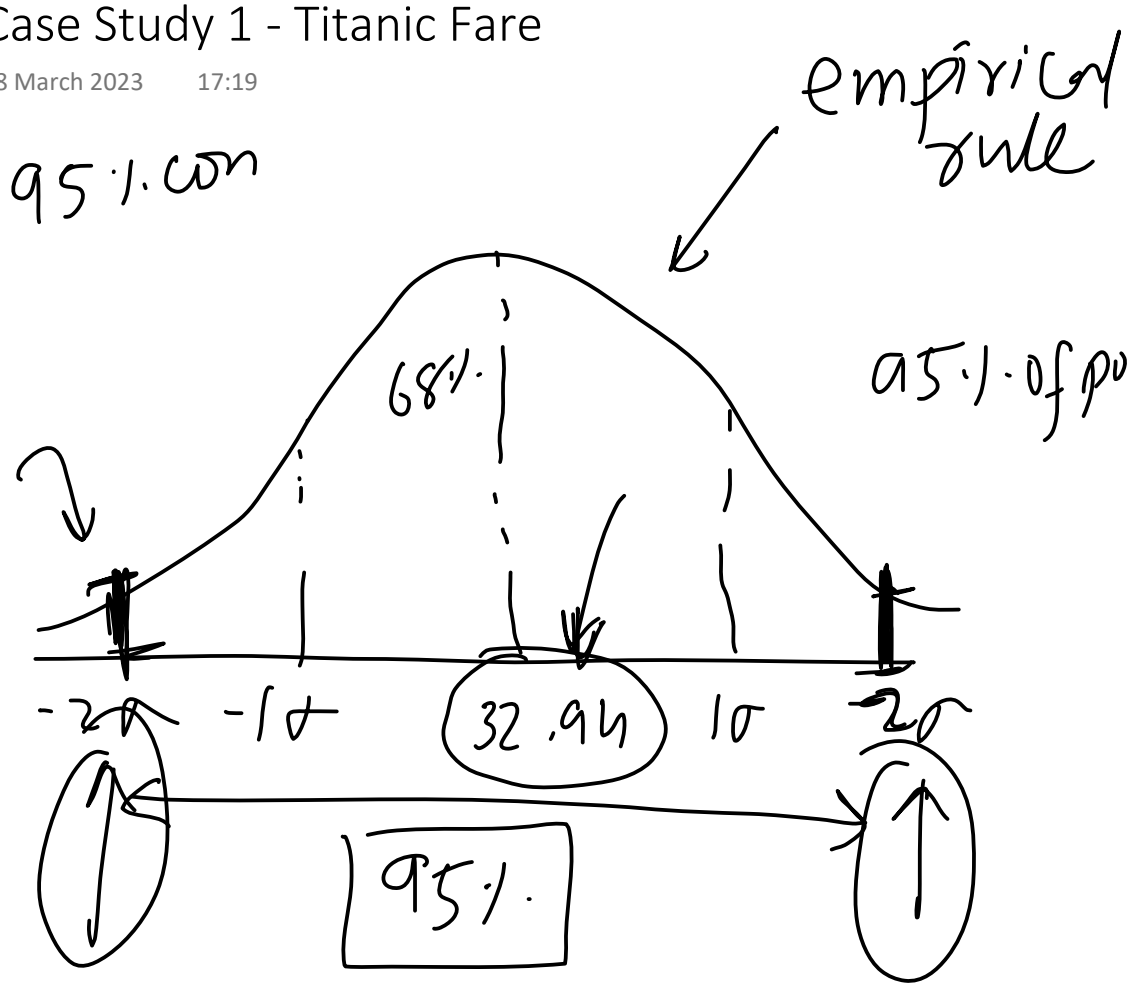
1. The sample size is large enough, typically greater than or equal to 30.
2. The sample is drawn from a finite population or an infinite population with a finite variance.
3. The random variables in the sample are independent and identically distributed.



The CLT is important in statistics and machine learning because it allows us to make probabilistic inferences about a population based on a sample of data. For example, we can use the CLT to construct confidence intervals, perform hypothesis tests, and make predictions about the population mean based on the sample data. The CLT also provides a theoretical justification for many commonly used statistical techniques, such as t-tests, ANOVA, and linear regression.

Case Study 1 - Titanic Fare

28 March 2023 17:19



Case Study - What is the average income of Indians

28 March 2023 15:49

Step-by-step process:

1. Collect multiple random samples of salaries from a representative group of Indians. Each sample should be large enough (usually, n > 30) to ensure the CLT holds. Make sure the samples are representative and unbiased to avoid skewed results.
2. Calculate the sample mean (average salary) and sample standard deviation for each sample.
3. Calculate the average of the sample means. This value will be your best estimate of the population mean (average salary of all Indians).
4. Calculate the standard error of the sample means, which is the standard deviation of the sample means divided by the square root of the number of samples.
5. Calculate the confidence interval around the average of the sample means to get a range within which the true population mean likely falls. For a 95% confidence interval:

   lower_limit = average_sample_means - 1.96 * standard_error
   upper_limit = average_sample_means + 1.96 * standard_error

6. Report the estimated average salary and the confidence interval.

Python code

Remember that the validity of your results depends on the quality of your data and the representativeness of your samples. To obtain accurate results, it's crucial to ensure that your samples are unbiased and representative.